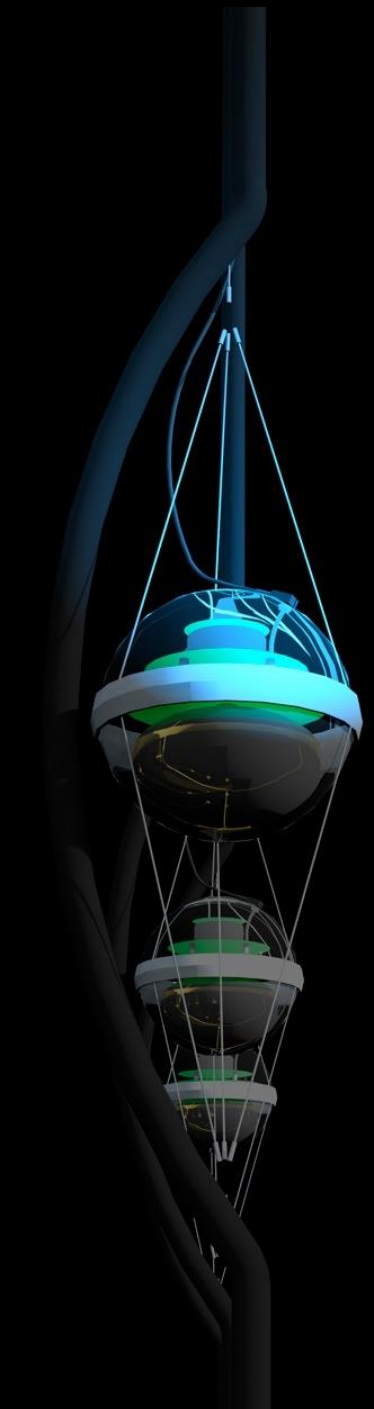


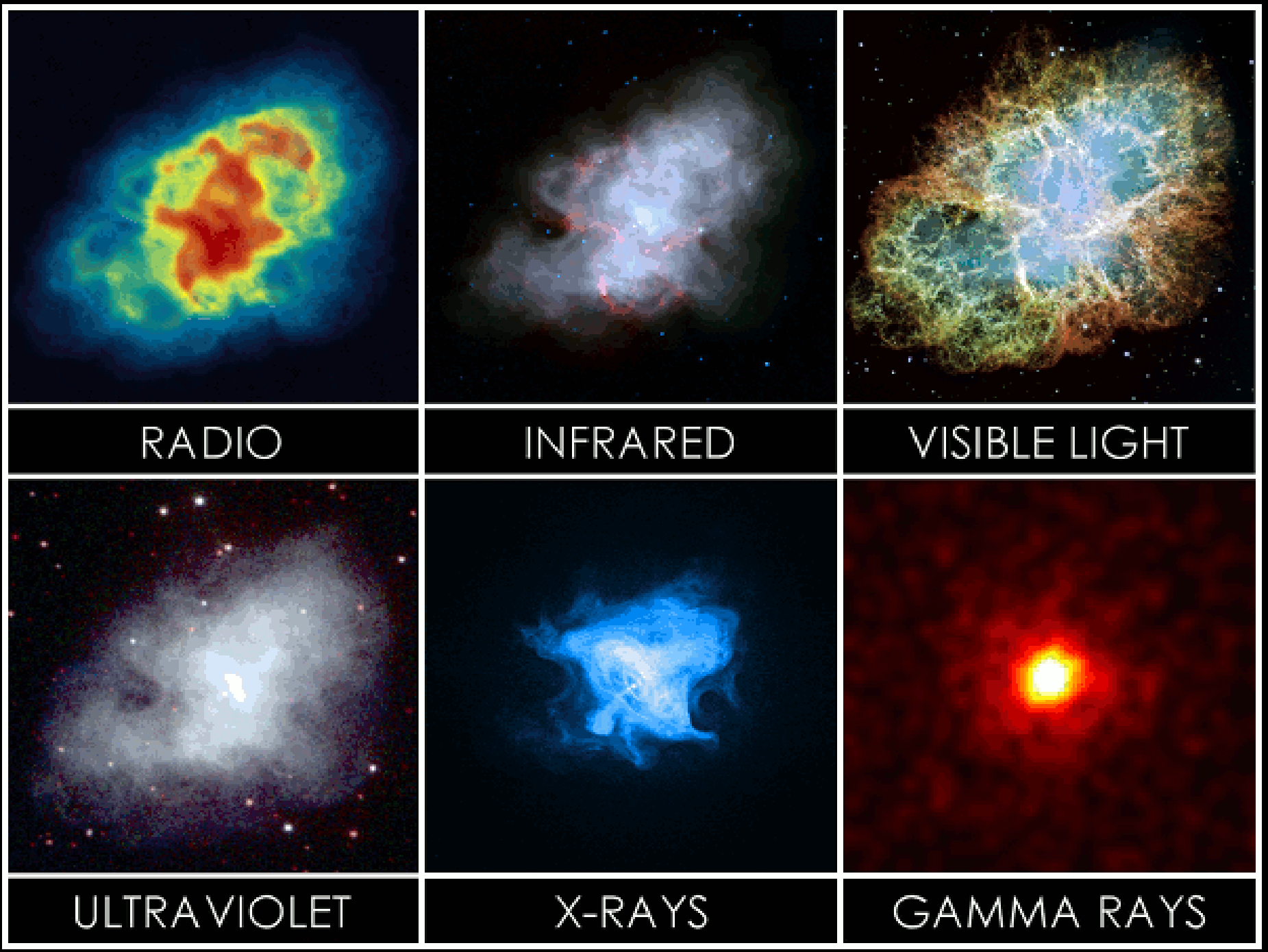
The First Decade of High-Energy Neutrino Astronomy and Neutrino Physics

francis halzen

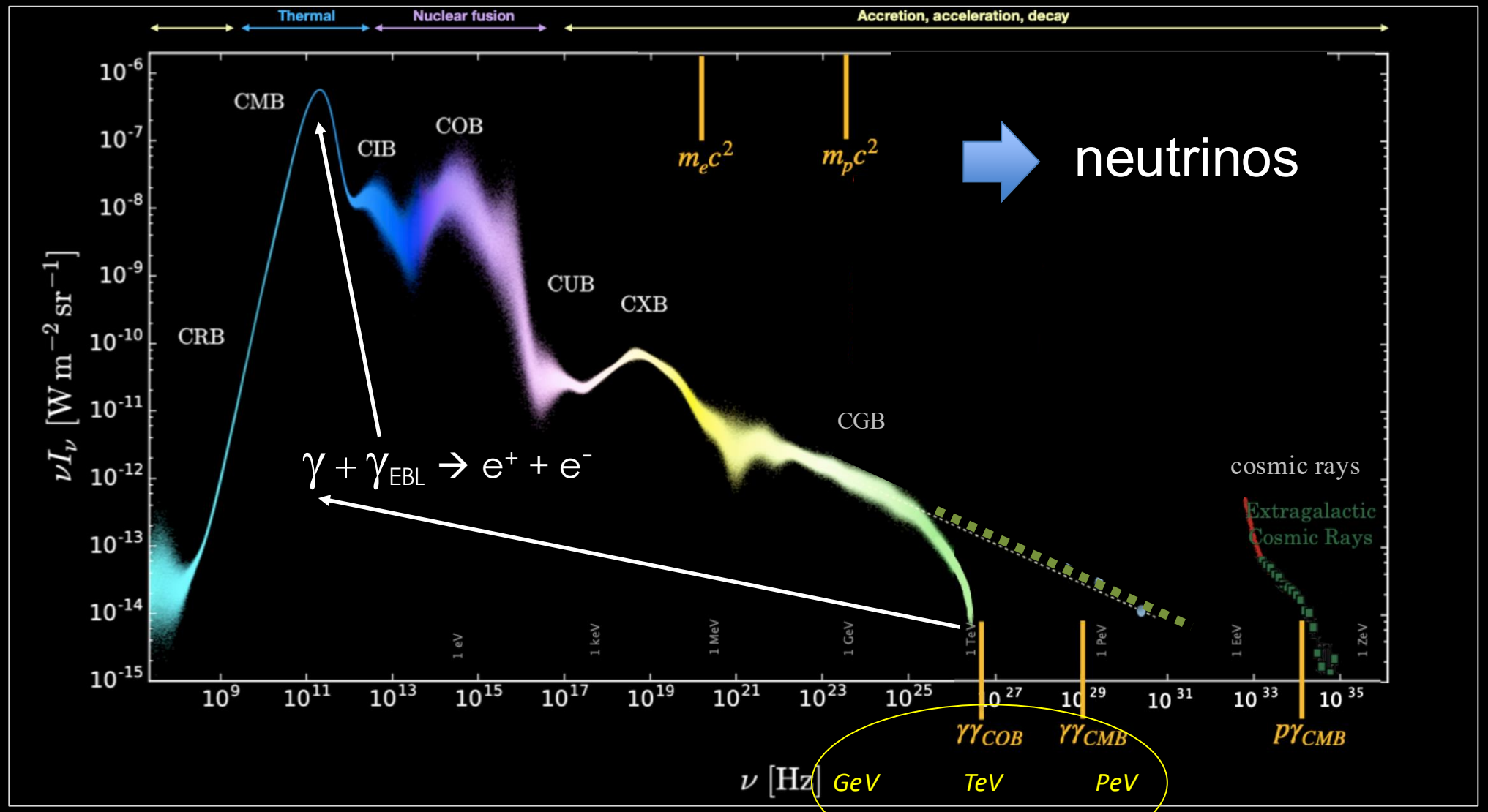
- High-Energy Neutrino Astronomy
- IceCube and the discovery of Cosmic Neutrinos: First Surprises
- The First Cosmic Accelerators: Active Galaxies

IceCube.wisc.edu

multiwavelenghts images of the Crab supernova remnant

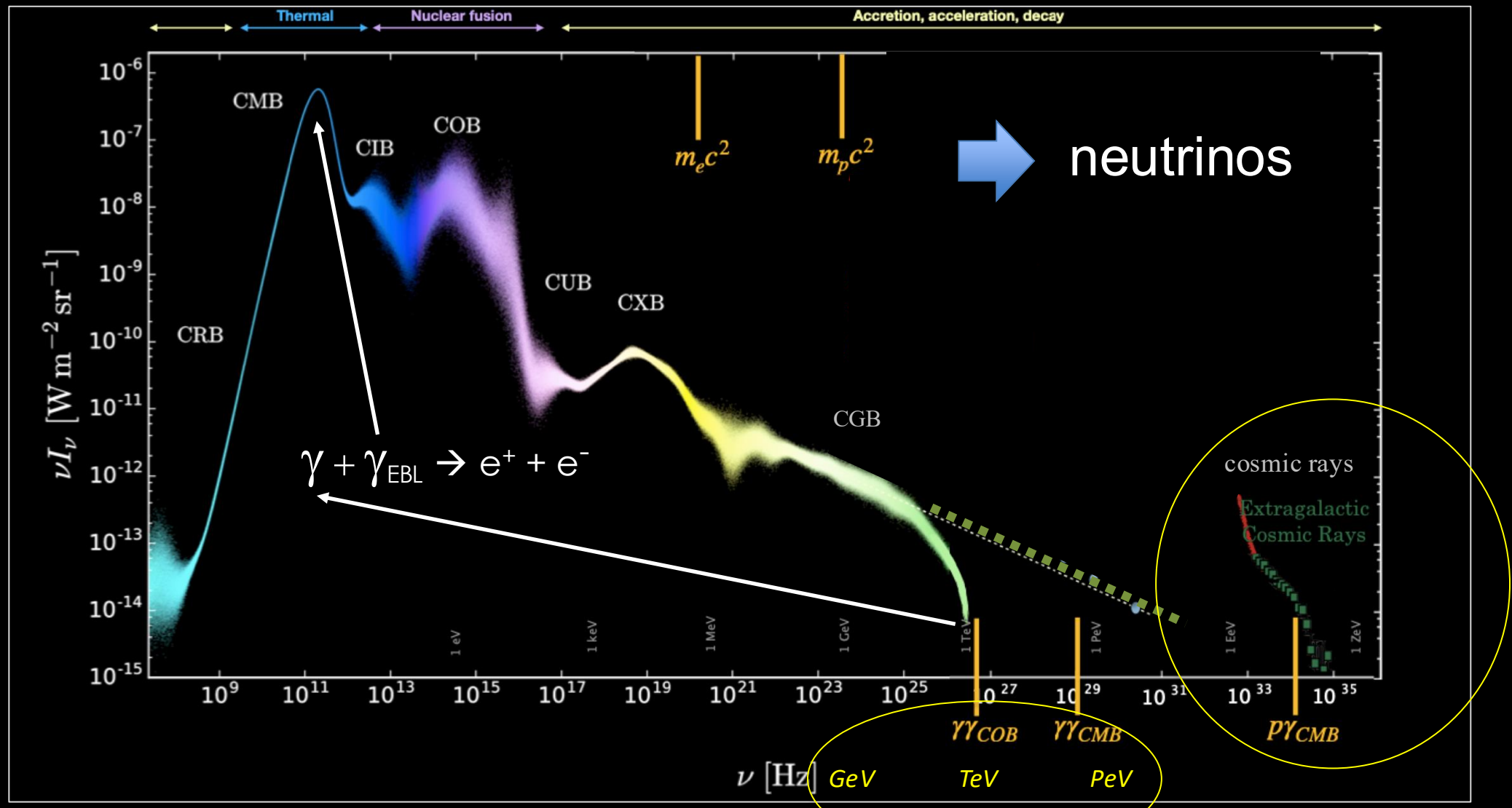


energy in photons in the Universe as a function of their frequency



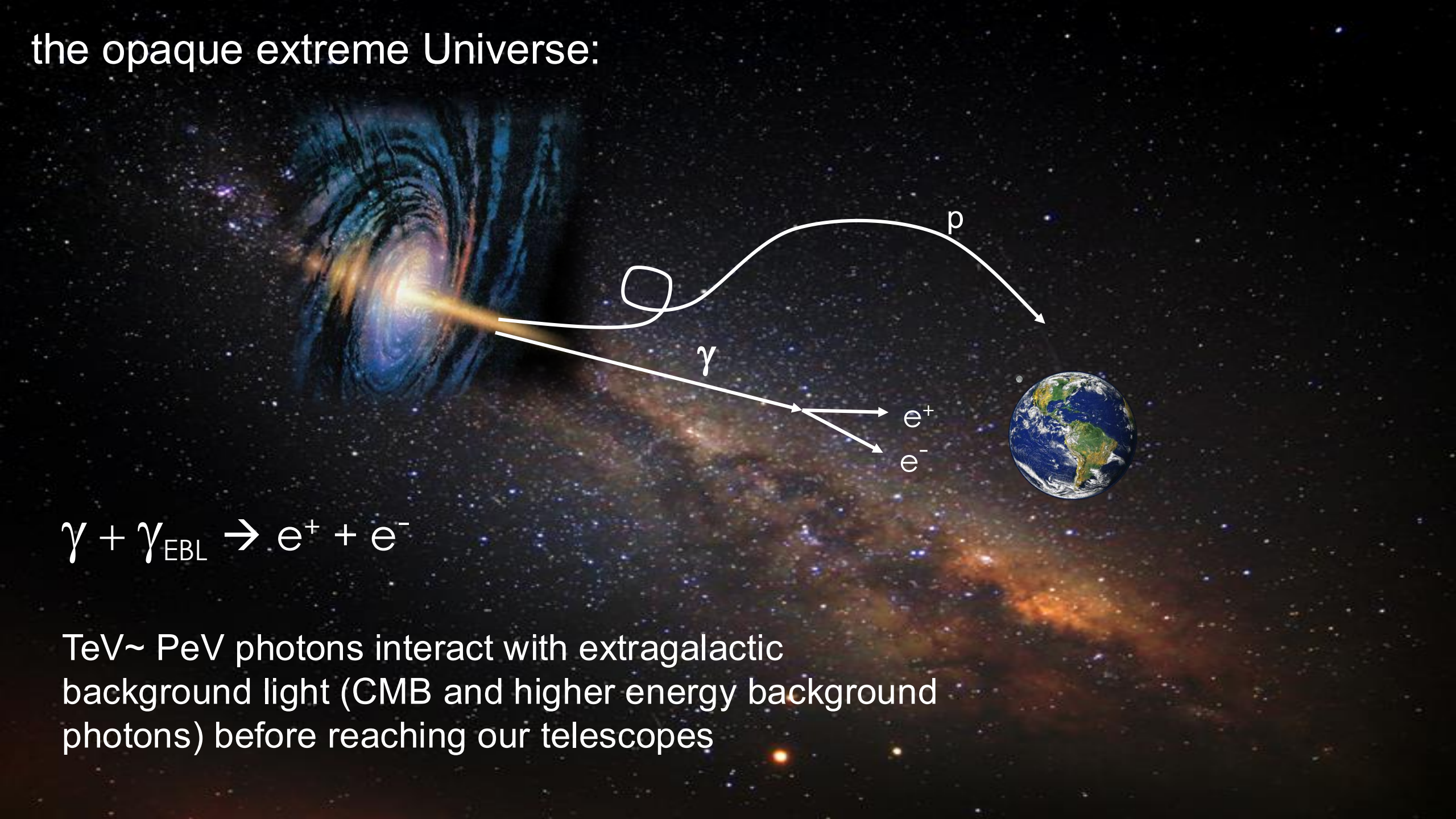
the Universe turns opaque to light at TeV energies: enter neutrinos

energy in photons in the Universe as a function of their frequency



the Universe turns opaque to light at TeV energies: enter neutrinos

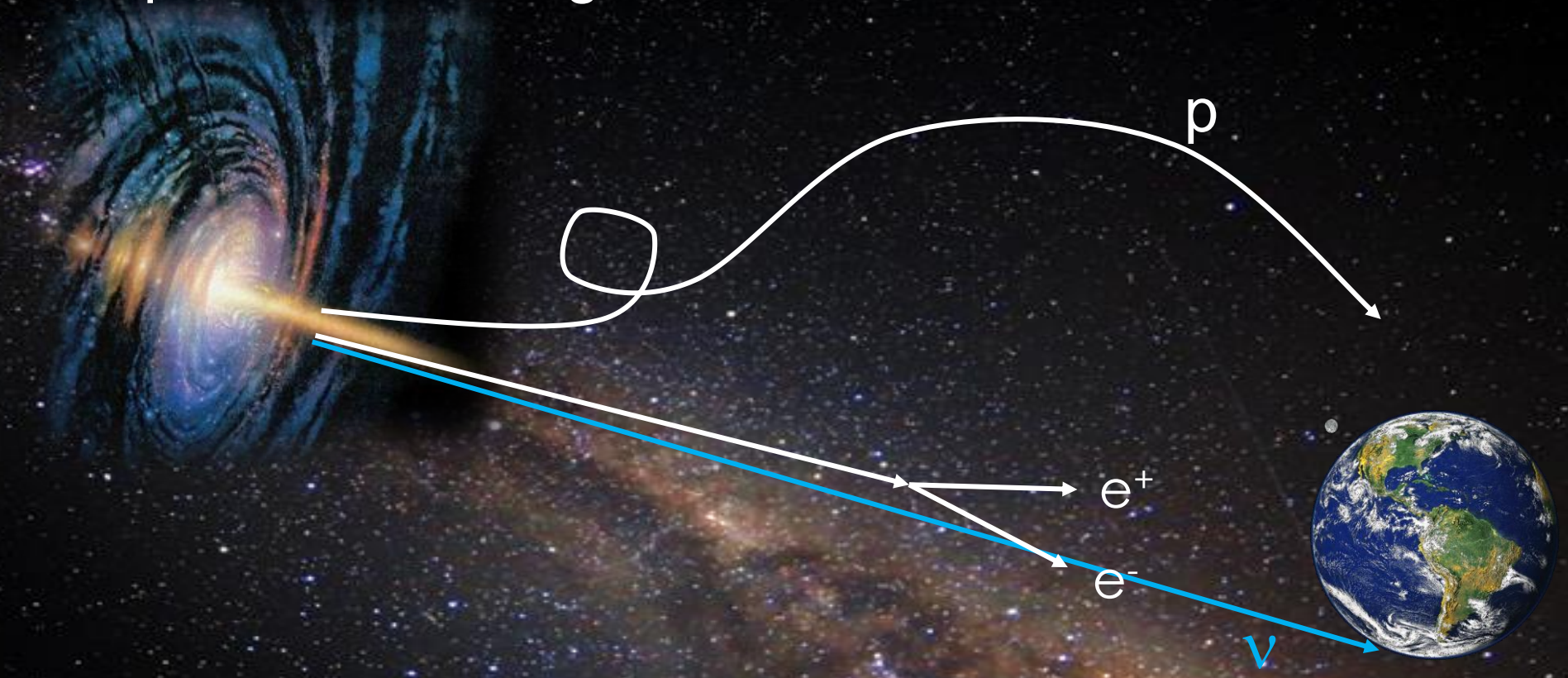
the opaque extreme Universe:



$\gamma + \gamma_{EBL} \rightarrow e^+ + e^-$

TeV~ PeV photons interact with extragalactic background light (CMB and higher energy background photons) before reaching our telescopes

neutrinos as perfect messengers

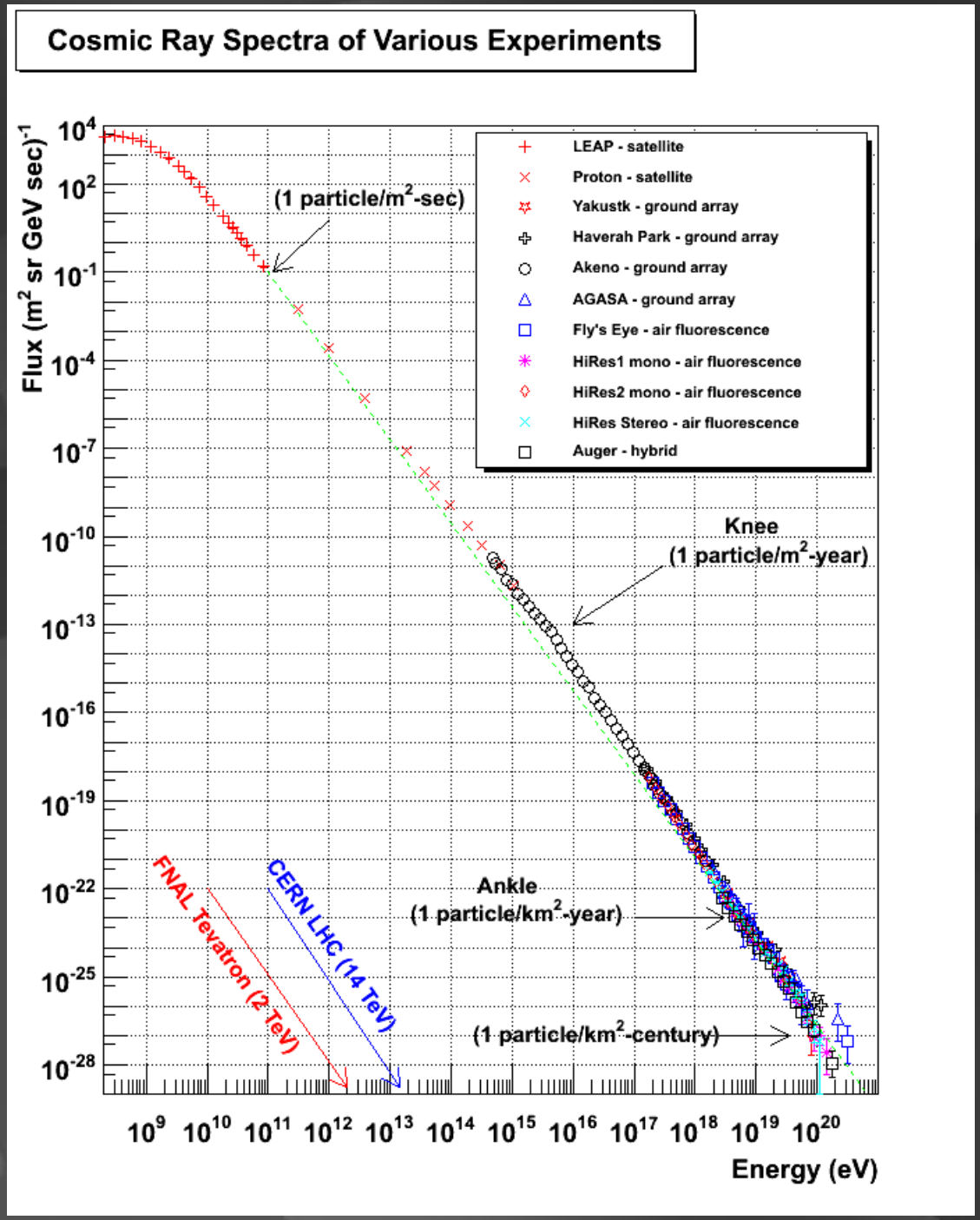


by Standard model particle physics neutrinos reach Earth from the edge of the Universe and from the beginning of time without absorption or deflection

cosmic ray challenge

both the energy of the particles and the total *luminosity* of the accelerators are large

gravitational energy from collapsing object is converted into particle acceleration?

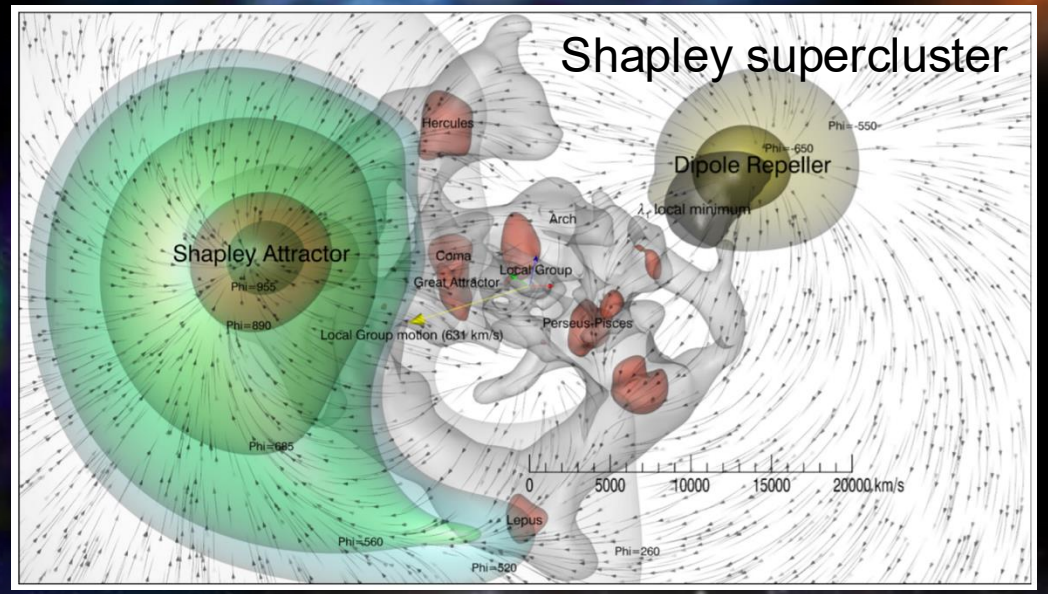


theoretical ideas….

active galactic nuclei

gamma ray burst

tidal disruption



starburst galaxies

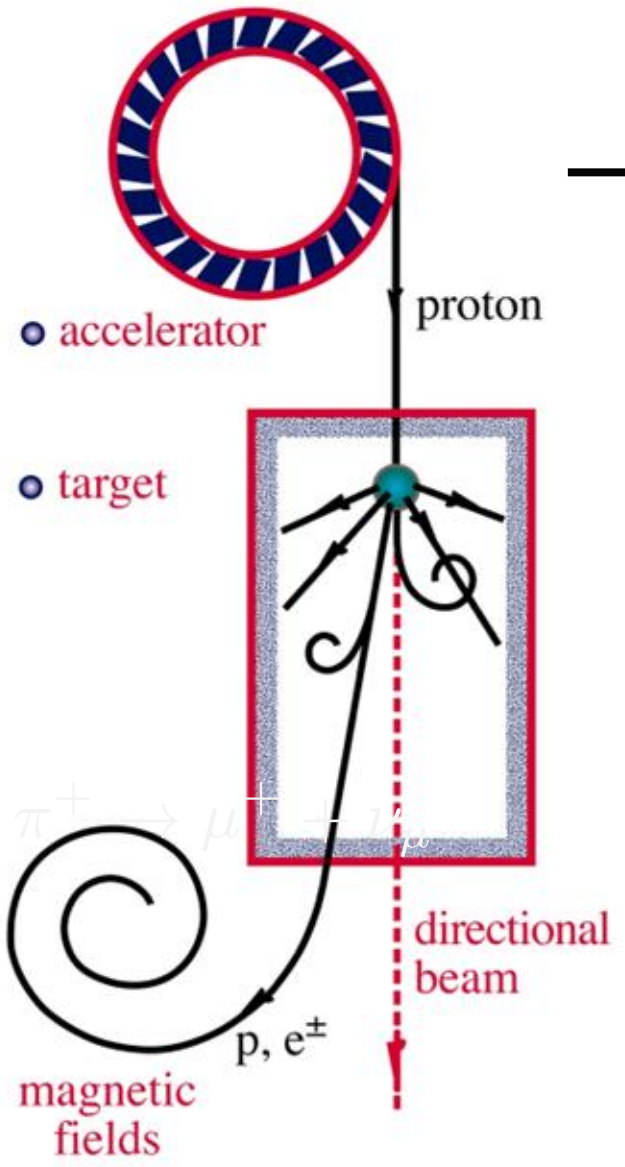
cosmic strings

magnetars

merging galaxies

Hess 1912: the sources of the cosmic rays?

ν and γ beams : heaven and earth



cosmic accelerators powered by gravity

supermassive black hole

nearby radiation

$$p + \gamma \rightarrow n + \pi^+$$

$$\pi^+ \rightarrow \mu^+ + \nu_\mu$$

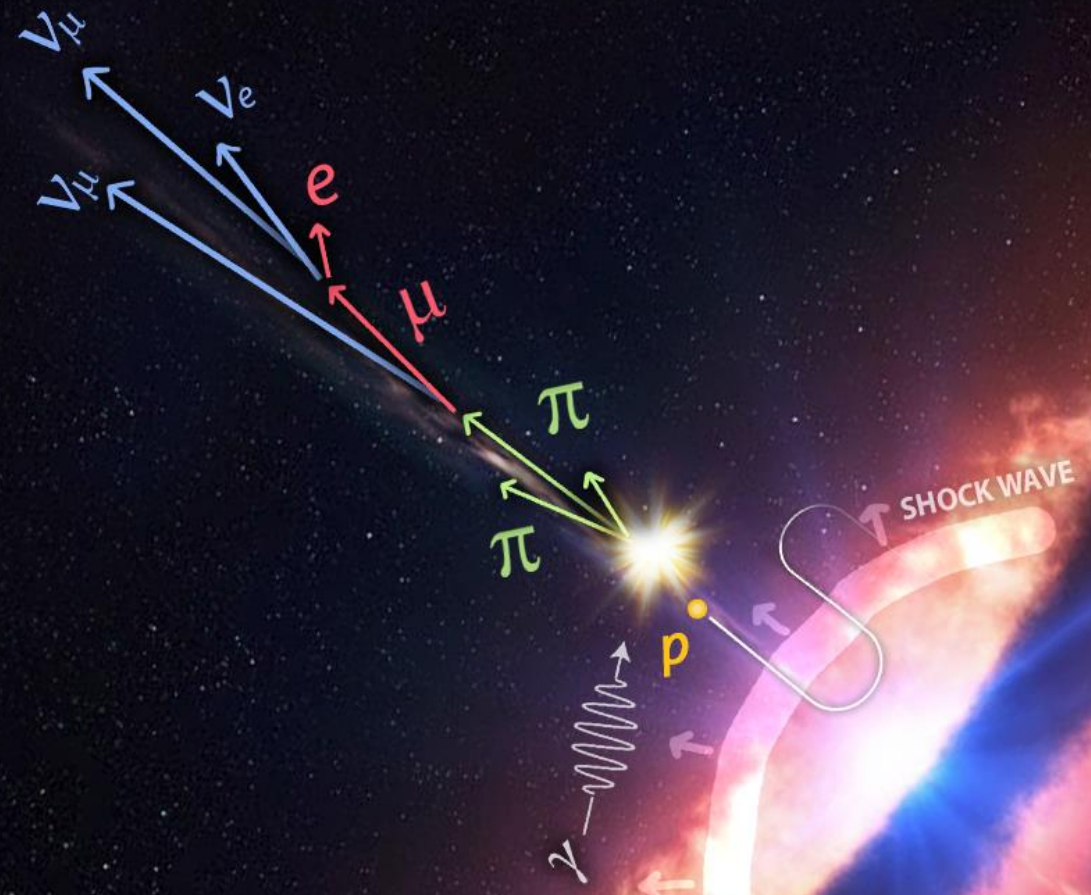
$$\mu^+ \rightarrow e^+ + \nu_e + \bar{\nu}_\mu$$

$$\rightarrow p + \pi^0$$

$$\pi^0 \rightarrow \gamma + \gamma$$

cosmic accelerators:

- protons accelerated near a black hole are submersed in a target of radiation and produce pions
- one photon for every neutrino
- neutrinos track cosmic ray (*proton*) accelerators making pions

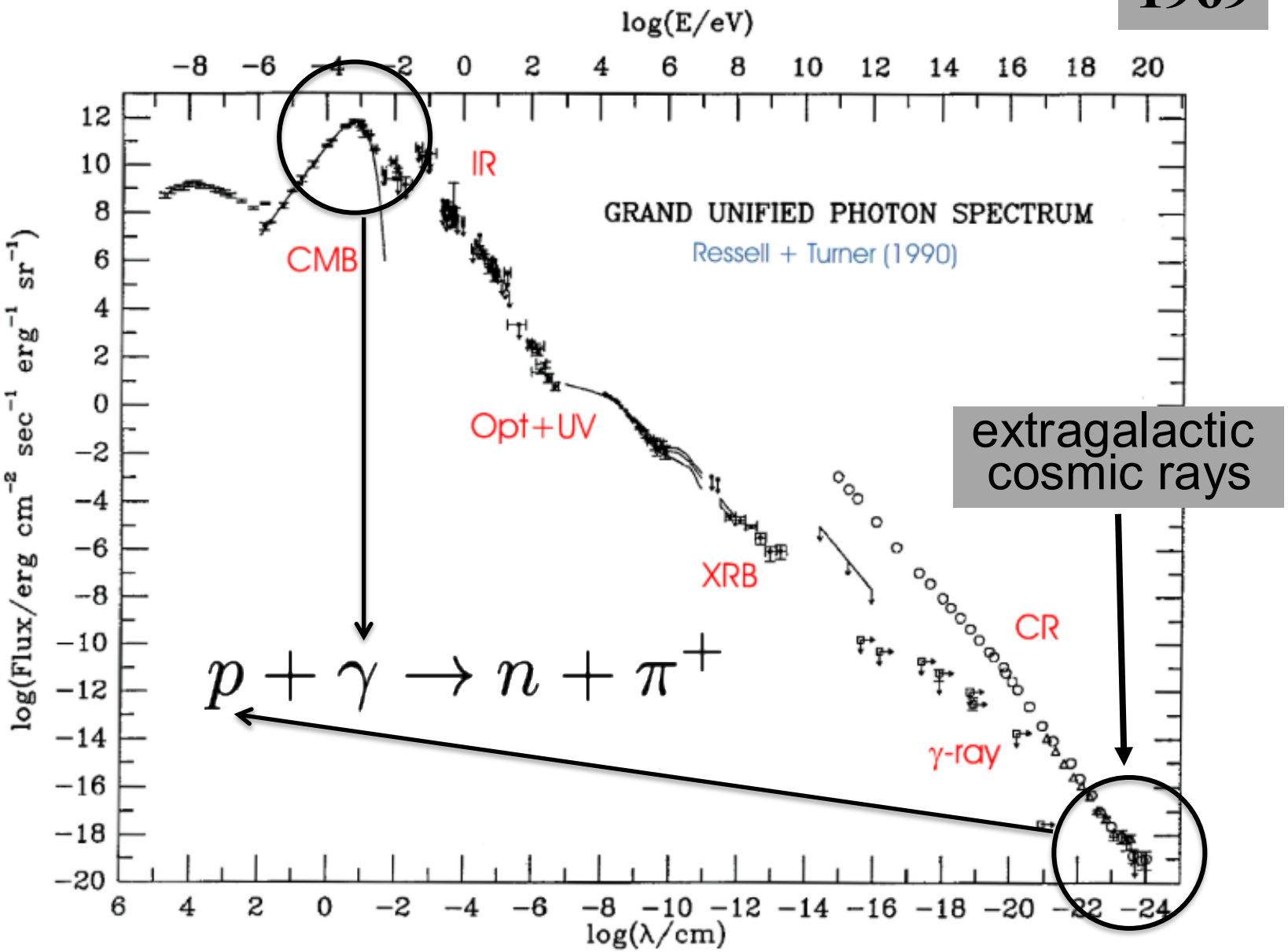


1969

flux

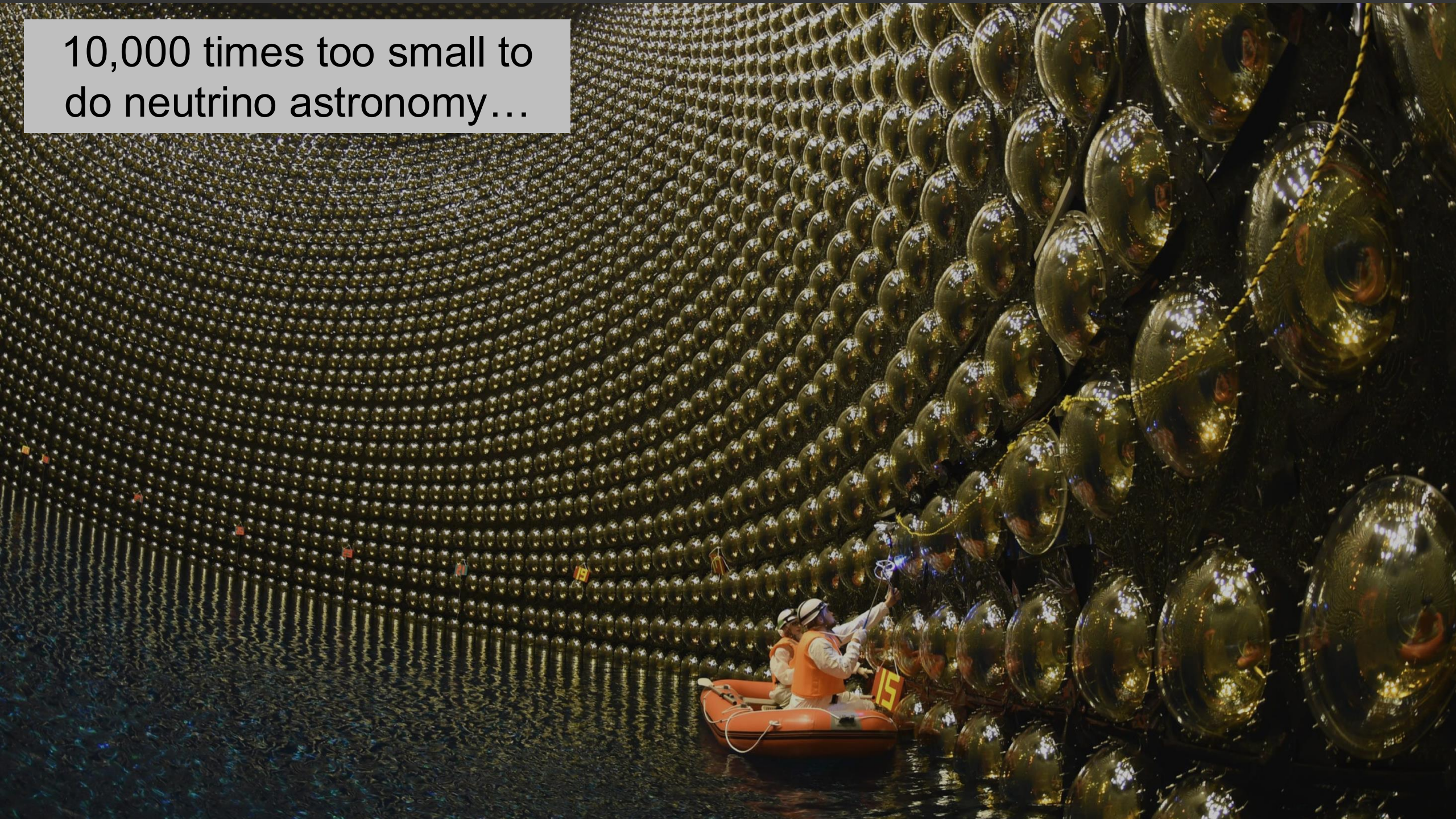
the guaranteed neutrinos

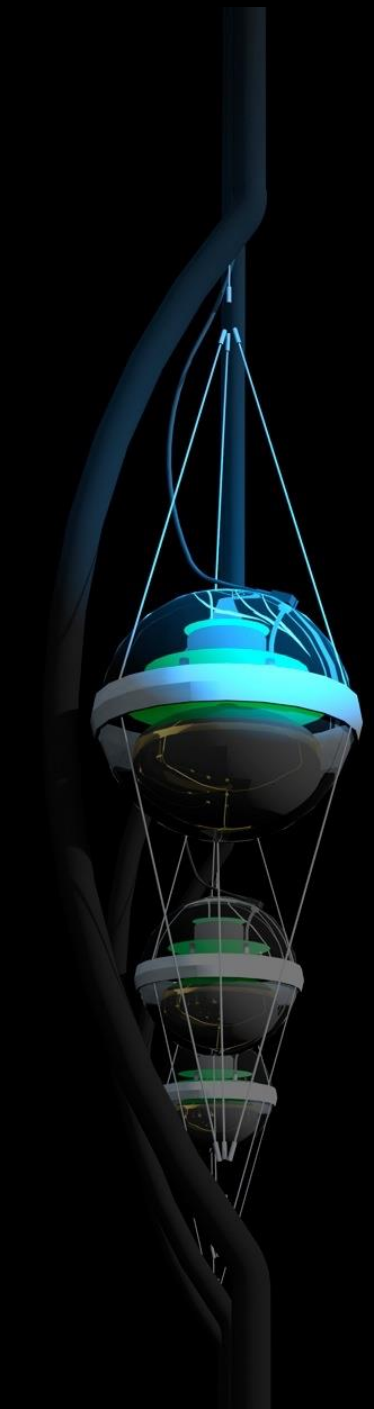
- the pions decay into “GZK” neutrinos
- observation requires cubic kilometer detector



extragalactic cosmic rays

energy (eV)


10,000 times too small to
do neutrino astronomy…

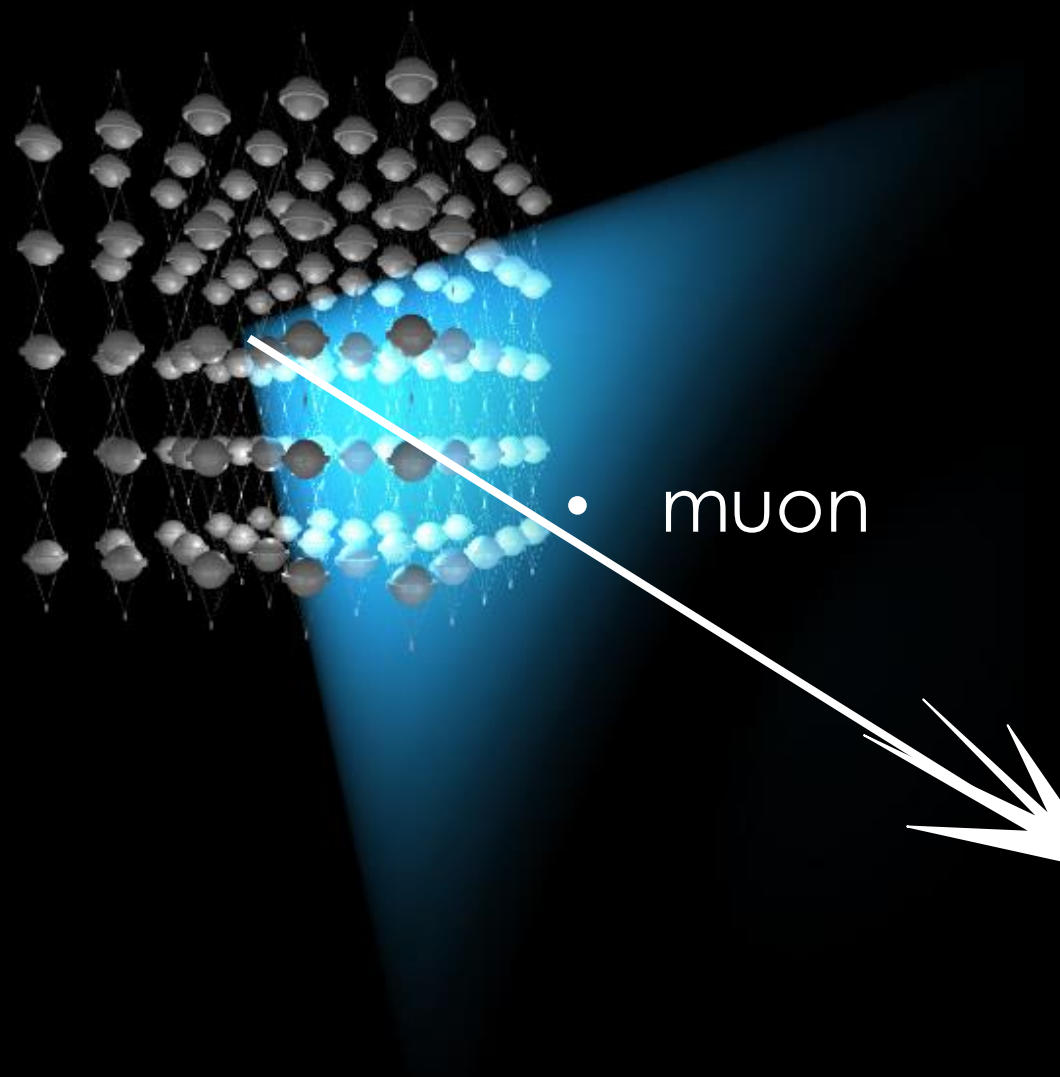


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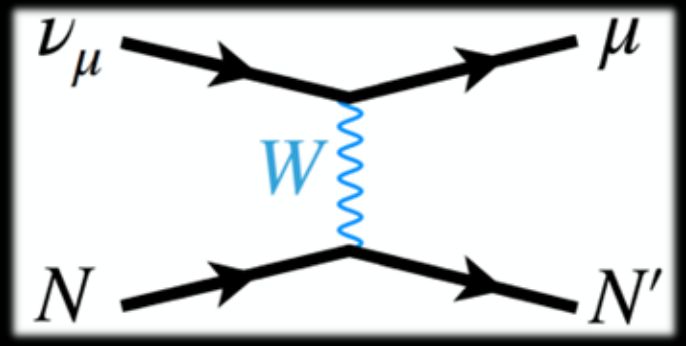
francis halzen

- High-Energy Neutrino Astronomy
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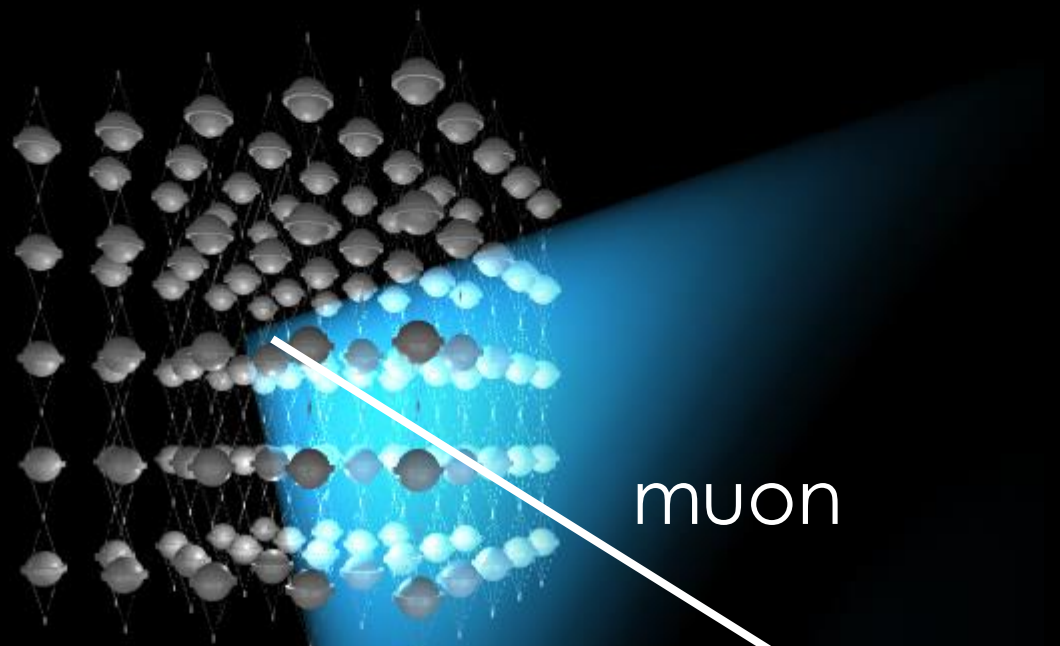
IceCube.wisc.edu



the invisible neutrino turns into charged particles that emit light



- muon
- nuclear interaction with ice nucleus
- lattice of photomultipliers
- neutrino



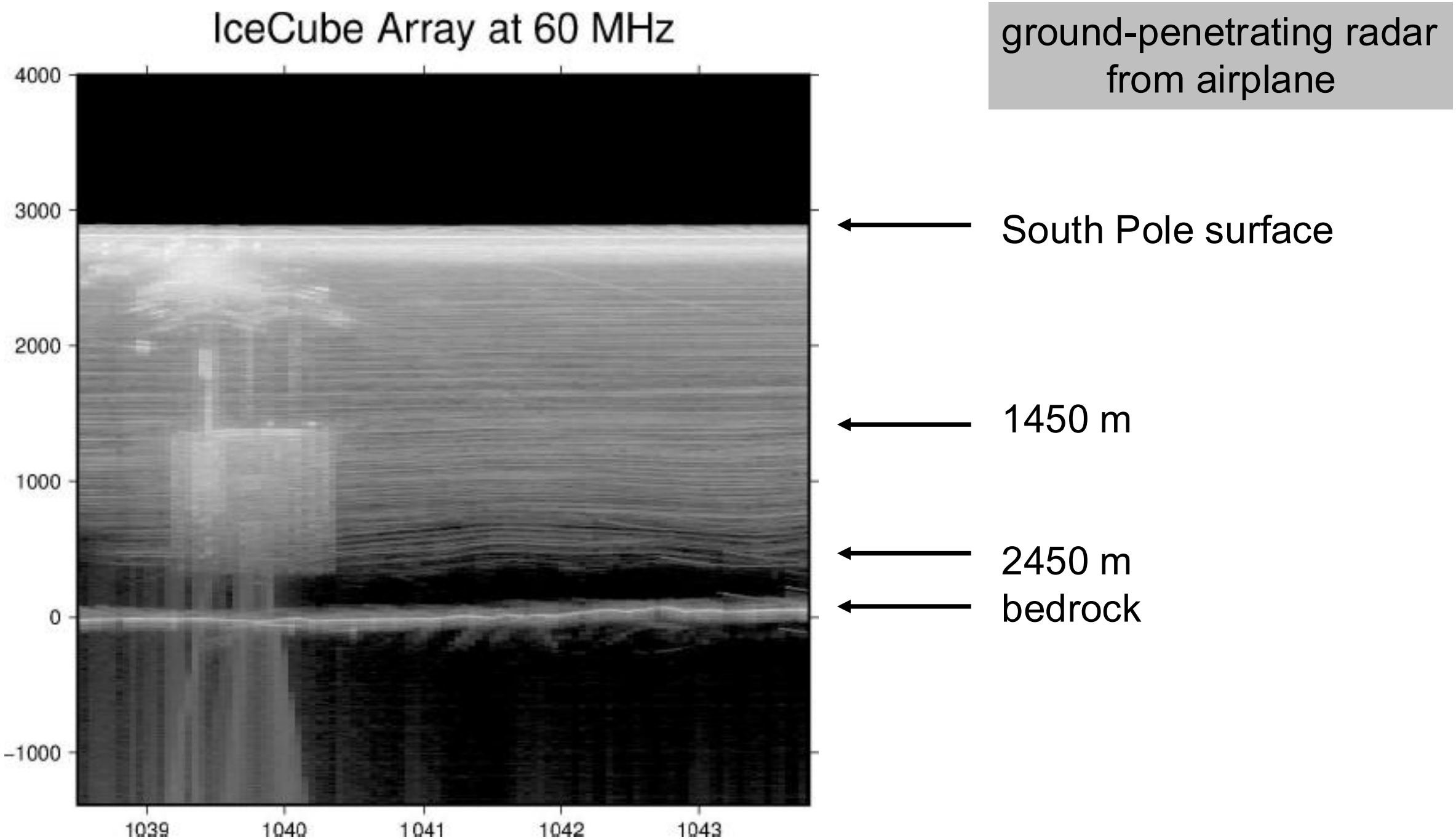
muon

- muon travels for kilometers through the water at the speed of light emitting blue light along its track



muon-neutrino

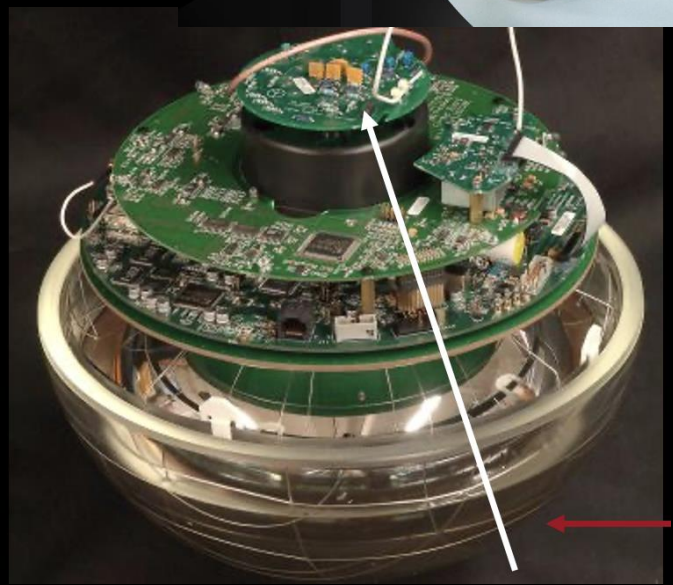
- speed of light in water is only three quarters of the speed of light → shockwave

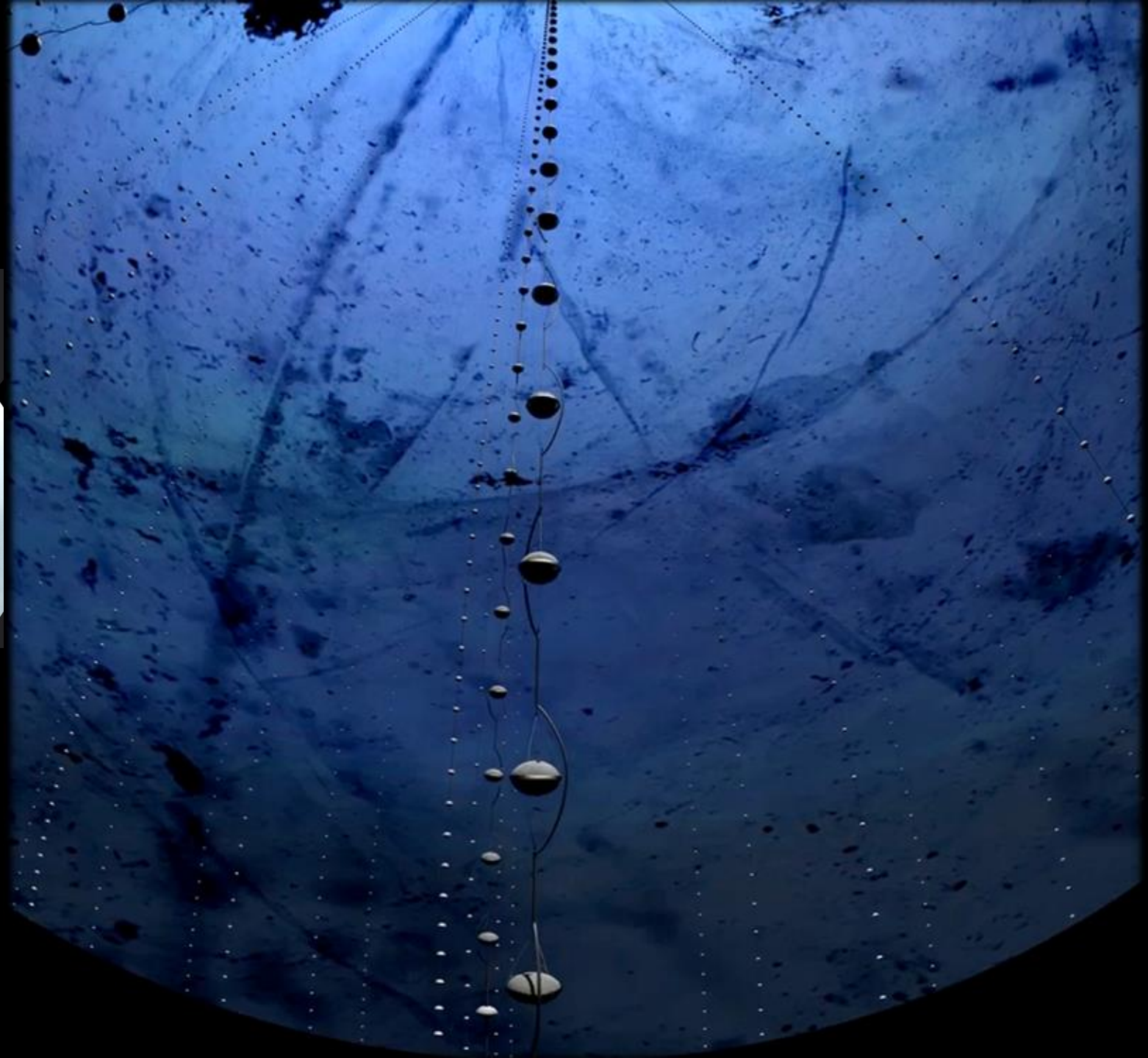

IceCube Array at 60 MHz
4000
3000
2000
1000
0
−1000
1039
1040
1041
1042
1043
ground-penetrating radar from airplane
South Pole surface
1450 m
2450 m
bedrock

IceCube:

5160 10-inch photomultipliers,
60 per string on 86 strings,
instrument one km^3 of
Antarctic ice between
1.4 and 2.4 km depth
as a Cherenkov detector

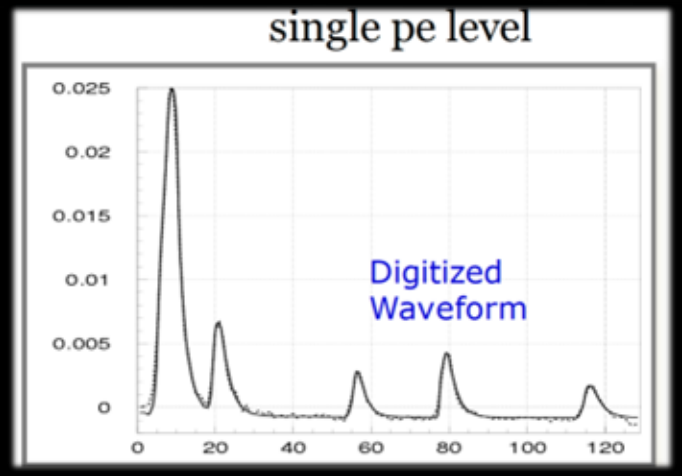






light signals from 10 inch photomultiplier embedded in the deep ice are digitized and sent to computers at surface

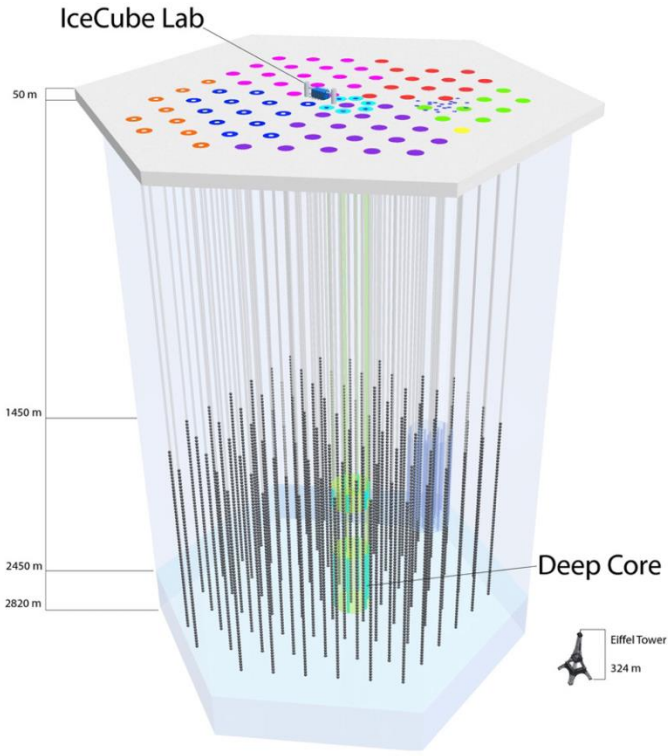






IceCube construction

(Upgrade 2026)





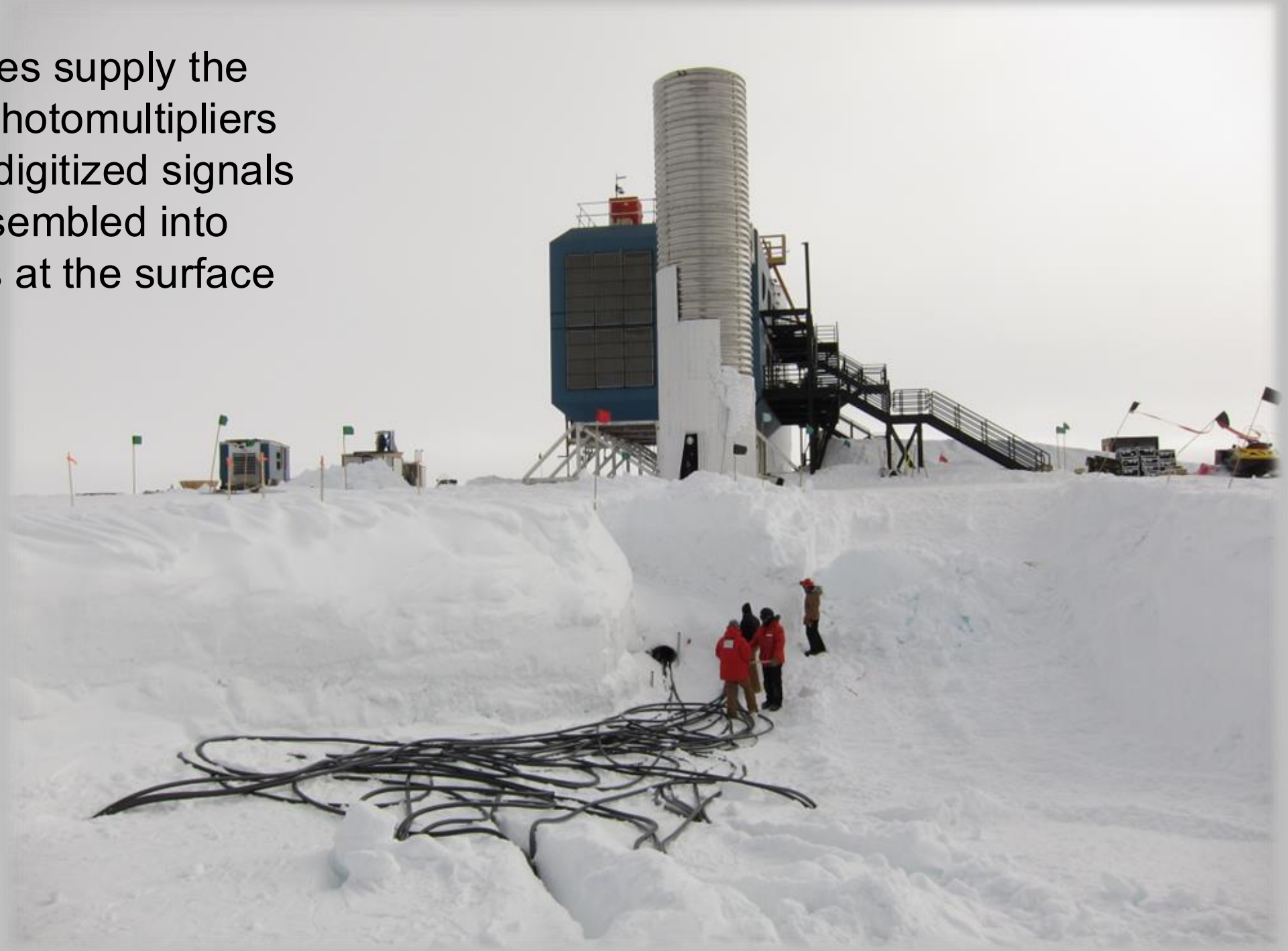






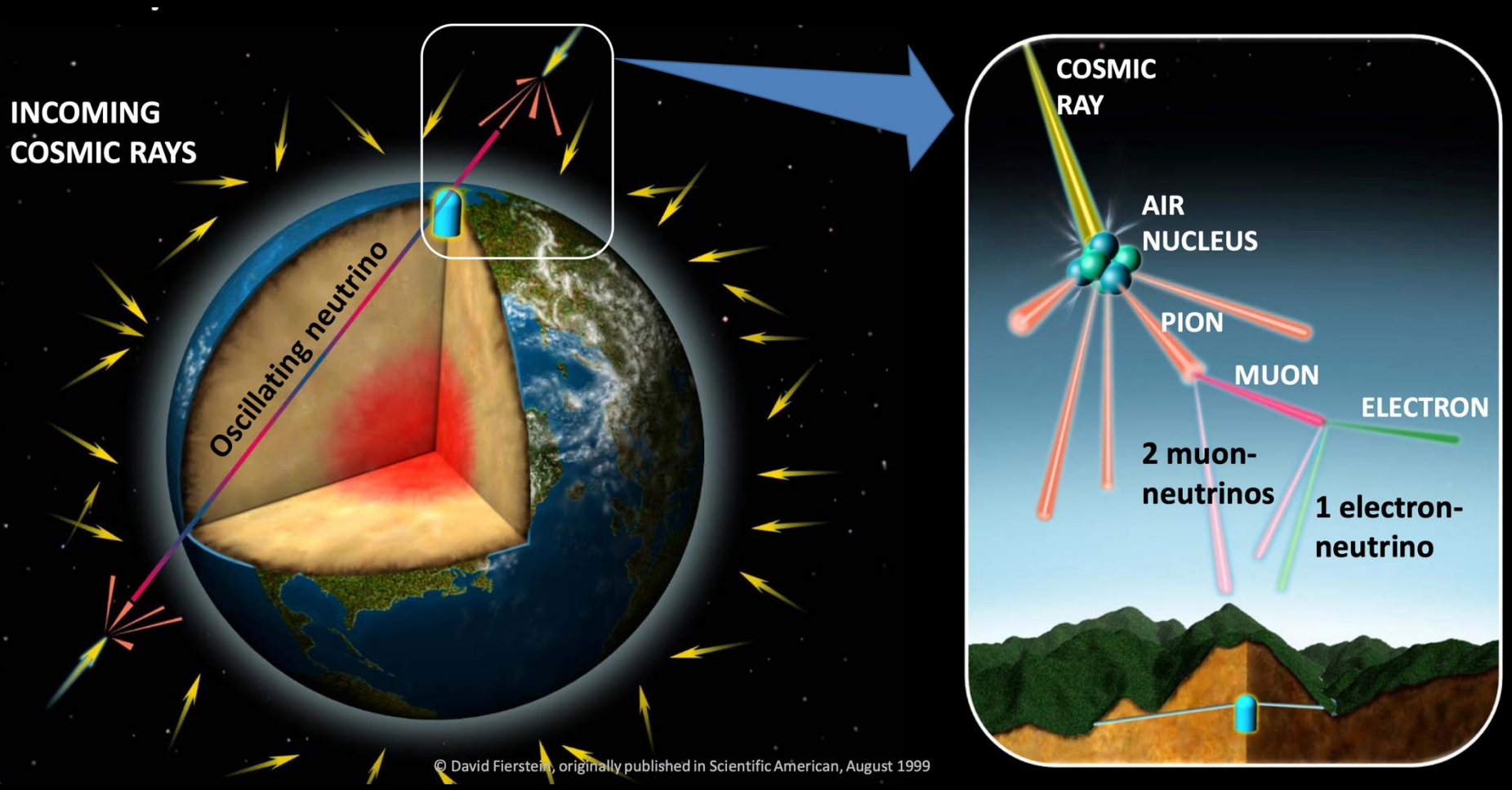
- instrument 1 cubic kilometer of natural ice below 1.45 km with 5160 10-inch photomultiplier tubes
- totally stable detector after deployment: 1 failure every 2 years

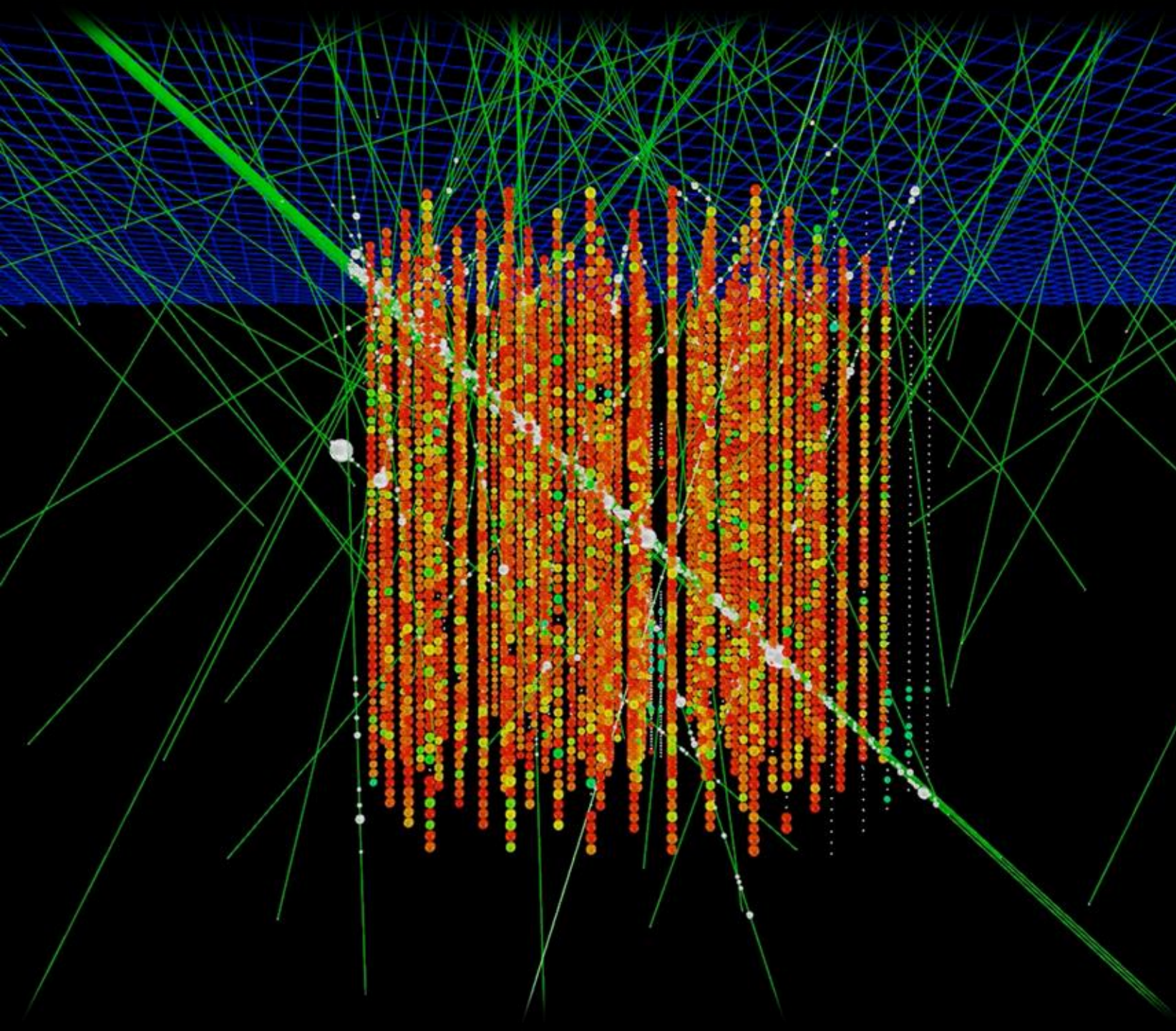
electrical cables supply the voltage to the photomultipliers and return their digitized signals which are assembled into neutrino events at the surface





What does IceCube “see”?





10 msec movie of IceCube taking data

muons detected per year

- cosmic ray muons ~ 10^{11}

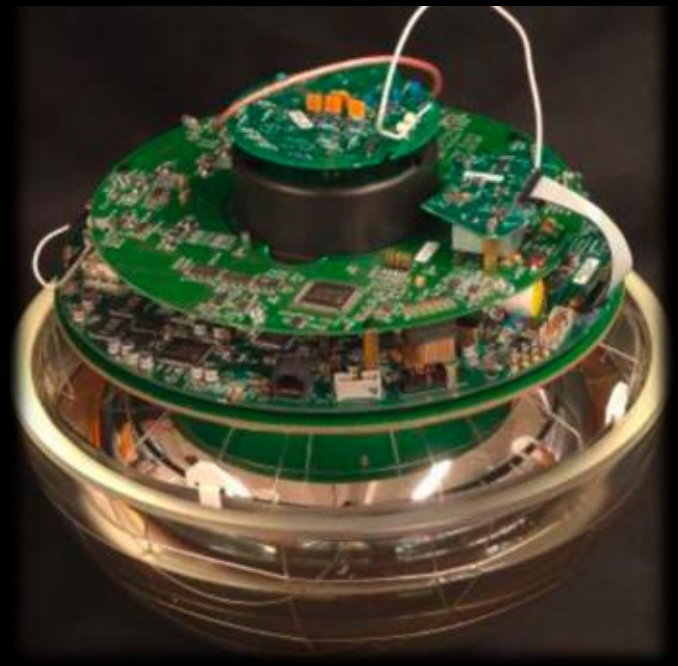
  (3000 per second)

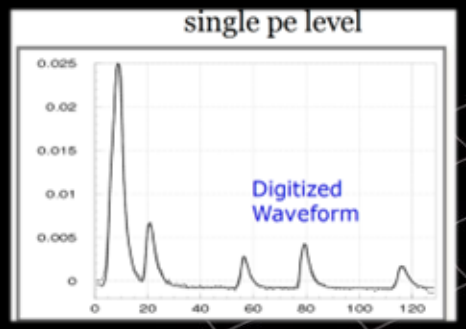
- atmospheric neutrinos ~ 10^{5}

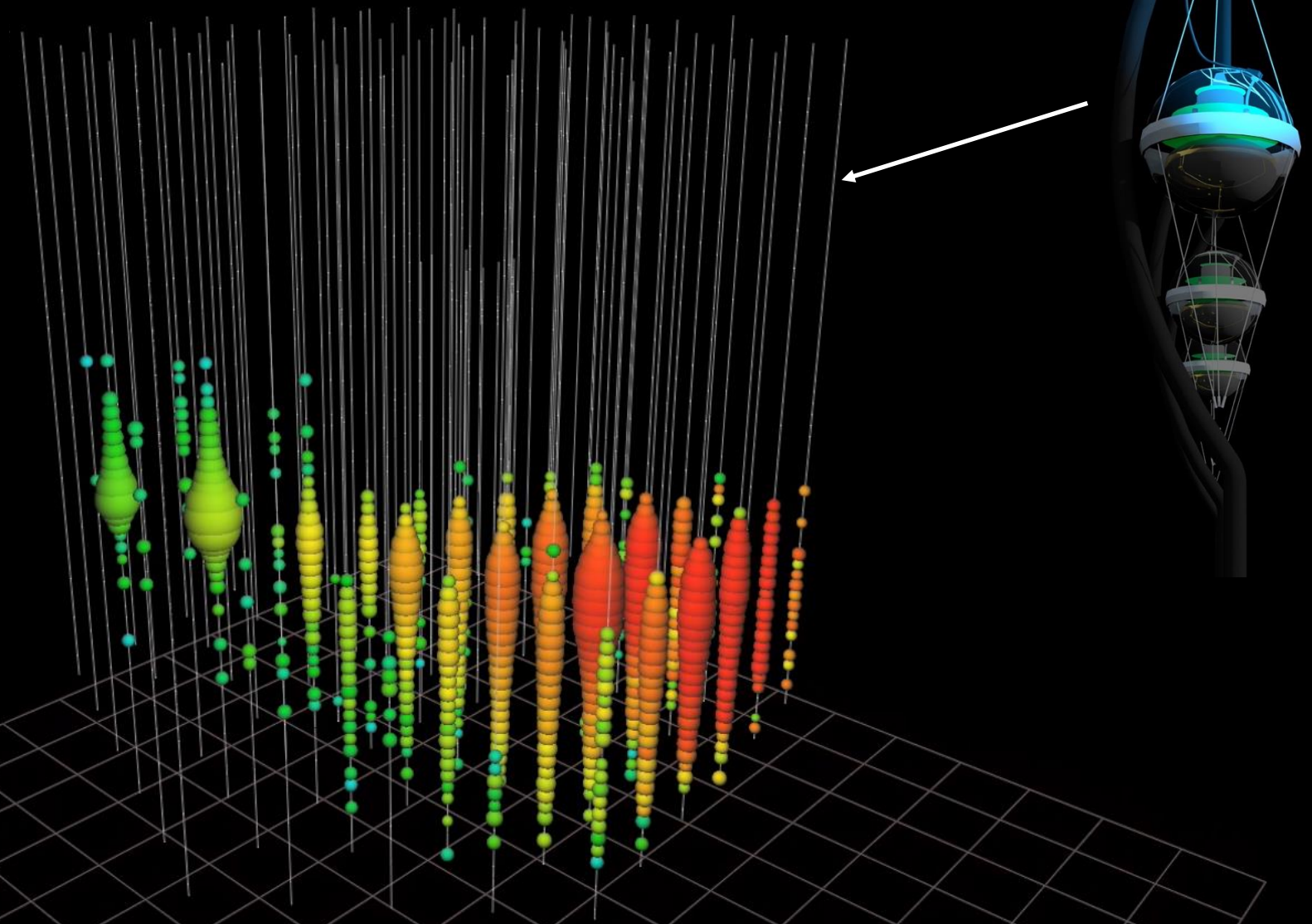
  (1 every 5 minutes)

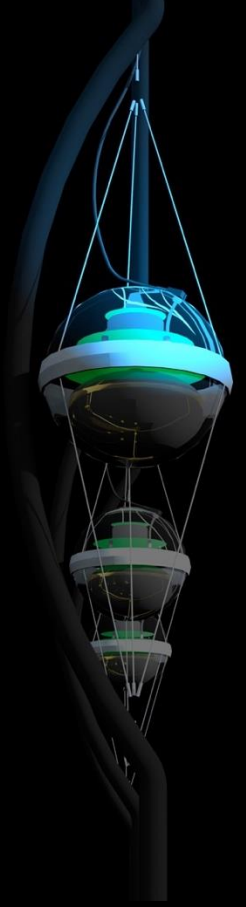
- cosmic neutrinos ~ 200

1 km^3 of Antarctic ice instrumented with 5160 PMT (10inch) below 1450m



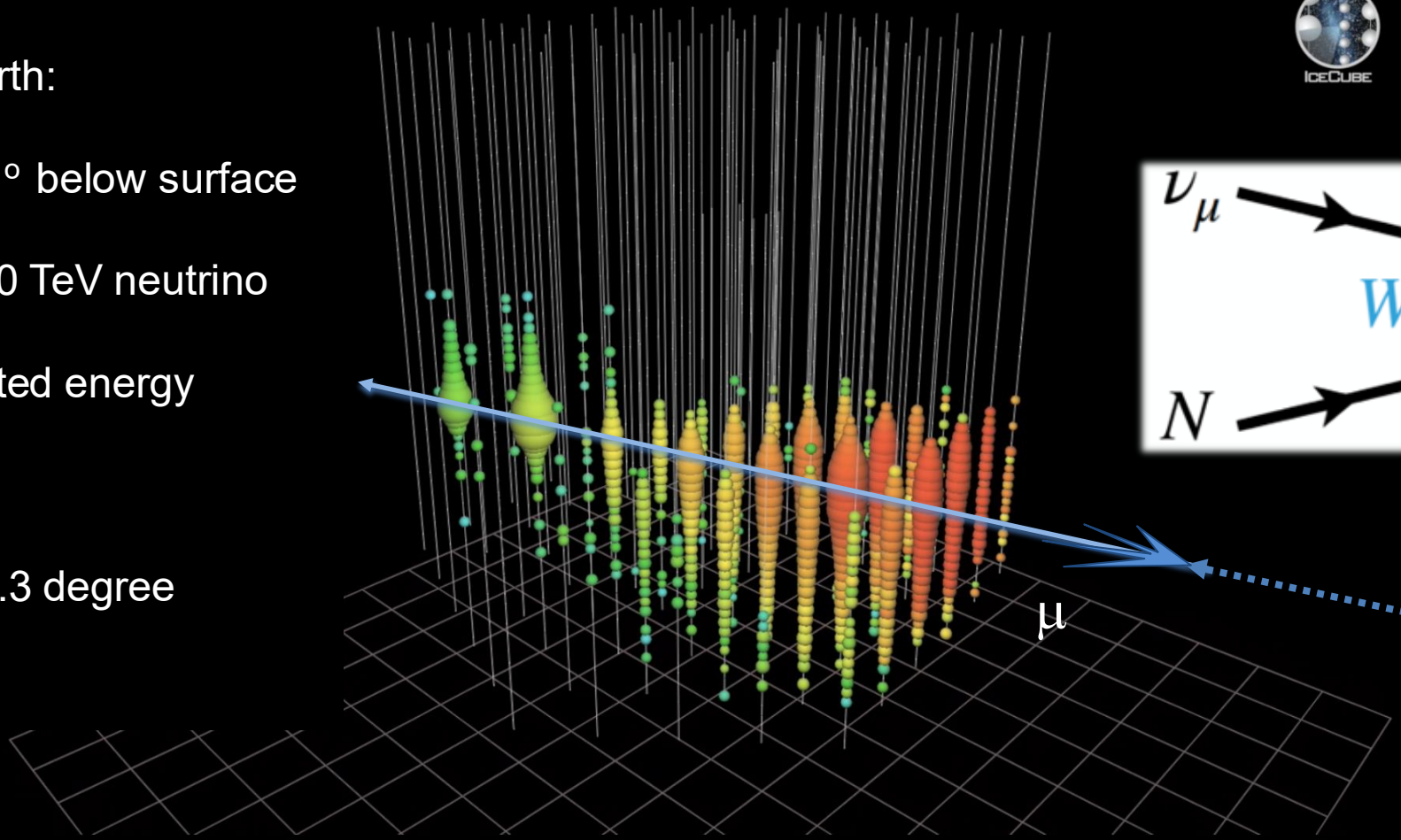


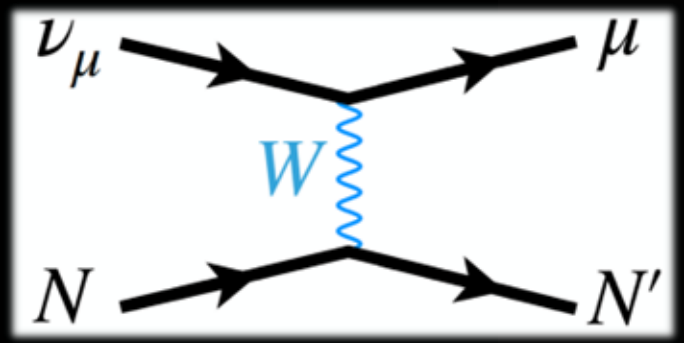




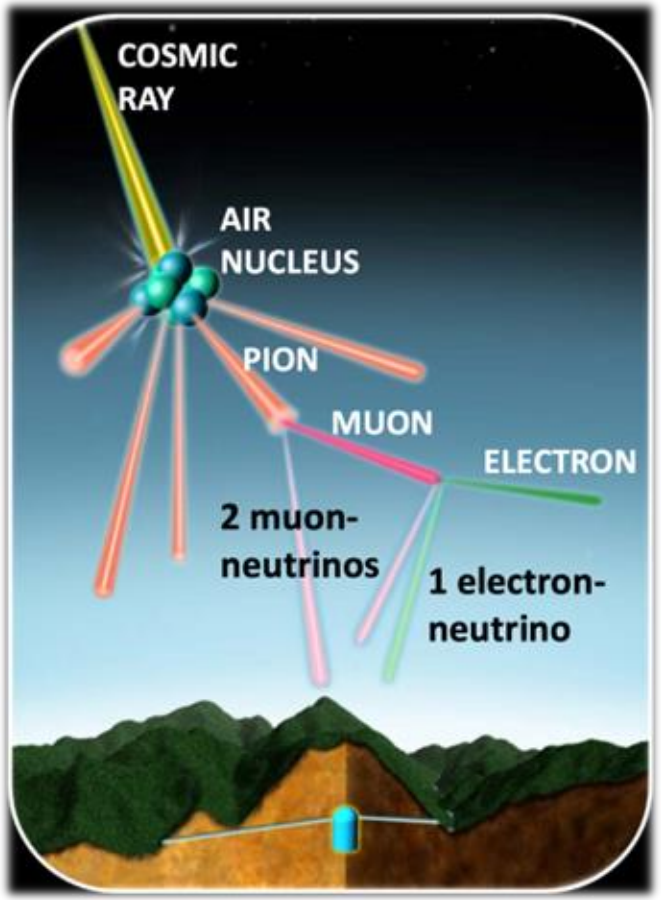
Look through the Earth:

- upgoing muon 11° below surface
- produced by 8,700 TeV neutrino
- 2,600 TeV deposited energy
- not atmospheric
- reconstructed < 0.3 degree
- astronomy





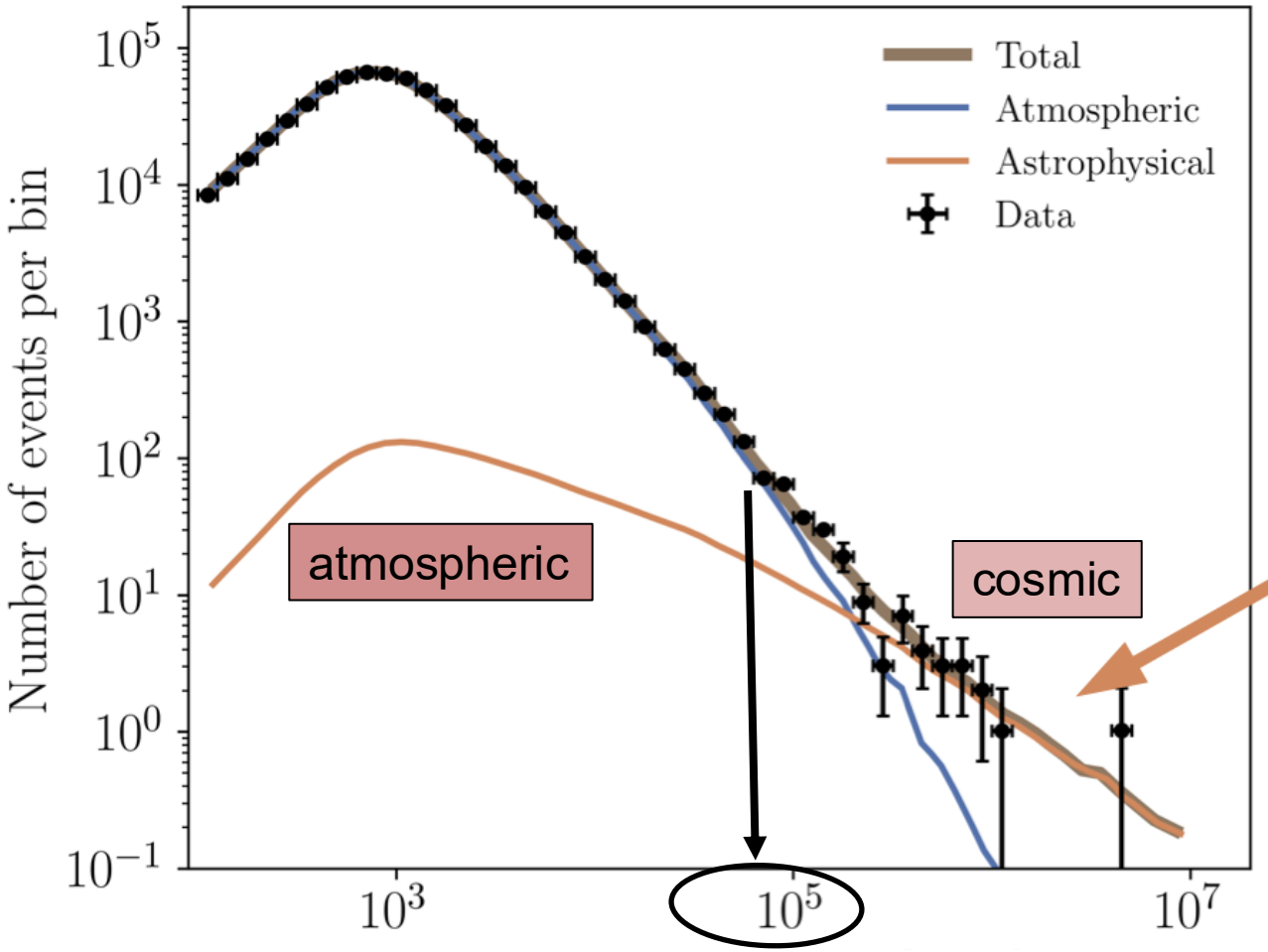
muon neutrino events [filtered by the Earth]: atmospheric vs cosmic



In 2013: Astrophysical diffuse neutrinos

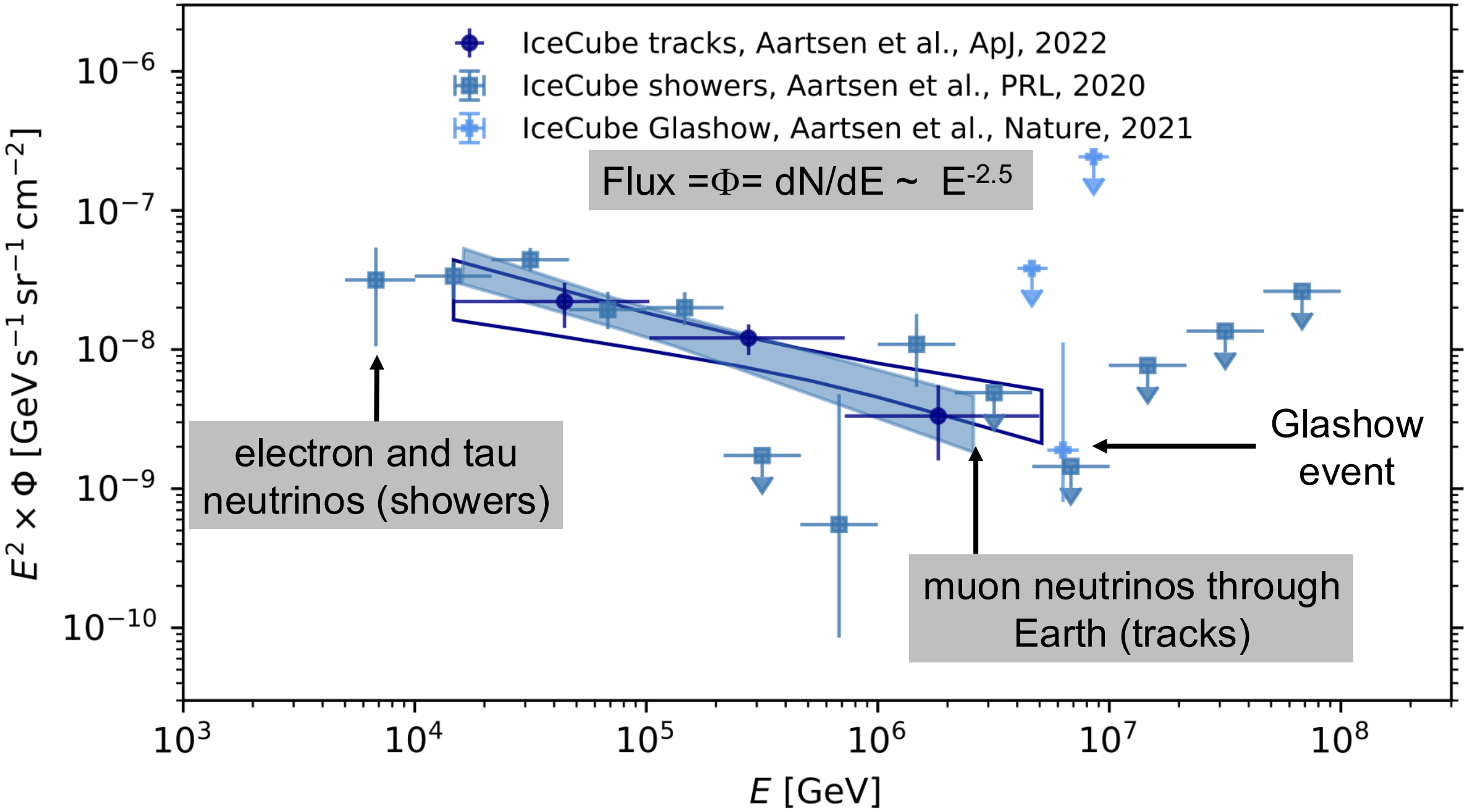
Data from arXiv:1908.09551 [astro-ph.HE]

number of events

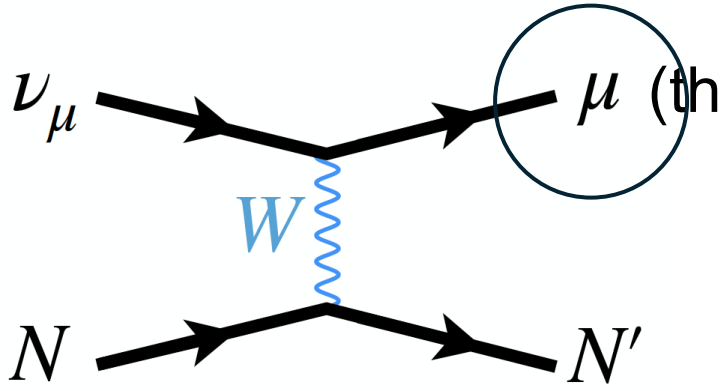


More abundant at higher energies

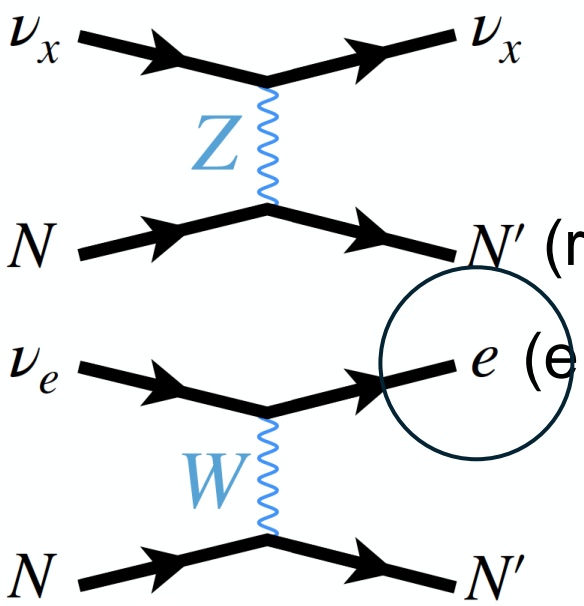
reconstructed energy (GeV)


IceCube tracks, Aartsen et al., ApJ, 2022
IceCube showers, Aartsen et al., PRL, 2020
IceCube Glashow, Aartsen et al., Nature, 2021
Flux =Φ= dN/dE ~ E^-2.5
electron and tau neutrinos (showers)
Glashow event
muon neutrinos through Earth (tracks)
E² x Φ [GeV s⁻¹ sr⁻¹ cm⁻²]
E [GeV]
10⁻⁶
10⁻⁷
10⁻⁸
10⁻⁹
10⁻¹⁰
10³
10⁴
10⁵
10⁶
10⁷
10⁸

other neutrino flavors

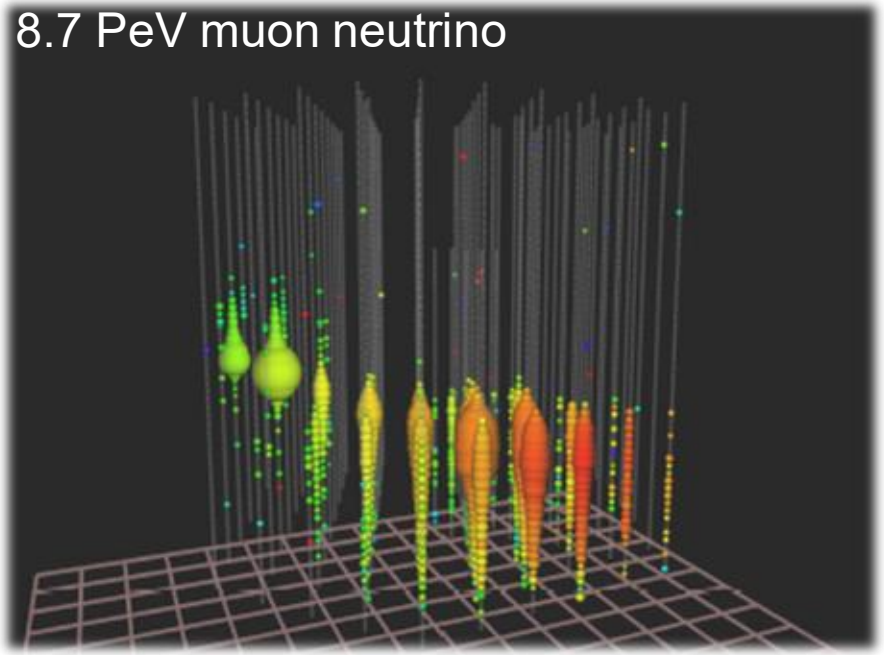


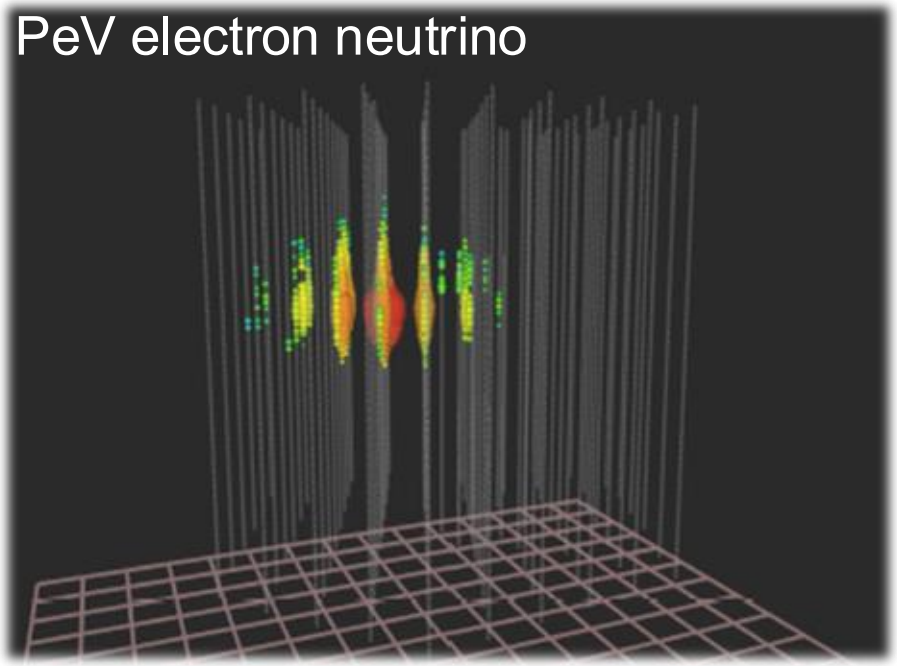
(μ throughgoing muon)



(N' recoil nucleus)

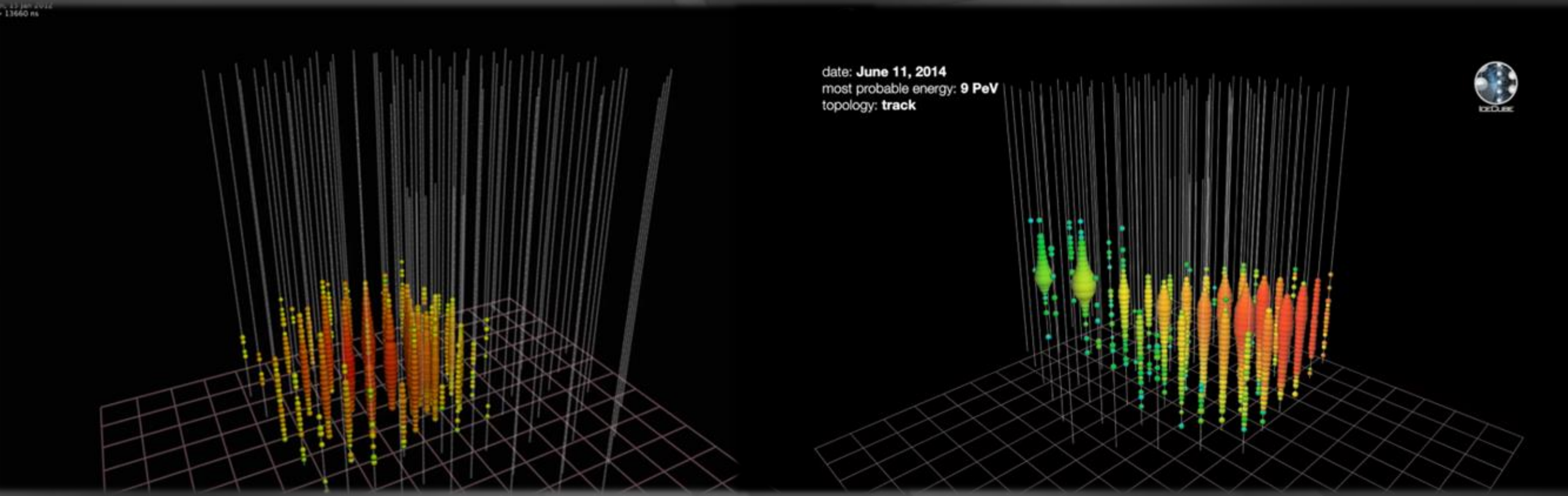
(e electron or tau shower)





neutrinos interacting
inside the detector

muon neutrinos
filtered by the Earth

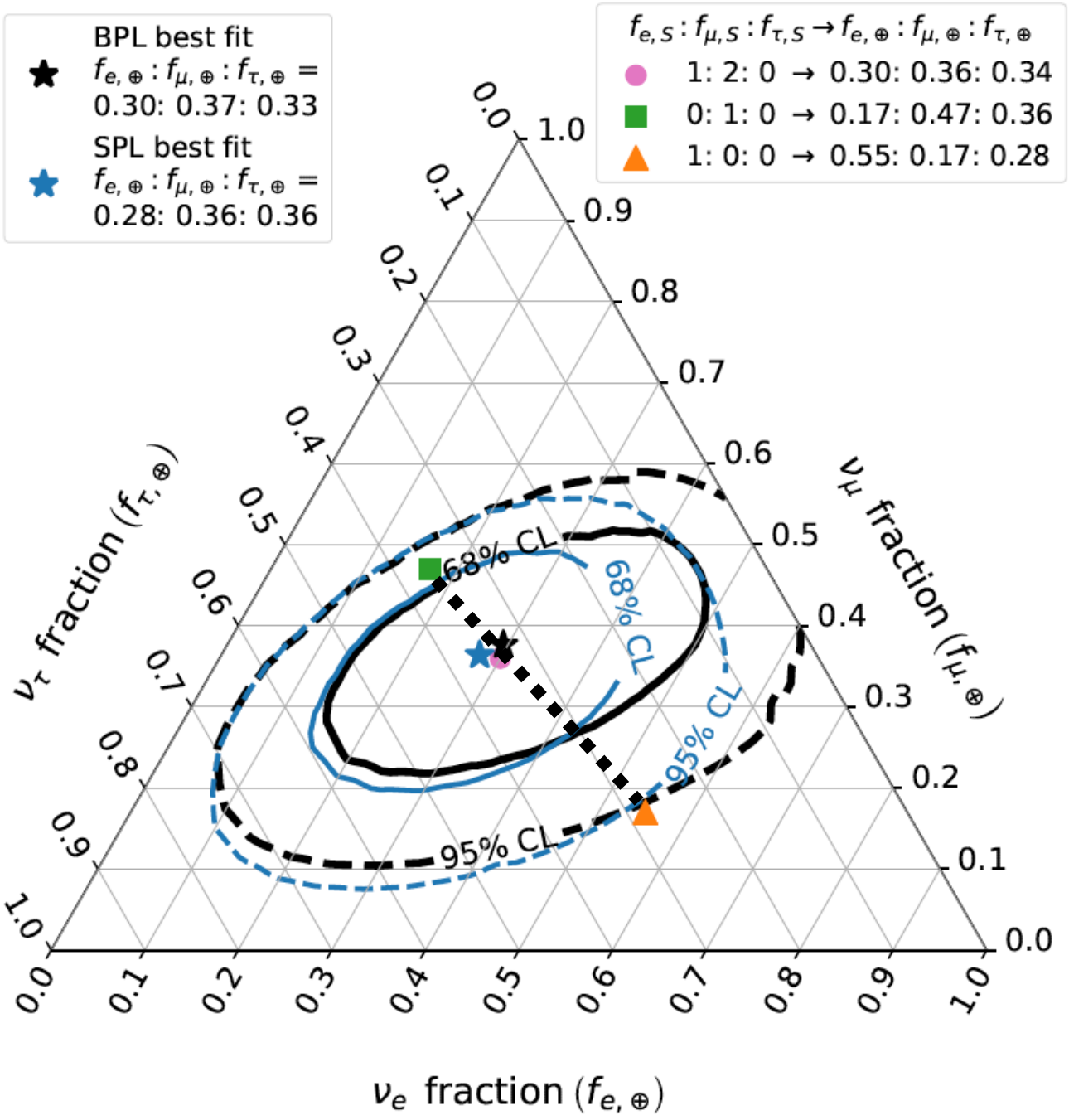


superior total energy
measurement
to 10%, all flavors, all sky

superior angular resolution < 0.3^{o}
including systematics

particle physics:

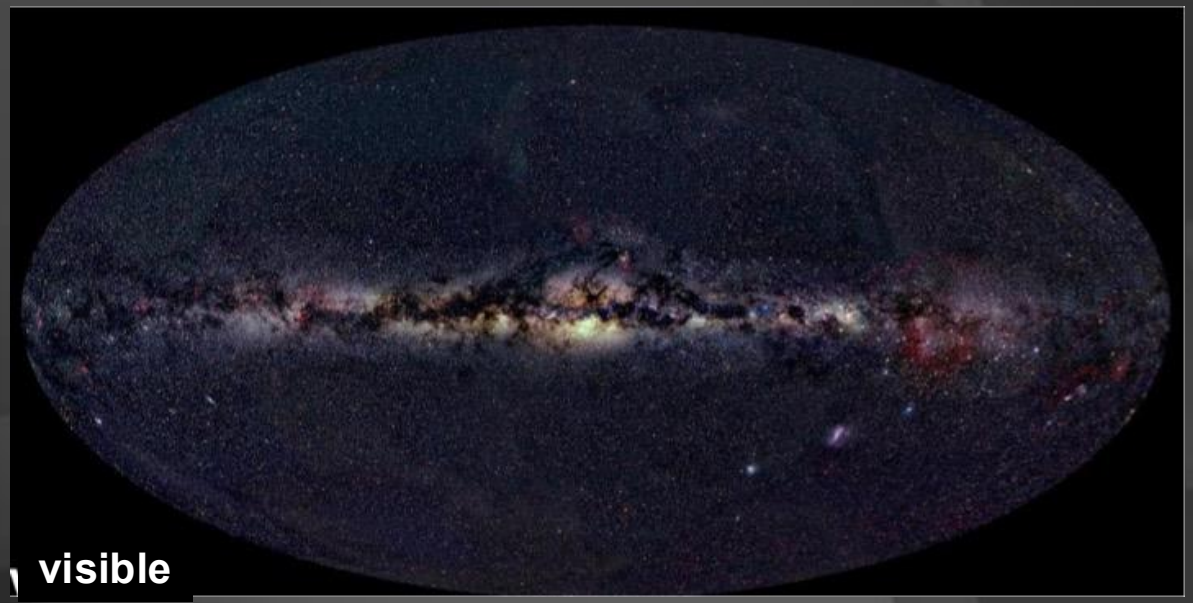
- neutrino oscillations over cosmic baselines: $\nu_\mu = \nu_e = \nu_\tau$
- dark matter
- next Galactic supernova
- quantum decoherence of very high energy neutrinos traveling over cosmic distances
- …

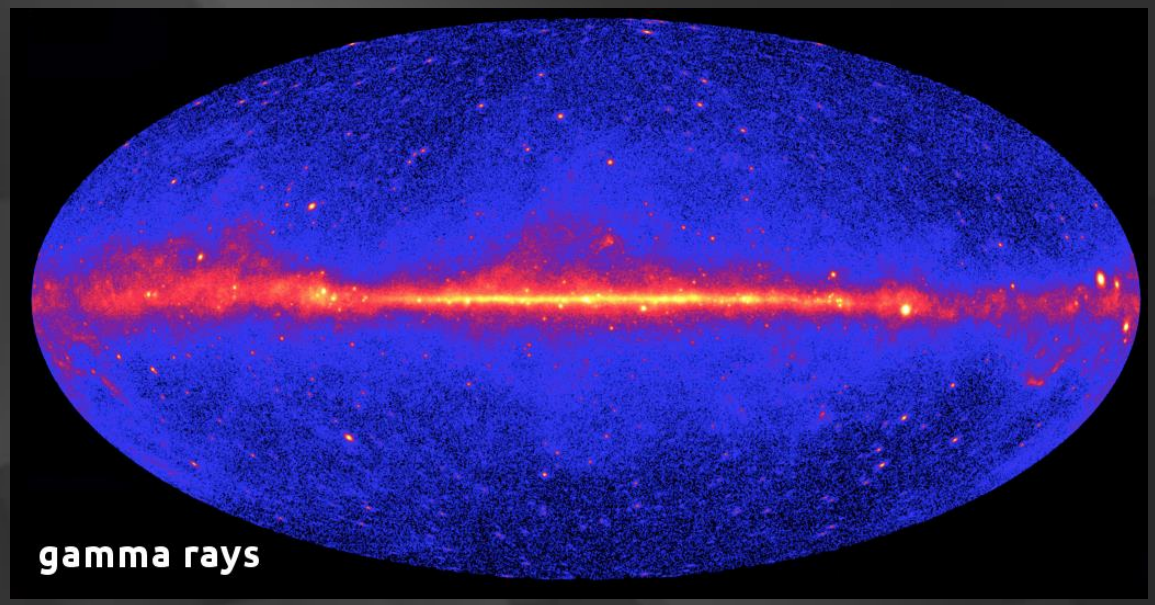


two surprises:

- we observe larger neutrino and associated gamma ray fluxes than anticipated

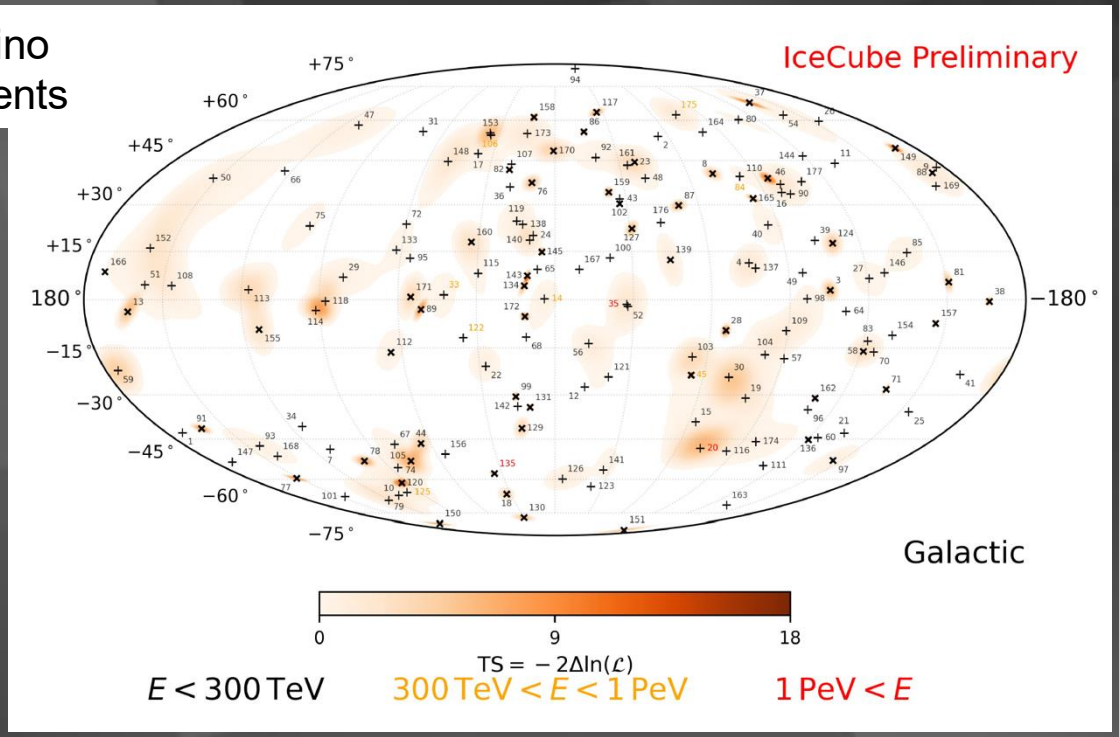
- the Milky Way is not a dominant feature in the neutrino sky (powerful accelerators produce neutrinos in other galaxies that do not exist in our own)





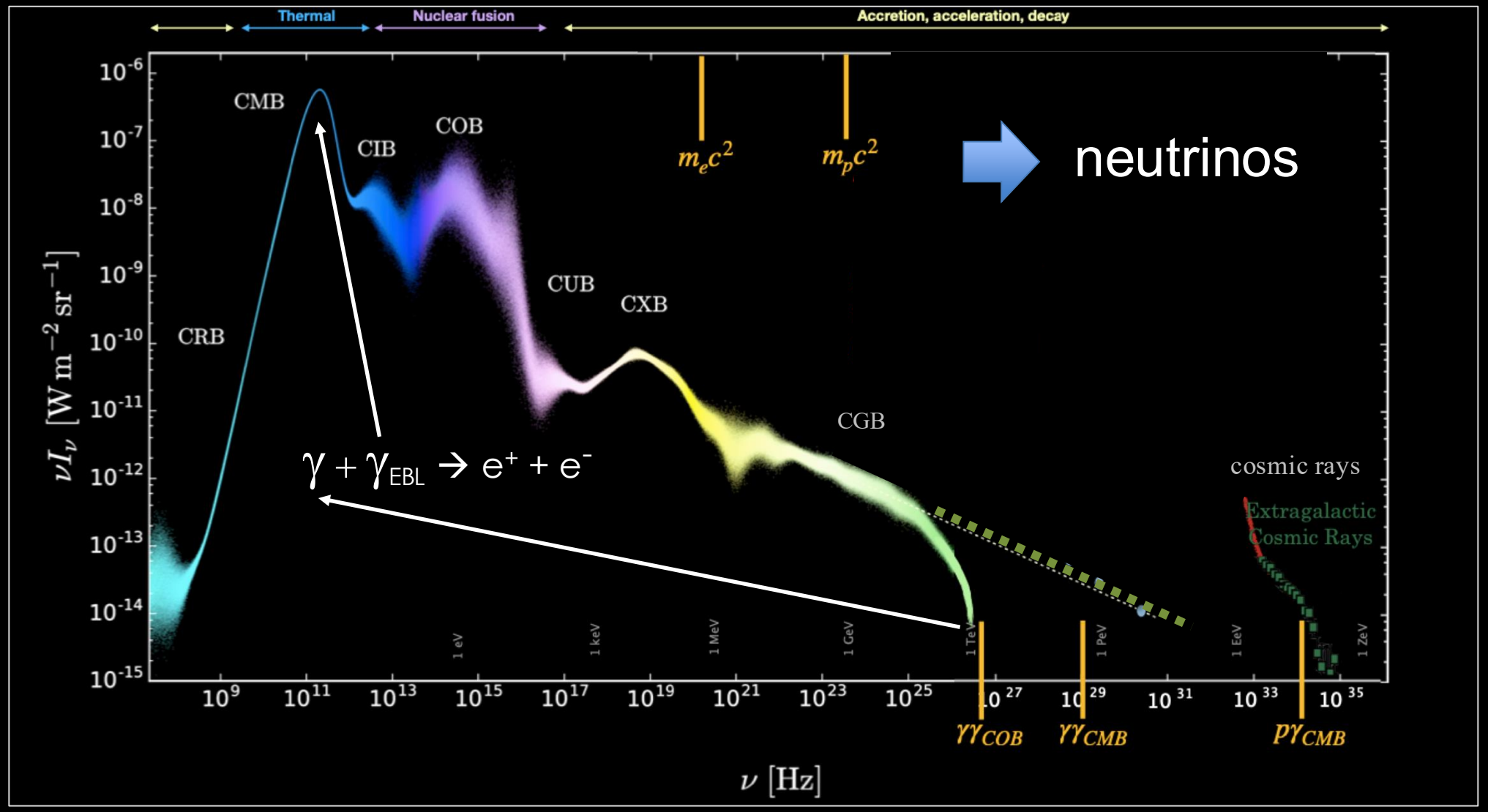
where is the neutrino Galactic plane?

166 neutrino starting events



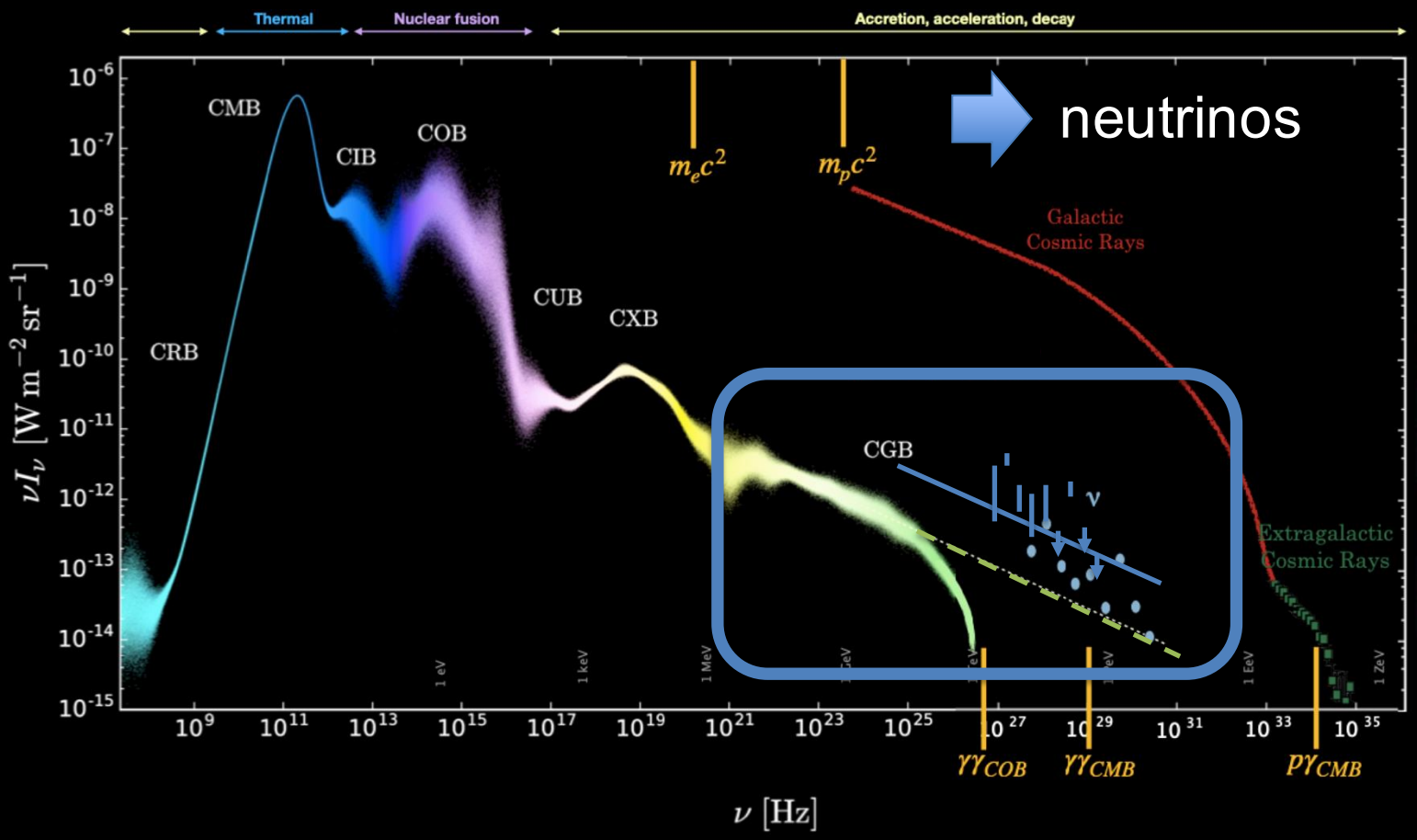
by geometry the flux from your own Galaxy should dominate the diffuse flux from all other galaxies combined!

energy in photons in the Universe as a function of their frequency

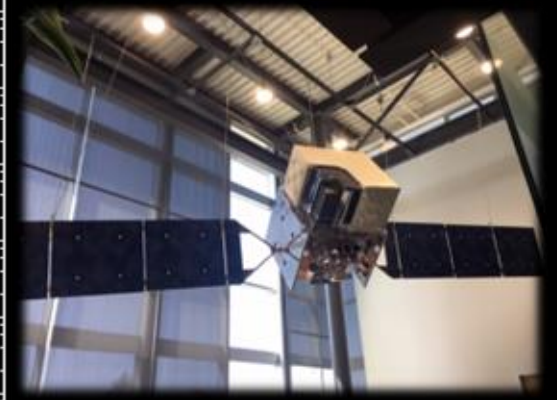


the Universe turns opaque to light at TeV energies: enter neutrinos

Fermi and IceCube do not observe the same gamma ray sources



Fermi NASA gamma-ray satellite



in the extreme universe the energy in neutrinos (and π^0 gamma rays)
is larger than the (cascaded) energy in GeV gamma rays observed by Fermi

two surprises:

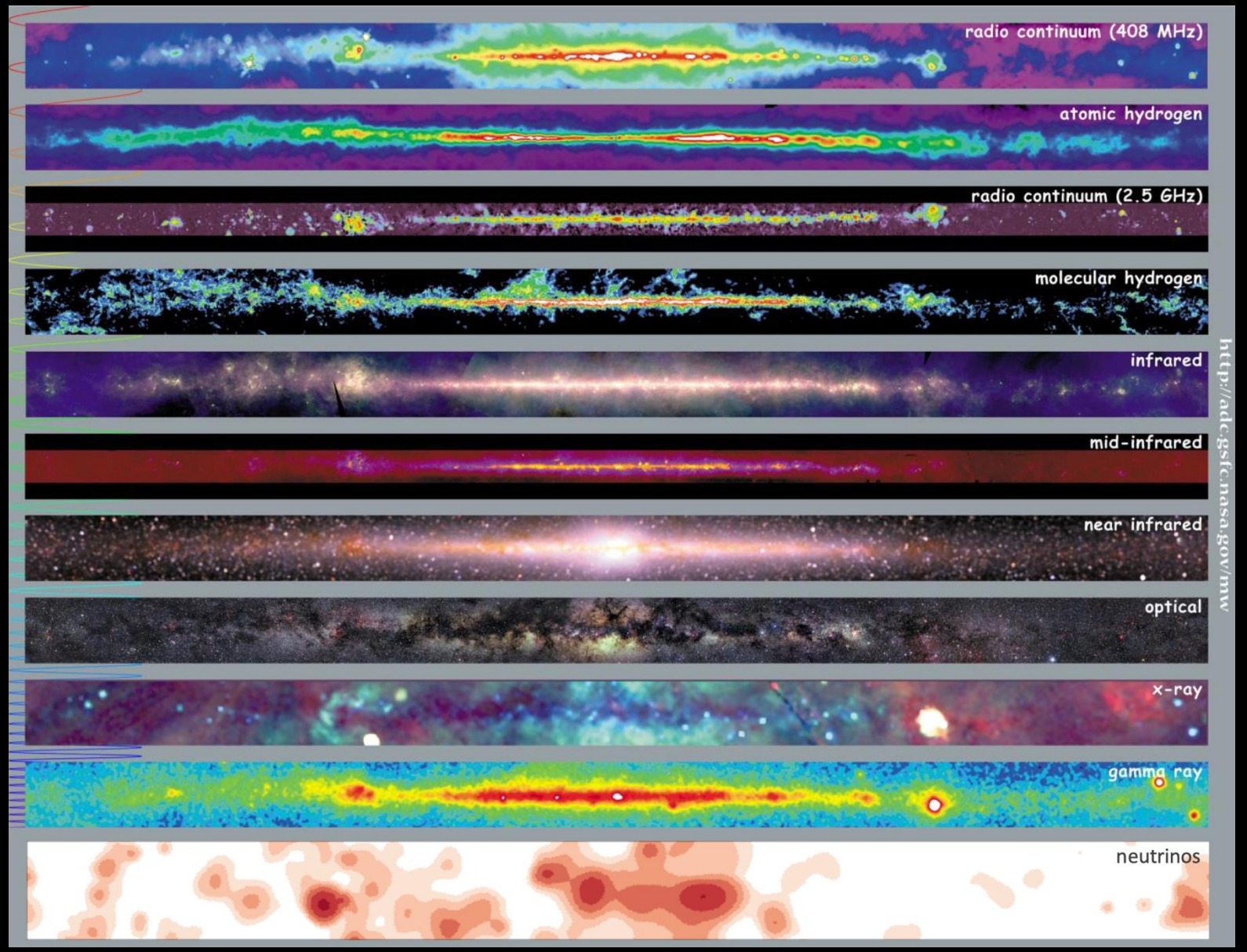
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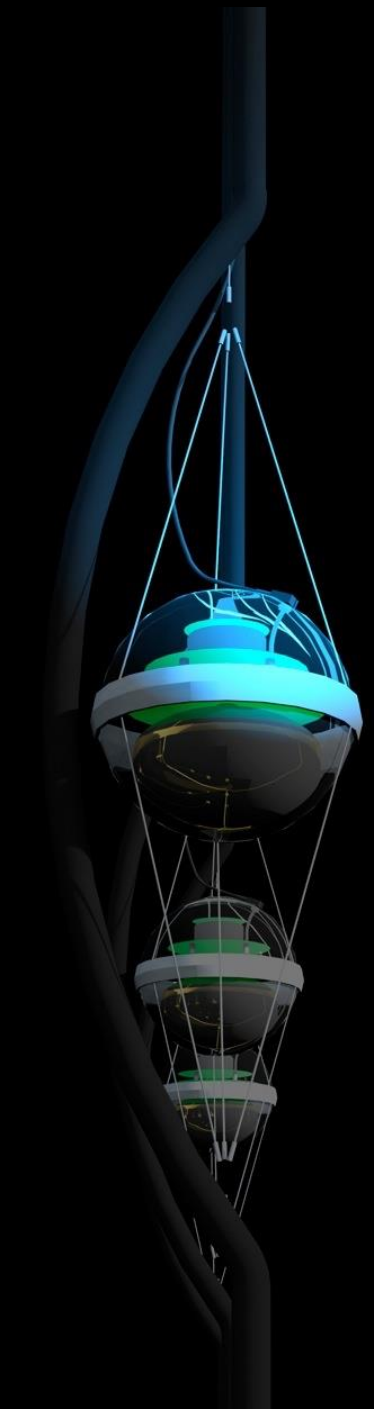
- the Milky Way is not a dominant feature in the neutrino sky (powerful accelerators produce neutrinos in other galaxies that do not exist in our own)

with the help of machine learning we did find the Galactic plane at a flux level of 10% of the astro-physical flux at 30 TeV



Milky Way viewed in neutrinos





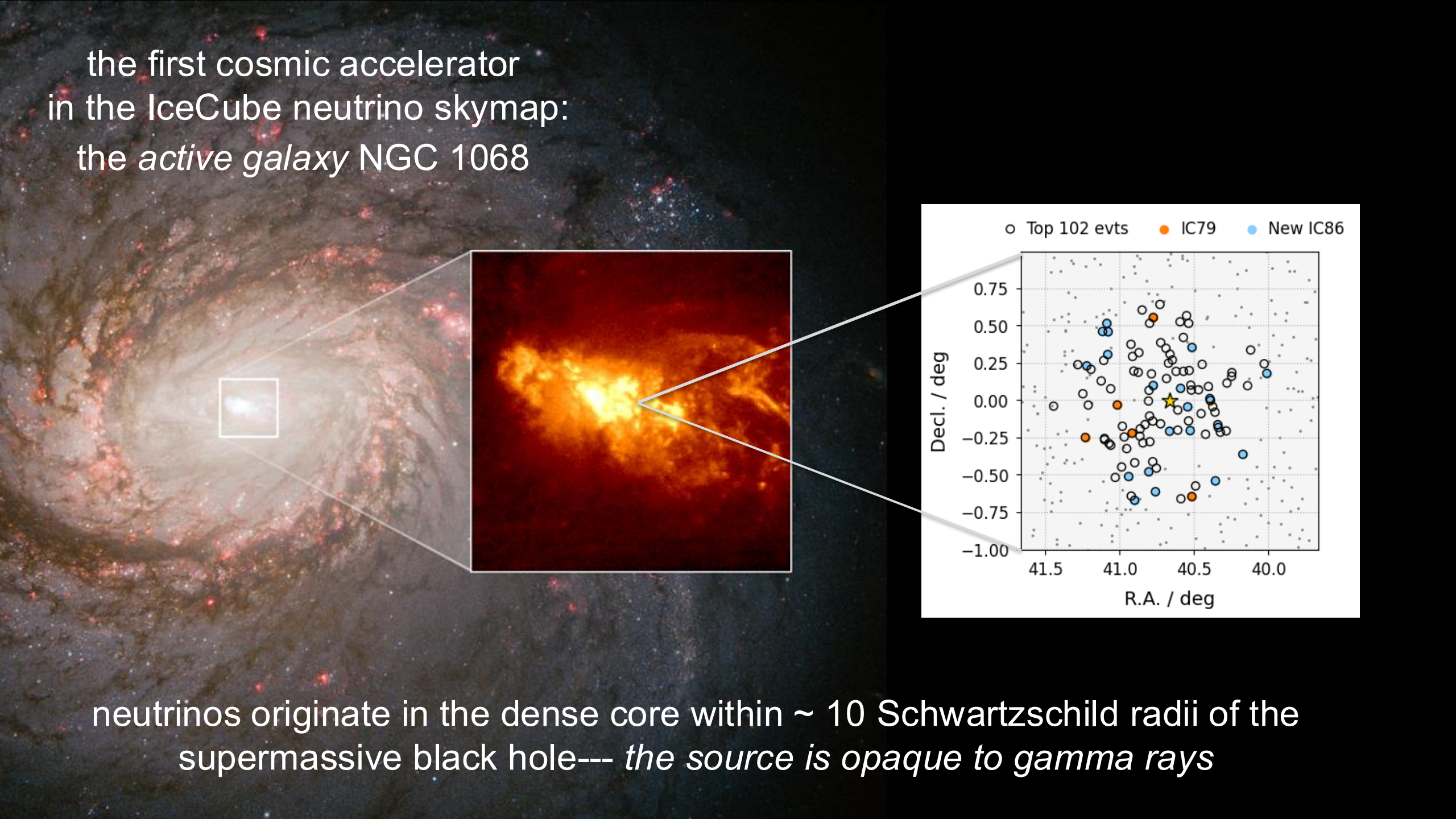
The First Decade of High-Energy Neutrino Astronomy and Neutrino Physics

francis halzen

- High-Energy Neutrino Astronomy
- IceCube and the discovery of Cosmic Neutrinos: First Surprises
- The First Cosmic Accelerators: Active Galaxies

→ resolves the puzzles associated with the diffuse flux

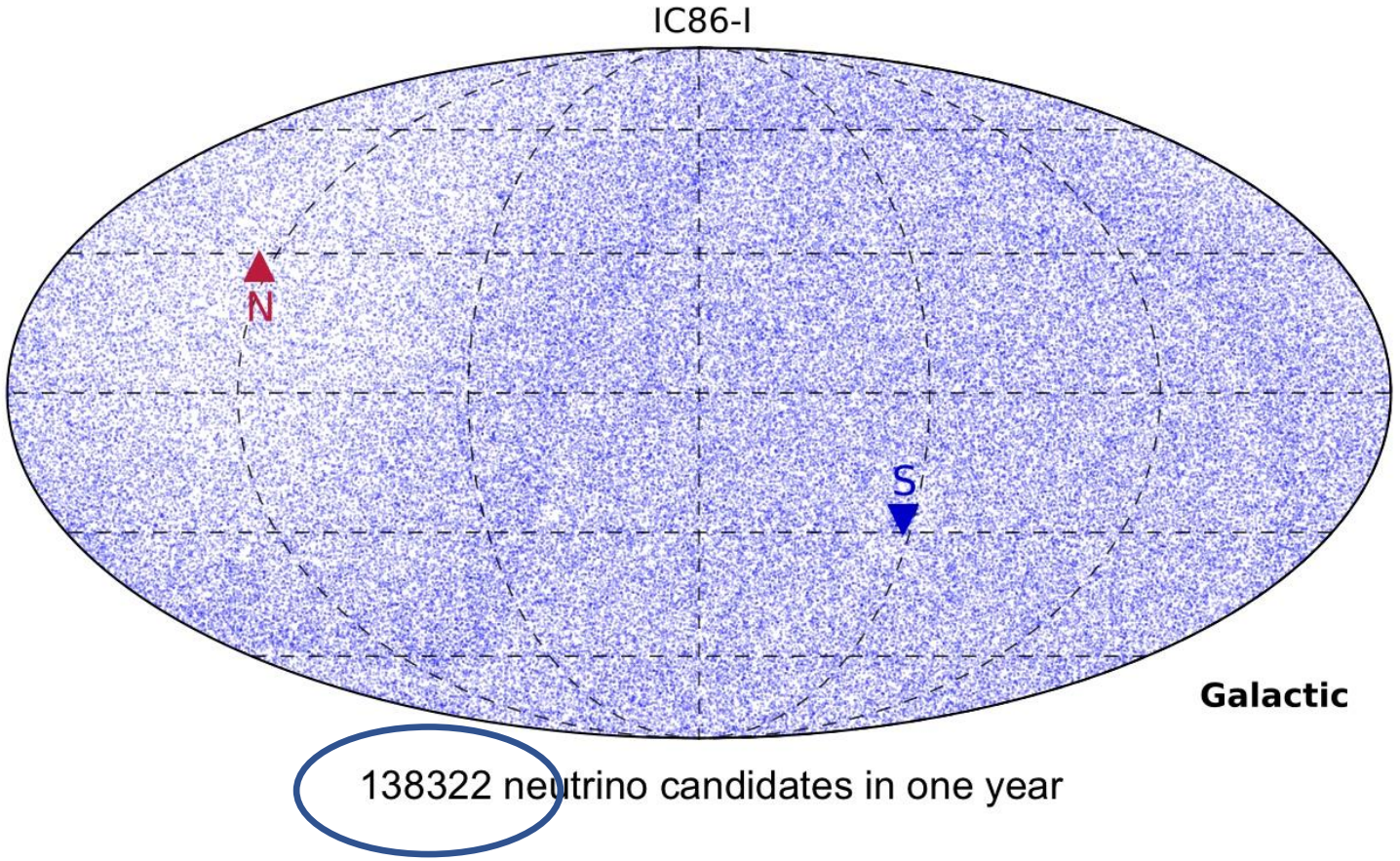
the first cosmic accelerator
in the IceCube neutrino skymap:
the *active galaxy* NGC 1068



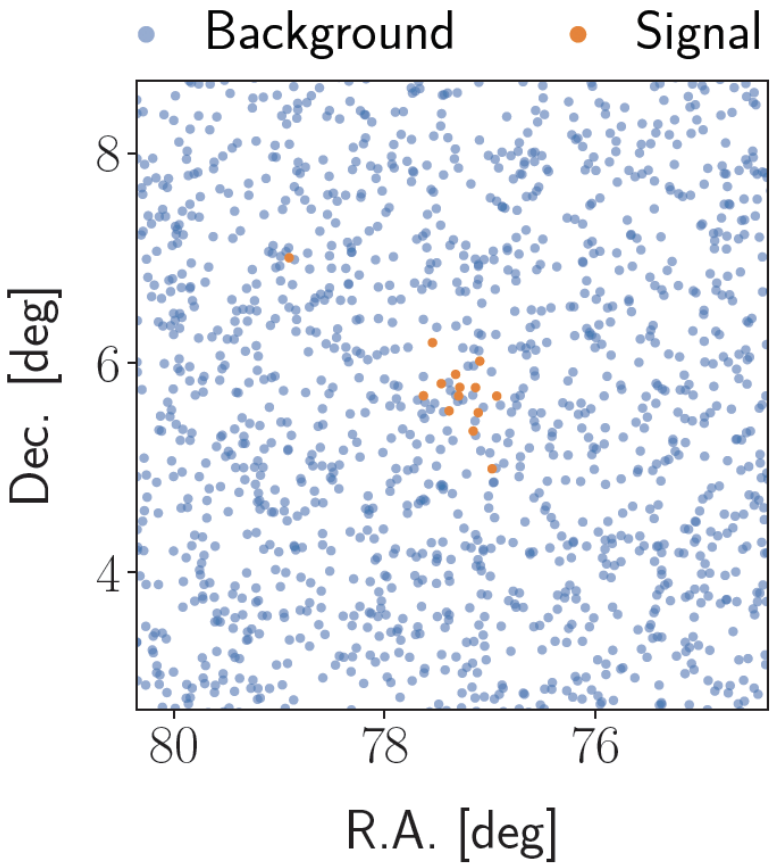
neutrinos originate in the dense core within ~ 10 Schwartzschild radii of the supermassive black hole--- *the source is opaque to gamma rays*

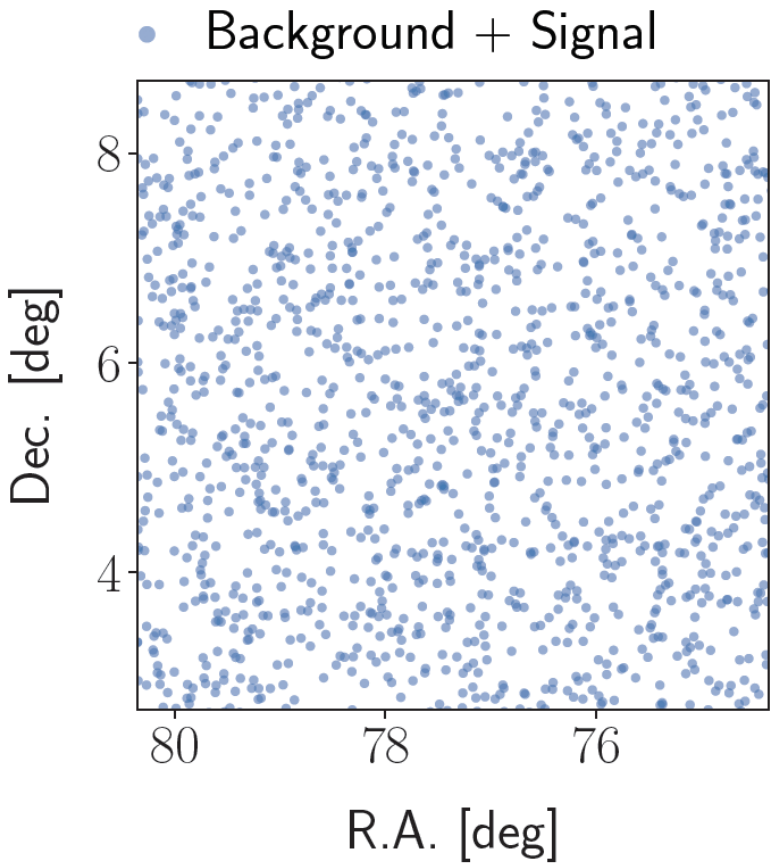
one year of IceCube neutrinos >100 GeV

(reaches neutrino purity of 97% but overwhelmingly atmospheric)

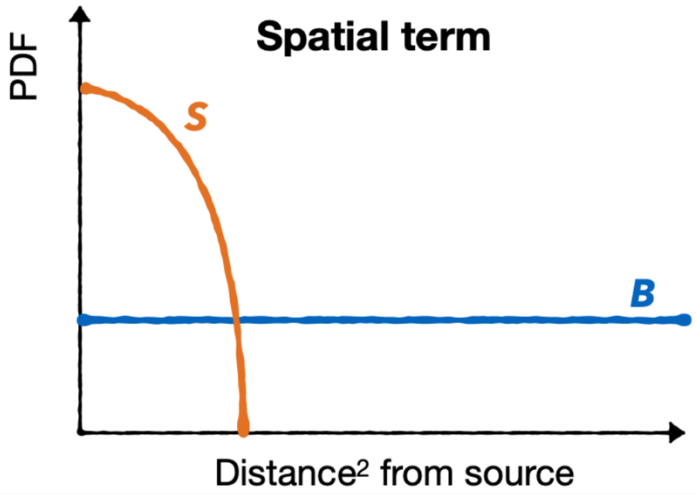


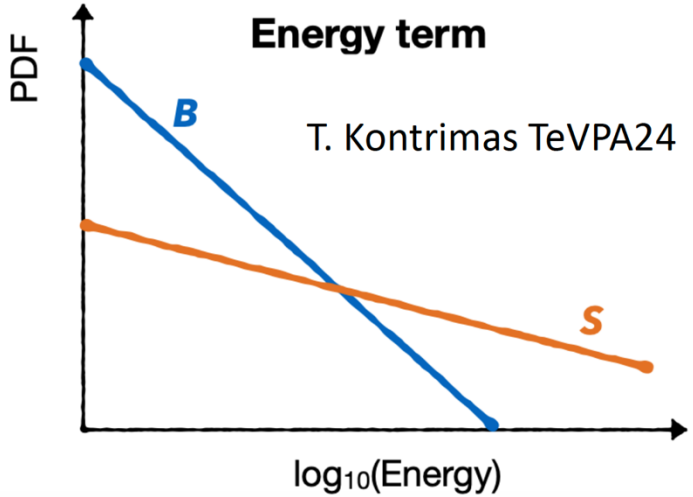
~200 cosmic neutrinos





- maximize the (model agnostic) likelihood *L* at each point in the sky
- usually, add energy term to the signal likelihood *S*

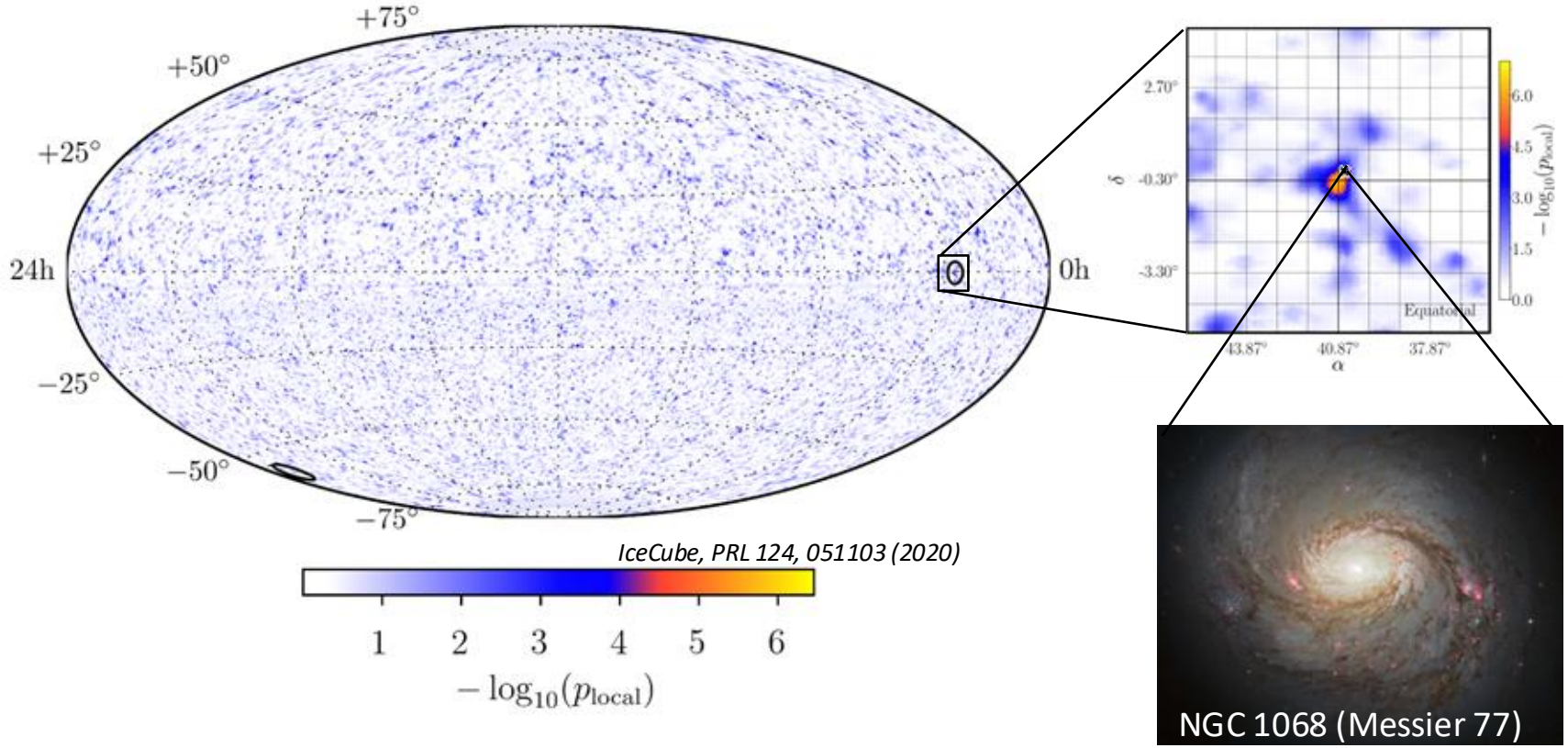




T. Kontrimas TeVPA24

NGC 1068

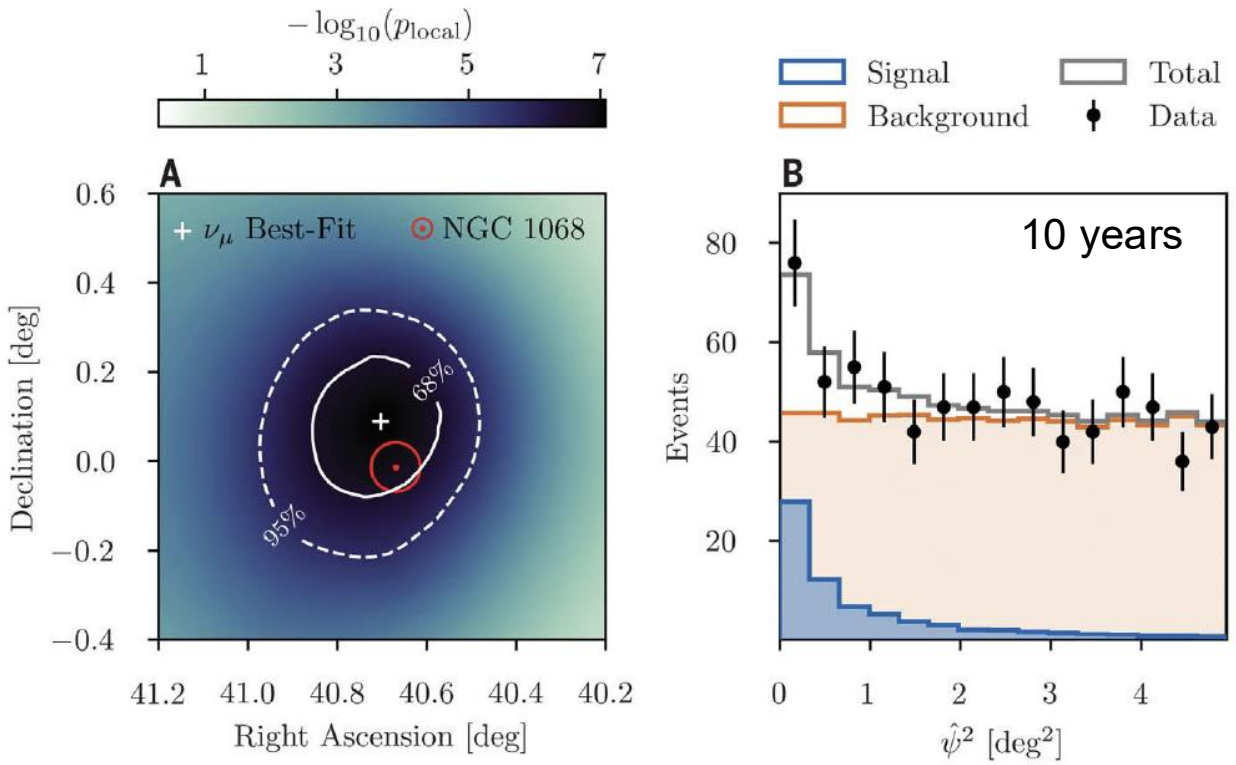
- most significant of 110 preselected sources
- leading hotspot in the neutrino sky

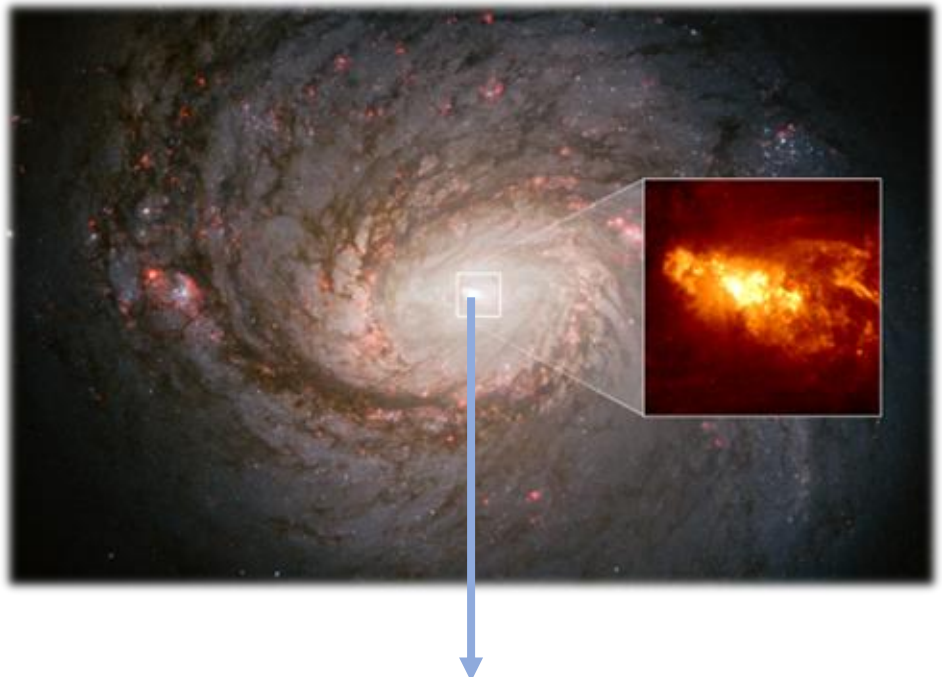


pre-trial p-value for clustering of high energy neutrinos

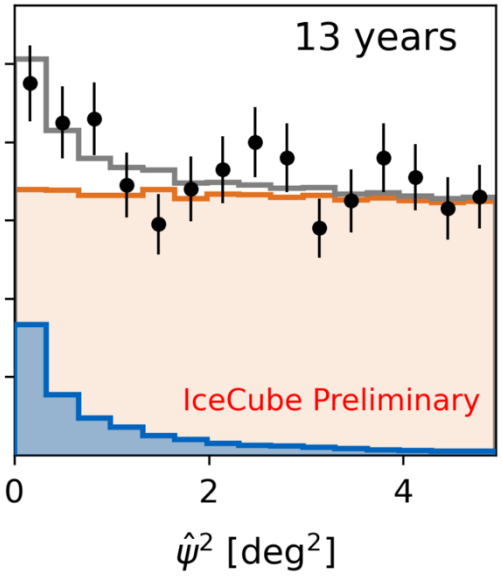
5.3(4.2)σ pre(post) trial [IceCube, PRL 124, 051103 (2020)]

80 high-energy neutrinos from the direction of the active galaxy NGC 1068



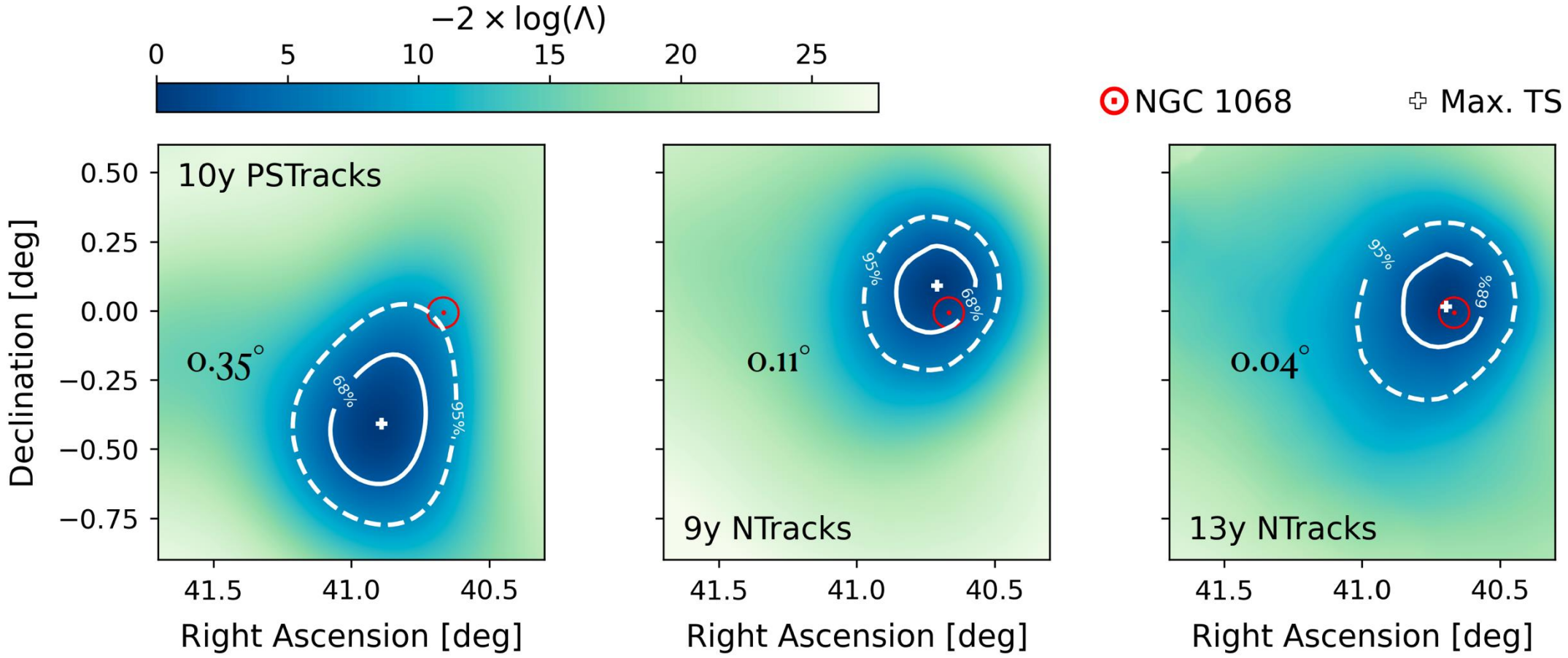


update: 80→100 neutrinos 10→13 years



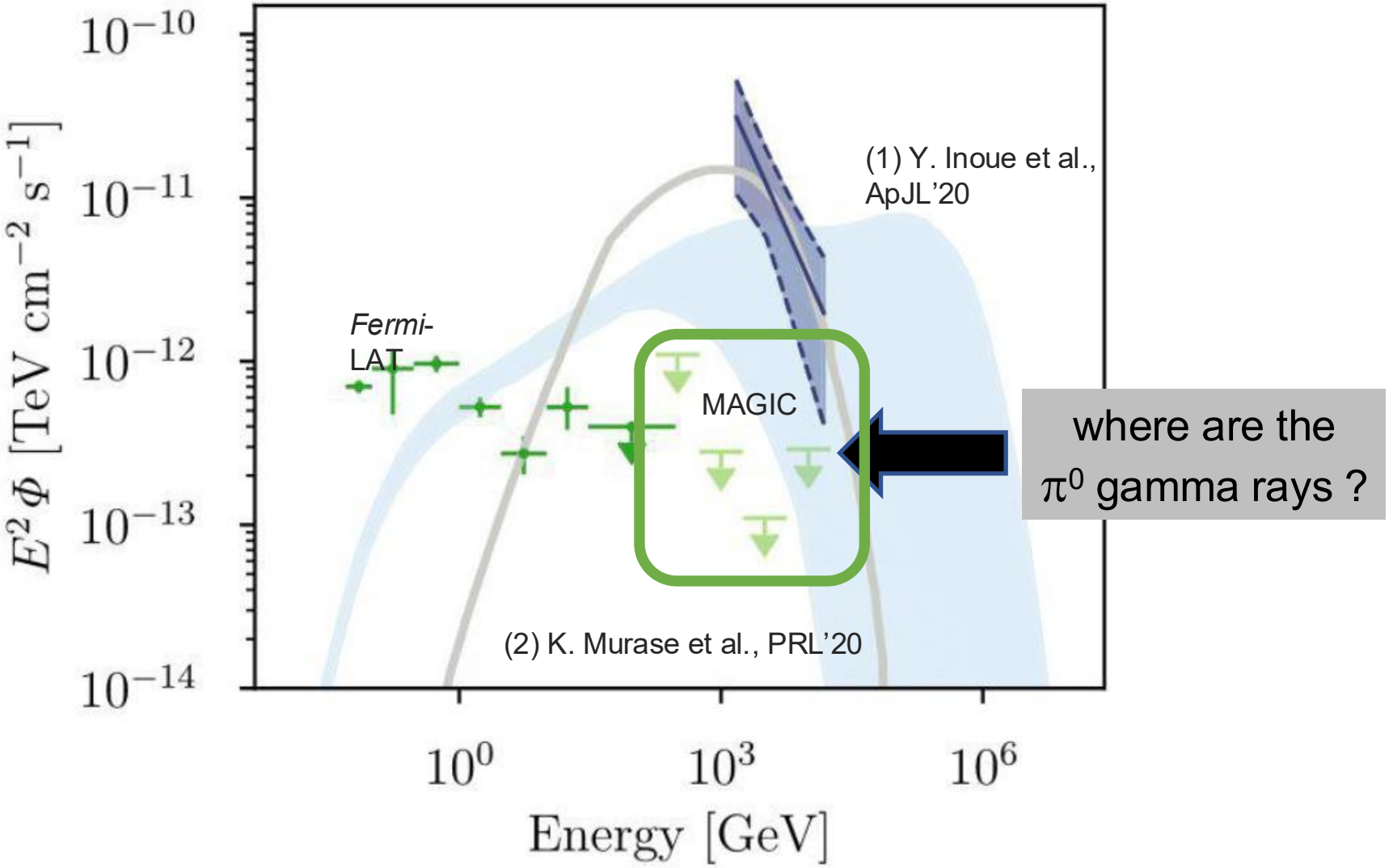
NGC 1068 comes into focus

0.35 deg → 0.1 deg → 0.04 deg



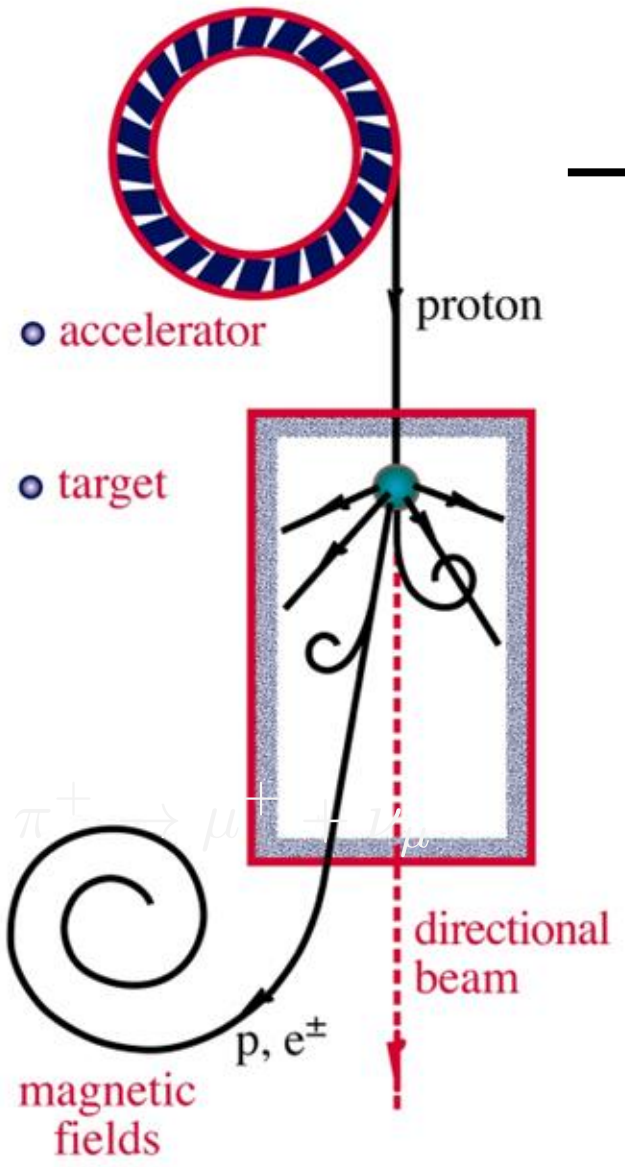
a gamma ray for every neutrino?

NGC 1068: an obscured cosmic accelerator



where are the π^0 gamma rays ?

ν and γ beams : heaven and earth



cosmic accelerators powered by gravity

supermassive black hole

nearby radiation

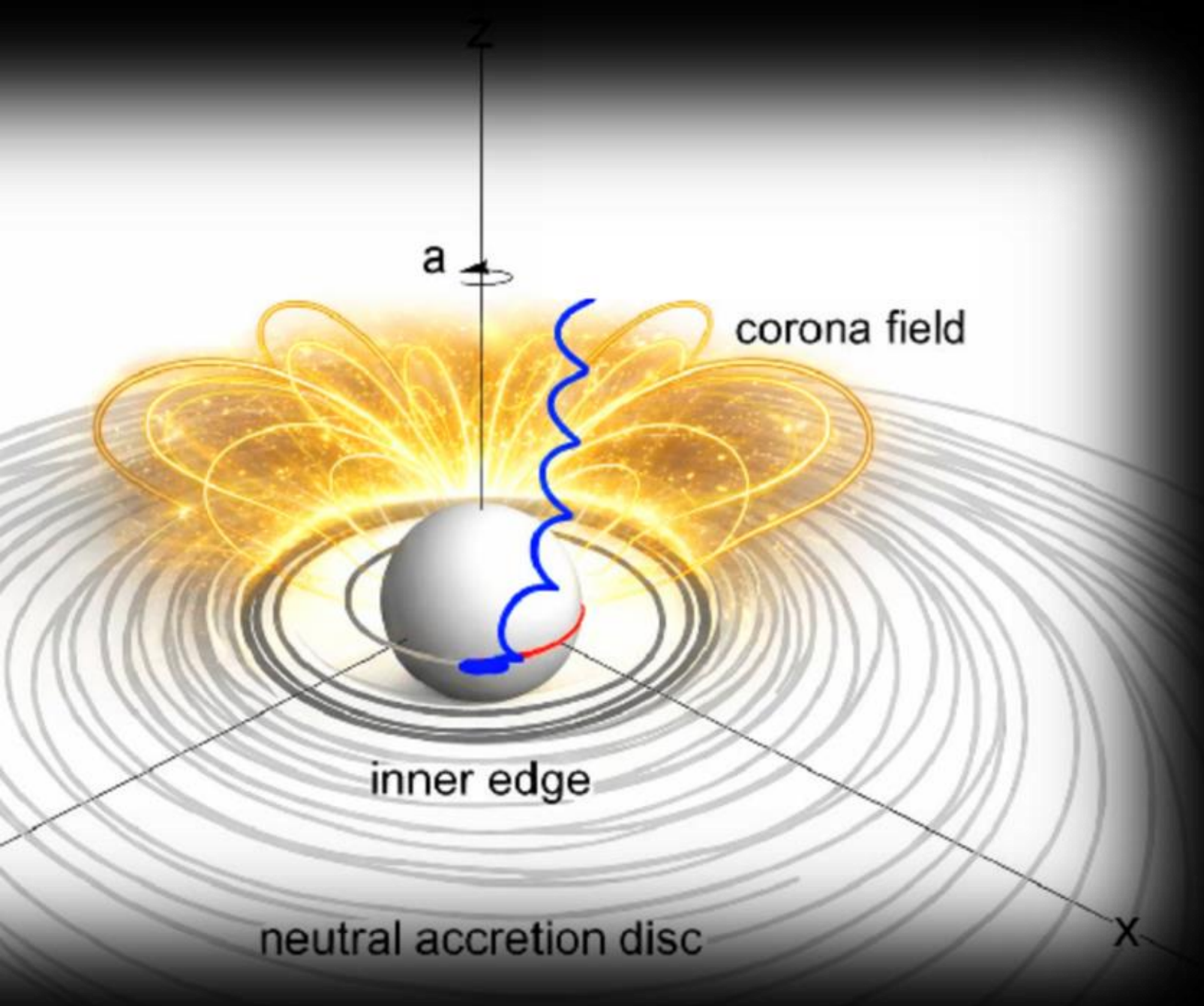
$$p + \gamma \rightarrow n + \pi^+$$

$$\pi^+ \rightarrow \mu^+ + \nu_\mu$$

$$\mu^+ \rightarrow e^+ + \nu_e + \bar{\nu}_\mu$$

$$\rightarrow p + \pi^0$$

$$\pi^0 \rightarrow \gamma + \gamma$$



- the accretion disk tabs the gravitational energy of the black hole very efficiently producing radiation with accretion rate

$$\dot{M}c^2 \sim 0.5\,M_{sun}/yr \sim 0.5\,M_{edd}$$

- velocity of particle falling into a black hole

$$v = \sqrt{\frac{2GM}{r}}$$

$$v = c\sqrt{\frac{R_g}{r}}$$

- magnetic reconnection

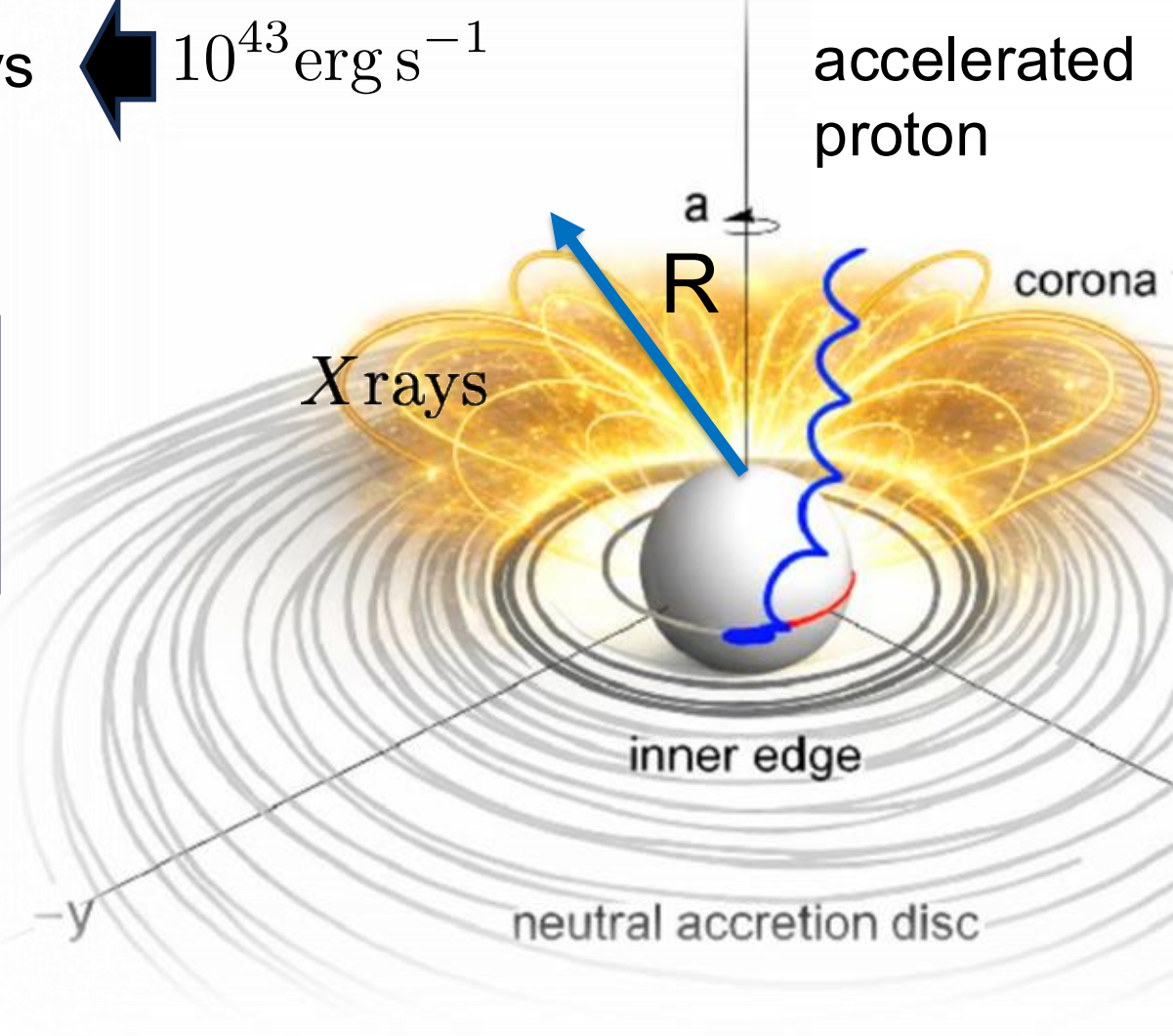
density of target X-rays $10^{43}\,\mathrm{erg\,s^{-1}}$

accelerated proton

$$L_\nu \sim L_p\,\sigma_{pX}\,n_{pX} \sim L_p\,\sigma_{pX}\,\frac{1}{R}\,\frac{L_X}{E_X}$$

size of the target

$$\mathrm{R} \simeq 10\,\mathrm{R_g}$$

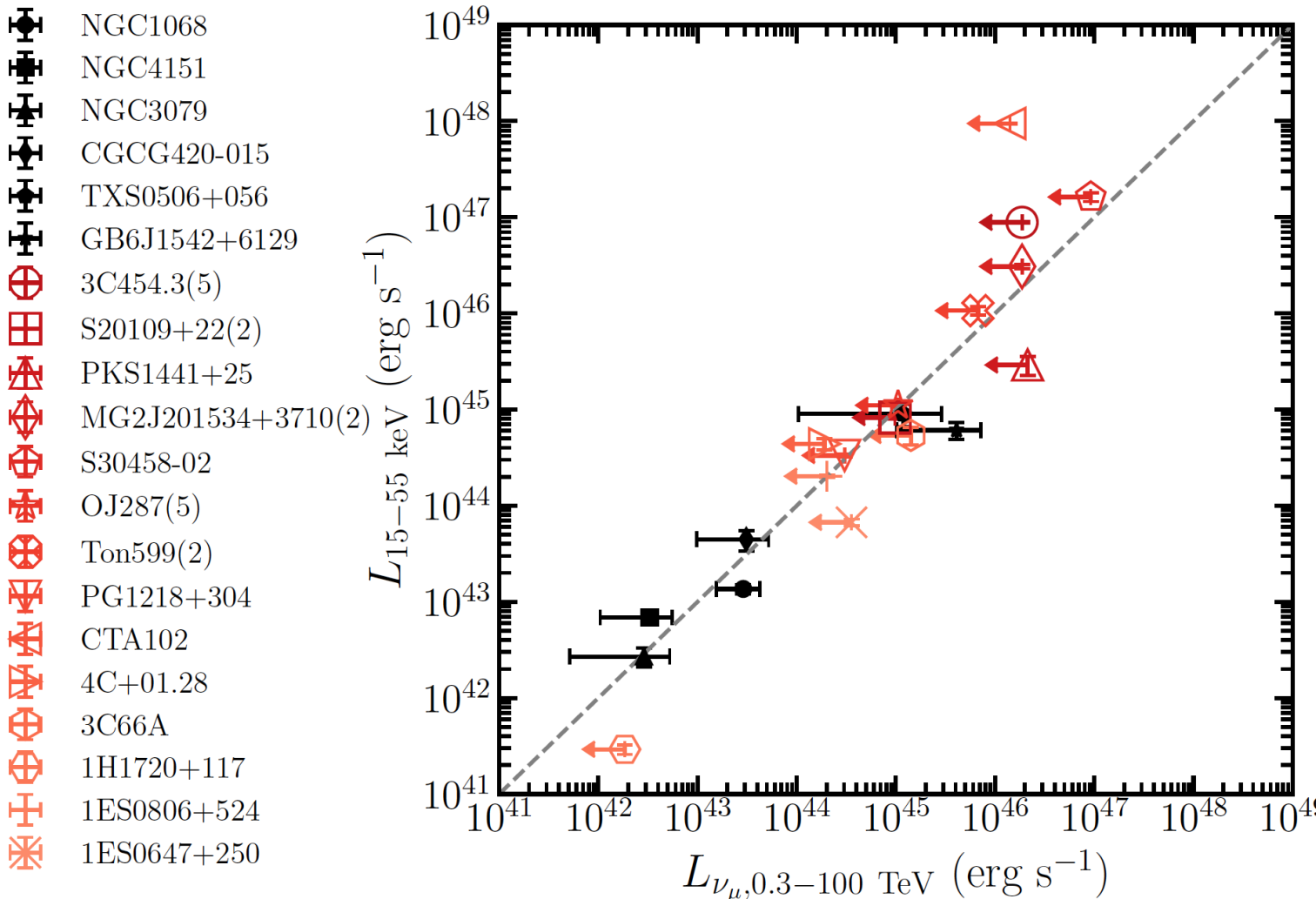


$$\tau_{\gamma\gamma} \sim [\sigma_{\gamma\gamma}/\sigma_{\mathrm{p}\gamma}]\,\tau_{\mathrm{p}\gamma} \sim 10^2 \rightarrow \pi^0 \text{ gamma rays absorbed}$$

M 87

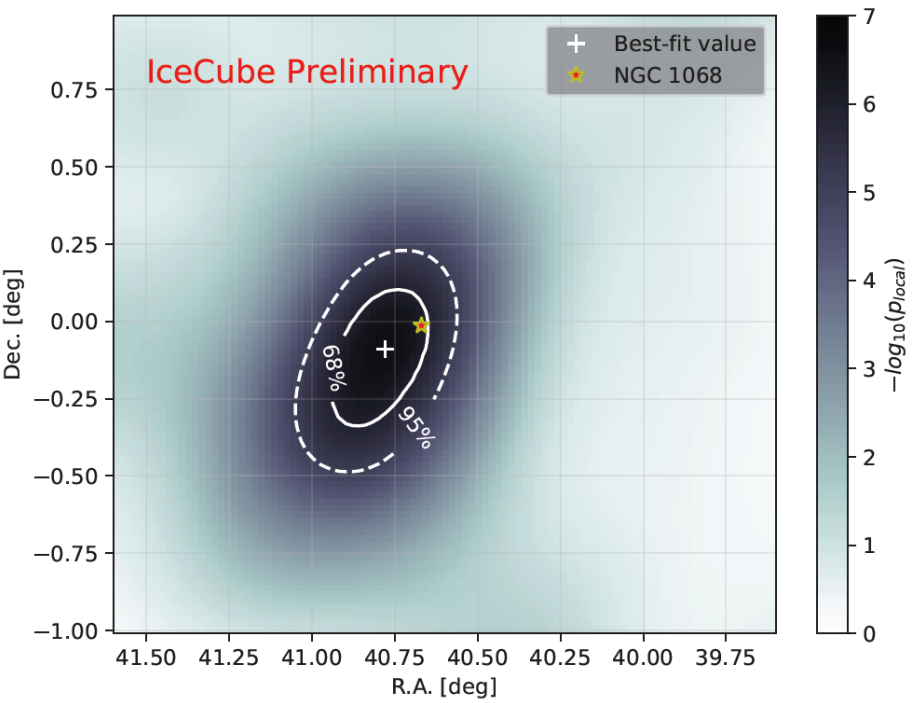


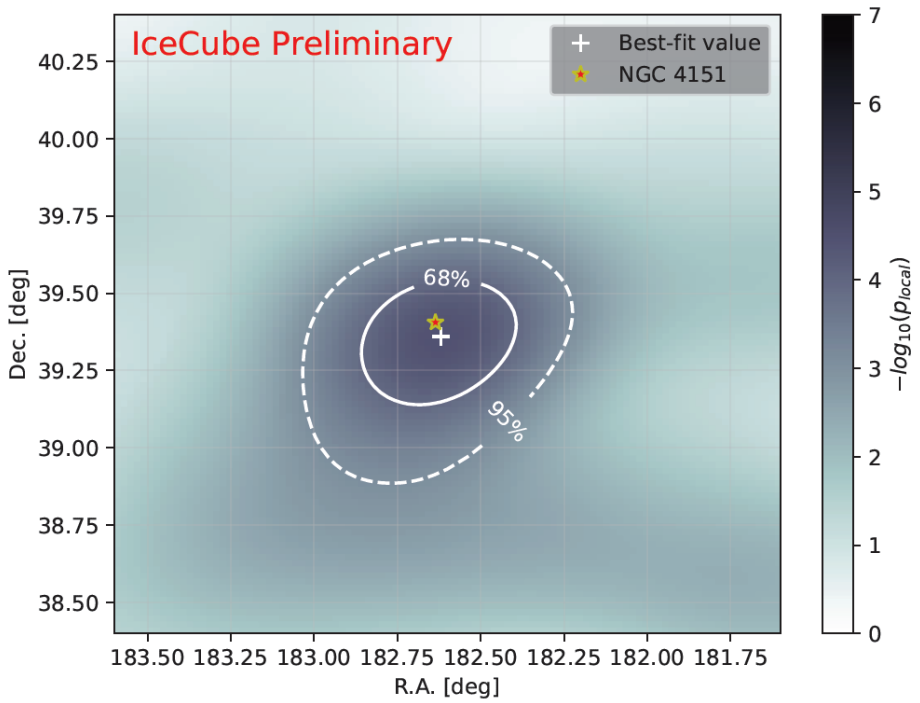
- hint from NGC 1068
- there is a correlation between the X-ray and neutrino flux
- select active galaxies with large X-ray fluxes



Emma Kun

look for X-ray (not gamma ray) bright sources …





- two brightest active galaxies discovered by Seyfert in 1943



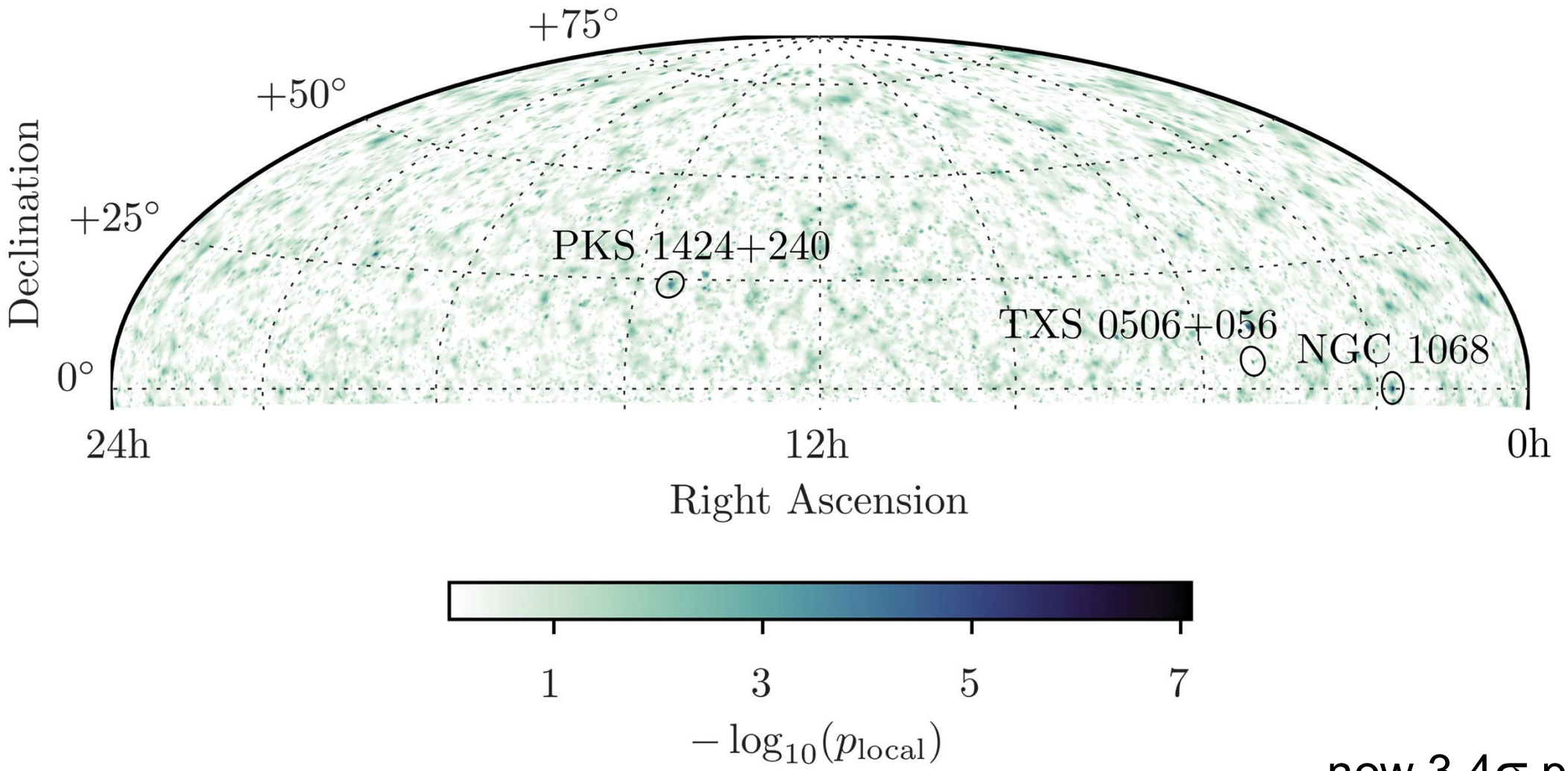
NUCLEAR EMISSION IN SPIRAL NEBULAE*

CARL K. SEYFERT† **1943**

ABSTRACT

Spectrograms of dispersion 37–200 A/mm have been obtained of six extragalactic nebulae with high-excitation nuclear emission lines superposed on a normal G-type spectrum. All the stronger emission lines from λ 3727 to λ 6731 found in planetaries like NGC 7027 appear in the spectra of the two brightest spirals observed, NGC 1068 and NGC 4151.

we observe an excess from an ensemble of 3 sources (also the 3 top sources)



now 3.4σ p-value

RESEARCH ARTICLE SUMMARY

NEUTRINO ASTROPHYSICS

Multimessenger observations of a flaring blazar coincident with high-energy neutrino IceCube-170922A

The IceCube Collaboration, *Fermi*-LAT, MAGIC, *AGILE*, ASAS-SN, HAWC, H.E.S.S, *INTEGRAL*, Kanata, Kiso, Kapteyn, Liverpool Telescope, Subaru, *Swift/NuSTAR*, VERITAS, and VLA/17B-403 teams*†

RESEARCH ARTICLE

NEUTRINO ASTROPHYSICS

Neutrino emission from the direction of the blazar TXS 0506+056 prior to the IceCube-170922A alert

IceCube Collaboration*†

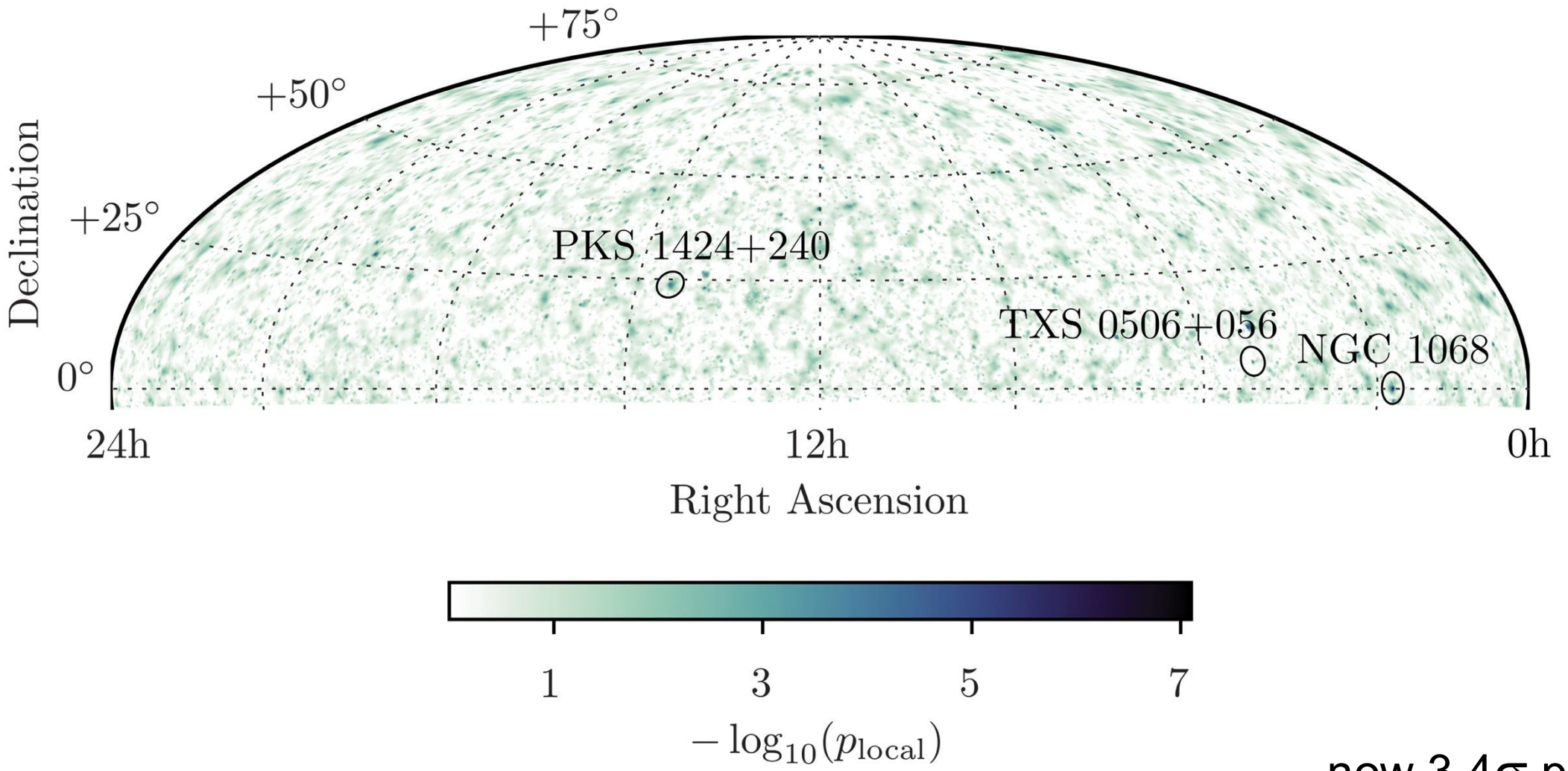
Optical Observations Reveal Strong Evidence for High Energy Neutrino Progenitor

V.M. Lipunov[1,2], V.G. Kornilov [1,2], K.Zhirkov[1], E. Gorbovskoy[2], N.M. Budnev[4], D.A.H.Buckley[3], R. Rebolo[5], M. Serra-Ricart[5], R. Podesta[9,10], N .Tyurina [2], O. Gress[4,2], Yu.Sergienko[8], V. Yurkov[8], A. Gabovich[8] , P.Balanutsa[2], I.Gorbunov[2], D.Vlasenko[1,2], F.Balakin[1,2], V.Topolev[1], A.Pozdnyakov[1], A.Kuznetsov[2], V.Vladimirov[2], A. Chasovnikov[1], D. Kuvshinov[1,2], V.Grinshpun[1,2], E.Minkina [1,2], V.B.Petkov [7], S.I.Svertilov[2,6] , C. Lopez[9], F. Podesta[9], H.Levato[10], A. Tlatov[11] B. Van Soelen[12], S. Razzaque[13], M. Böttcher[14]

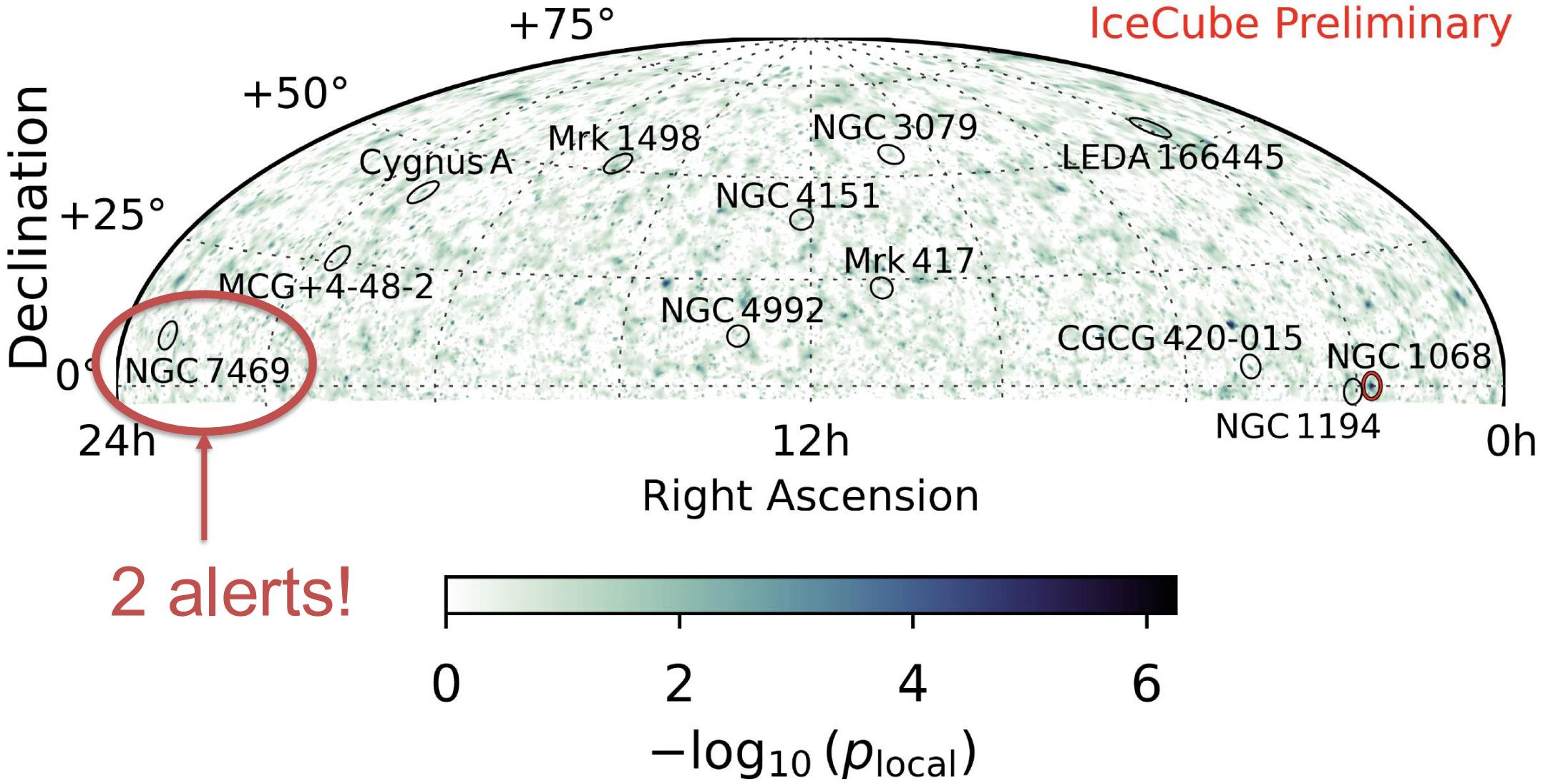
TXS 0506+056 detected in 2017

- multimessenger
- IceCube archival data
- observation of optical flash 2hours after IC170922 associates the neutrino with the source at the 50σ level

we observe an excess from an ensemble of 3 (*gamma ray*) sources

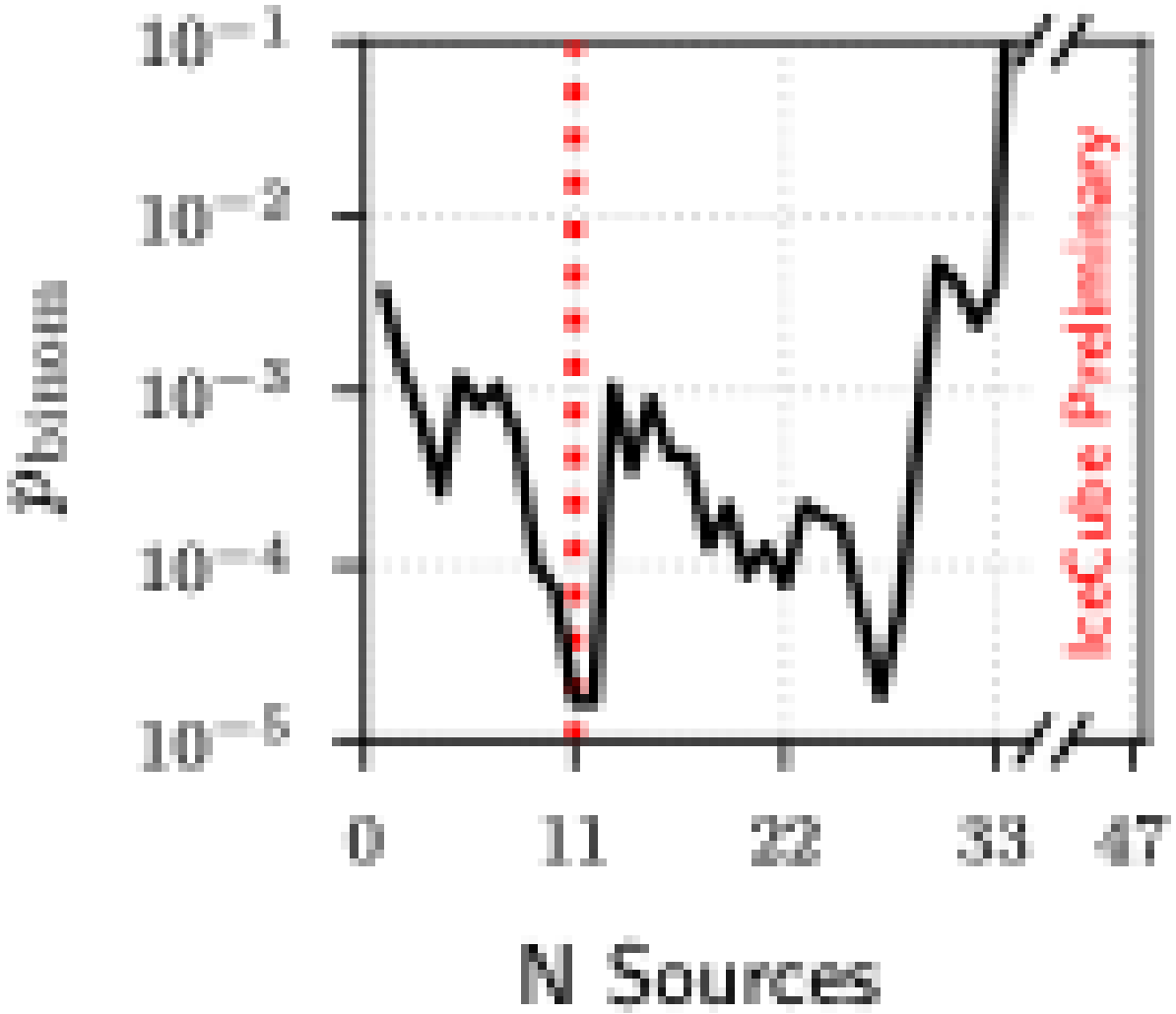


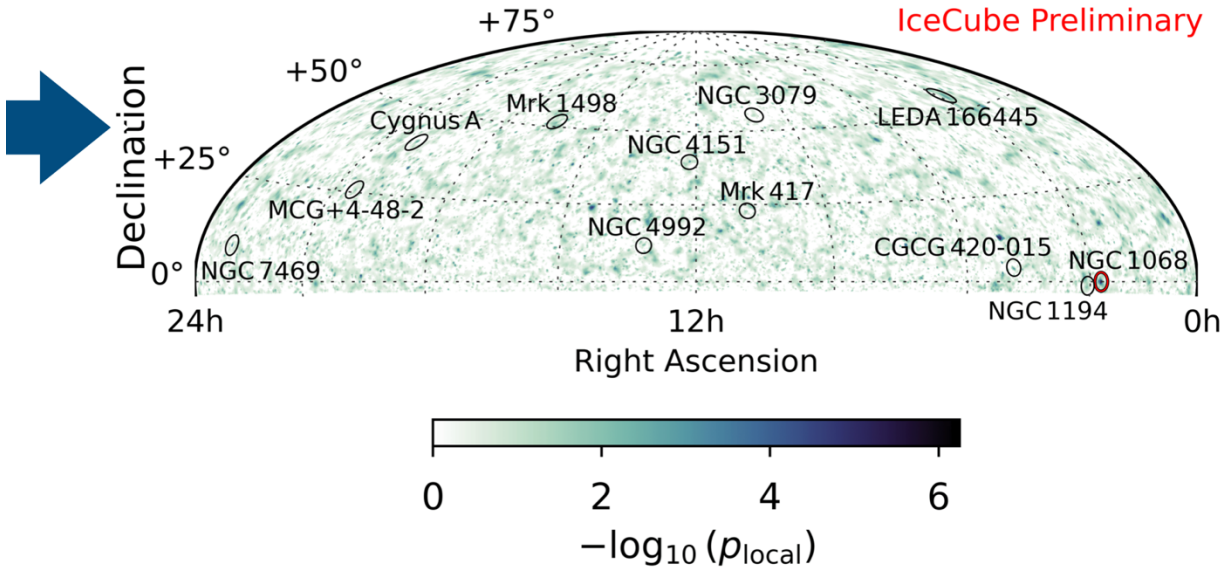
now 3.4σ p-value



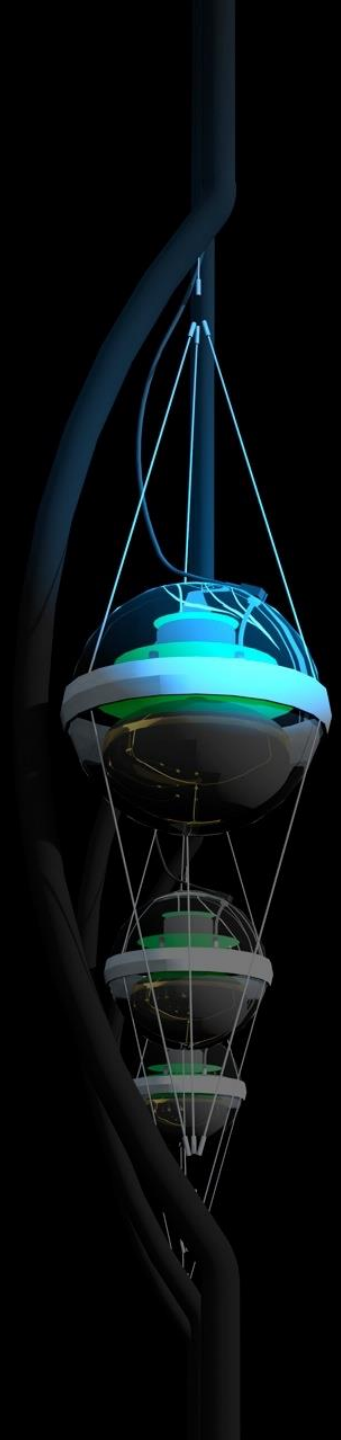
excess from an ensemble of *X-ray bright (non-jetted)* active galaxies

Binomial test of non-jetted sources





- Binomial Test: Probability of finding a signal from 47 AGNs too weak to be identified individually
- Result: 3.3σ excess for 11 sources (excluding NGC1068)



neutrino astronomy 2025

- it exists
- more neutrinos, better neutrinos, more telescopes
- closing in on cosmic ray sources a century after their discovery
- Galactic PeVatrons?



icecube.wisc.edu

THE ICECUBE COLLABORATION



ALIA 1

1

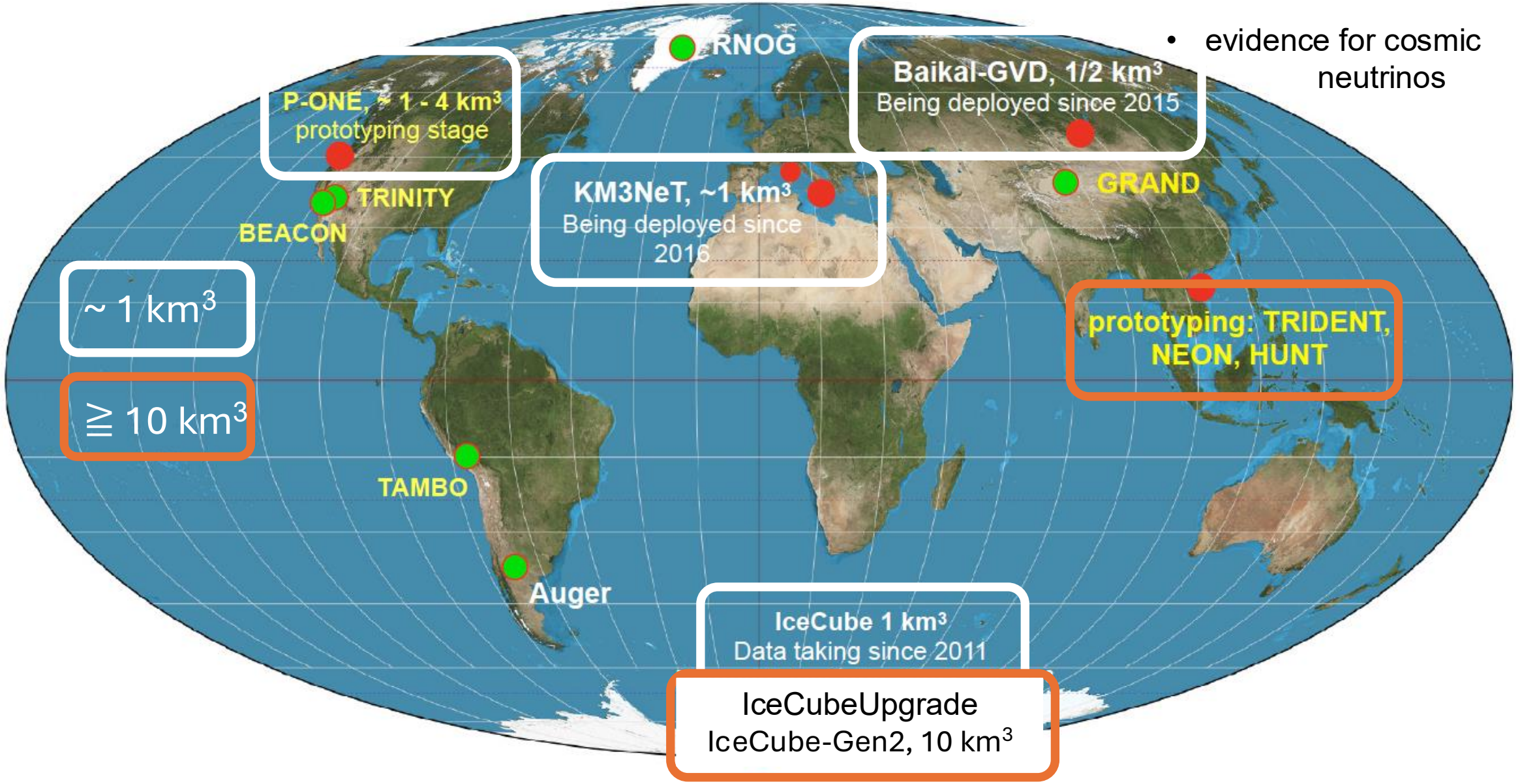
ED KINGDOM 1

UNITED STATES 25

IceCube
South Pole Neutrino Observatory

and there is more…

- from discovery to astronomy and neutrino physics: rapid progress!

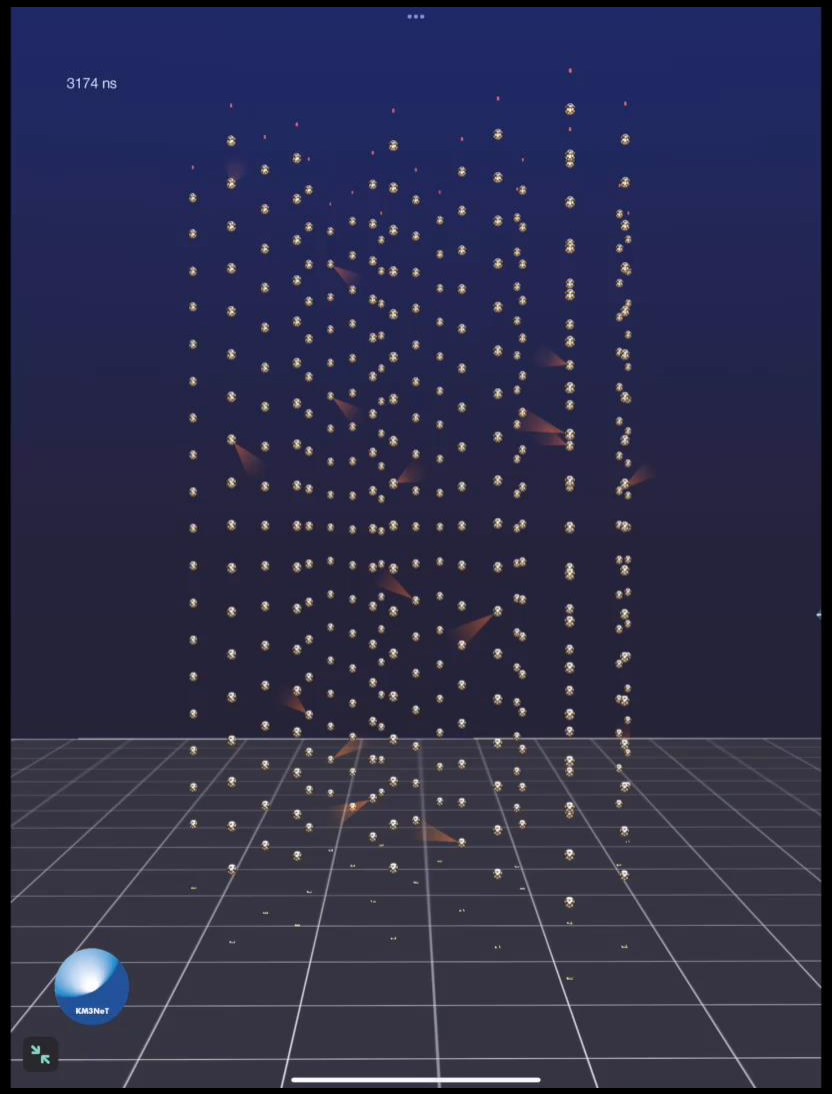


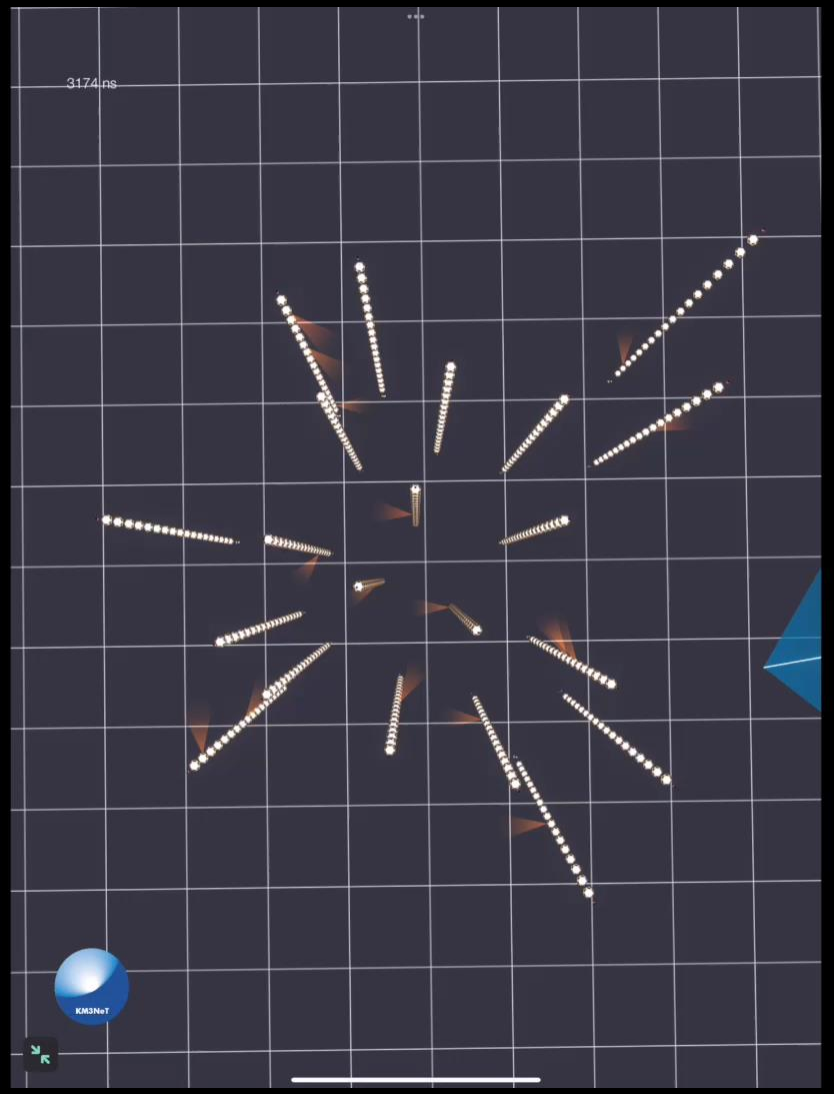
- evidence for cosmic neutrinos

- megaton GeV detectors: IceCube Upgrade, ORCA, HUNT

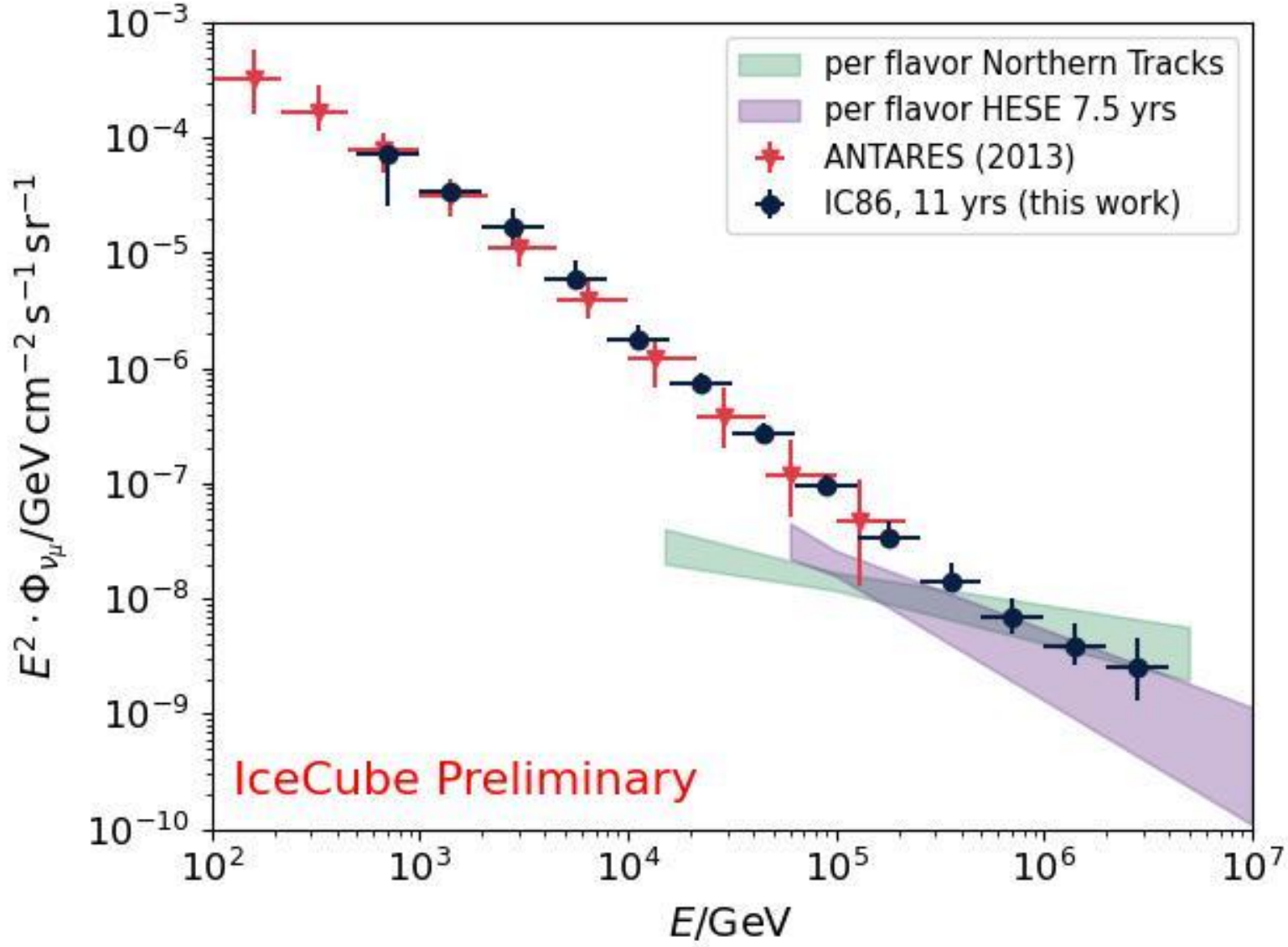
- EeV detectors

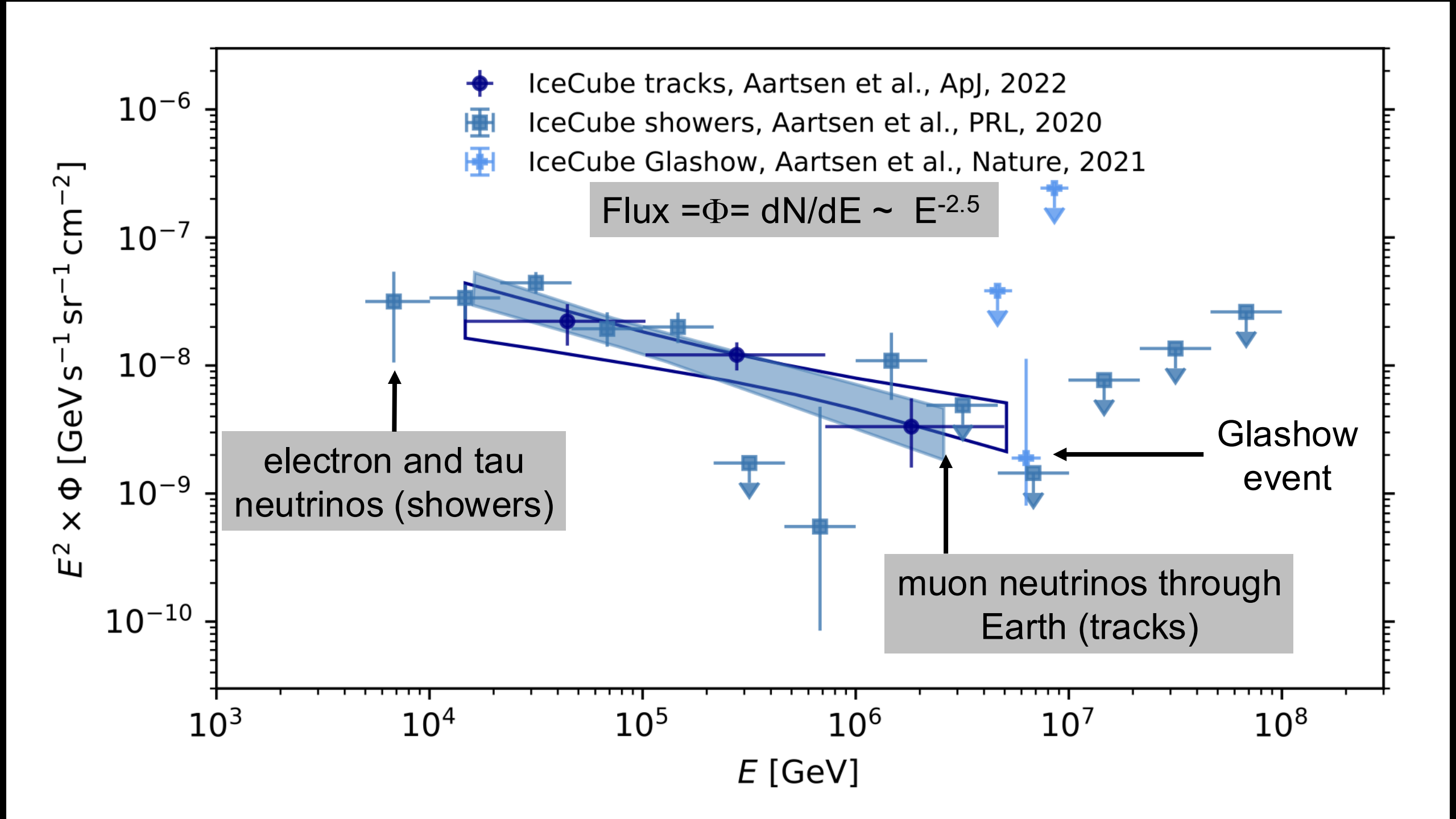
Uncharted Territory: E_ν = 220 PeV

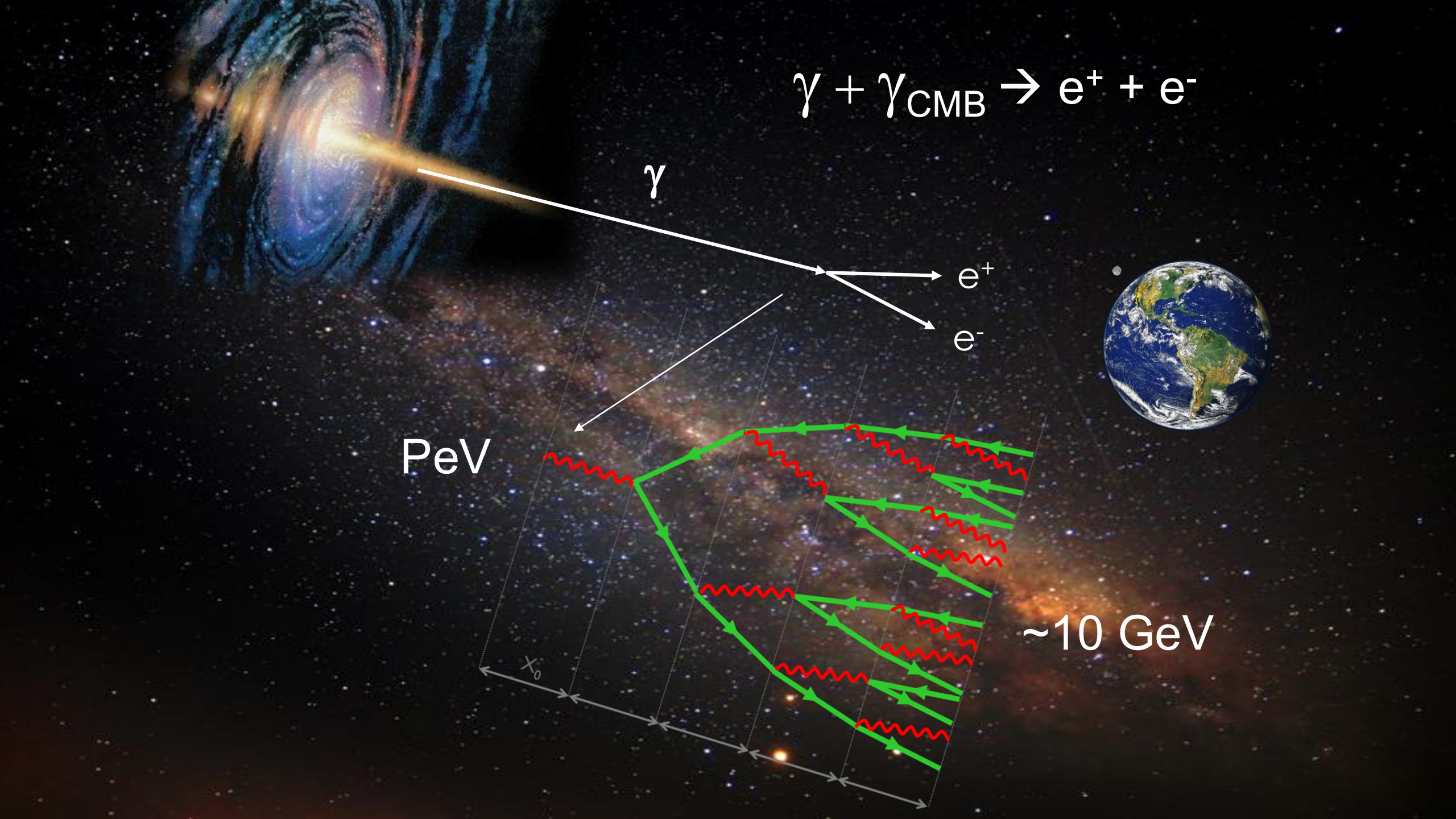




techniques mature (and improving): detectors agree on the muon neutrino spectrum

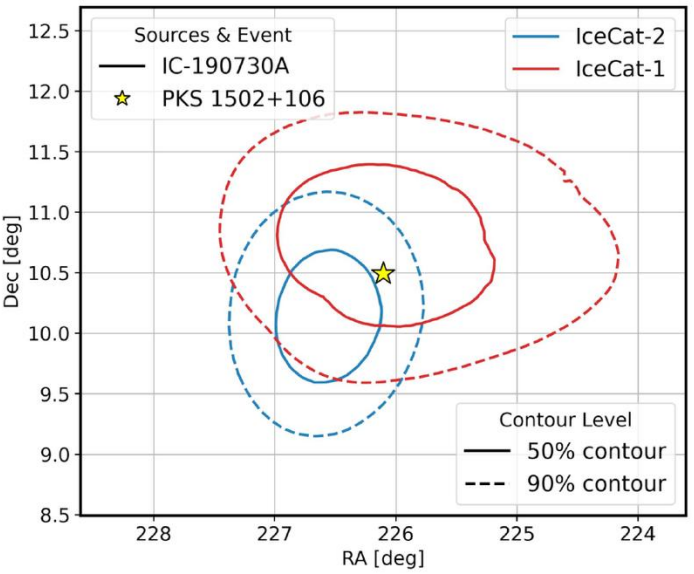



IceCube tracks, Aartsen et al., ApJ, 2022
IceCube showers, Aartsen et al., PRL, 2020
IceCube Glashow, Aartsen et al., Nature, 2021
Flux =Φ= dN/dE ~ E^-2.5
electron and tau neutrinos (showers)
Glashow event
muon neutrinos through Earth (tracks)
E² x Φ [GeV s⁻¹ sr⁻¹ cm⁻²]
E [GeV]
10⁻⁶
10⁻⁷
10⁻⁸
10⁻⁹
10⁻¹⁰
10³
10⁴
10⁵
10⁶
10⁷
10⁸


$\gamma + \gamma_{CMB} \rightarrow e^+ + e^-$
γ
e^+
e^-
PeV
~10 GeV
X_0

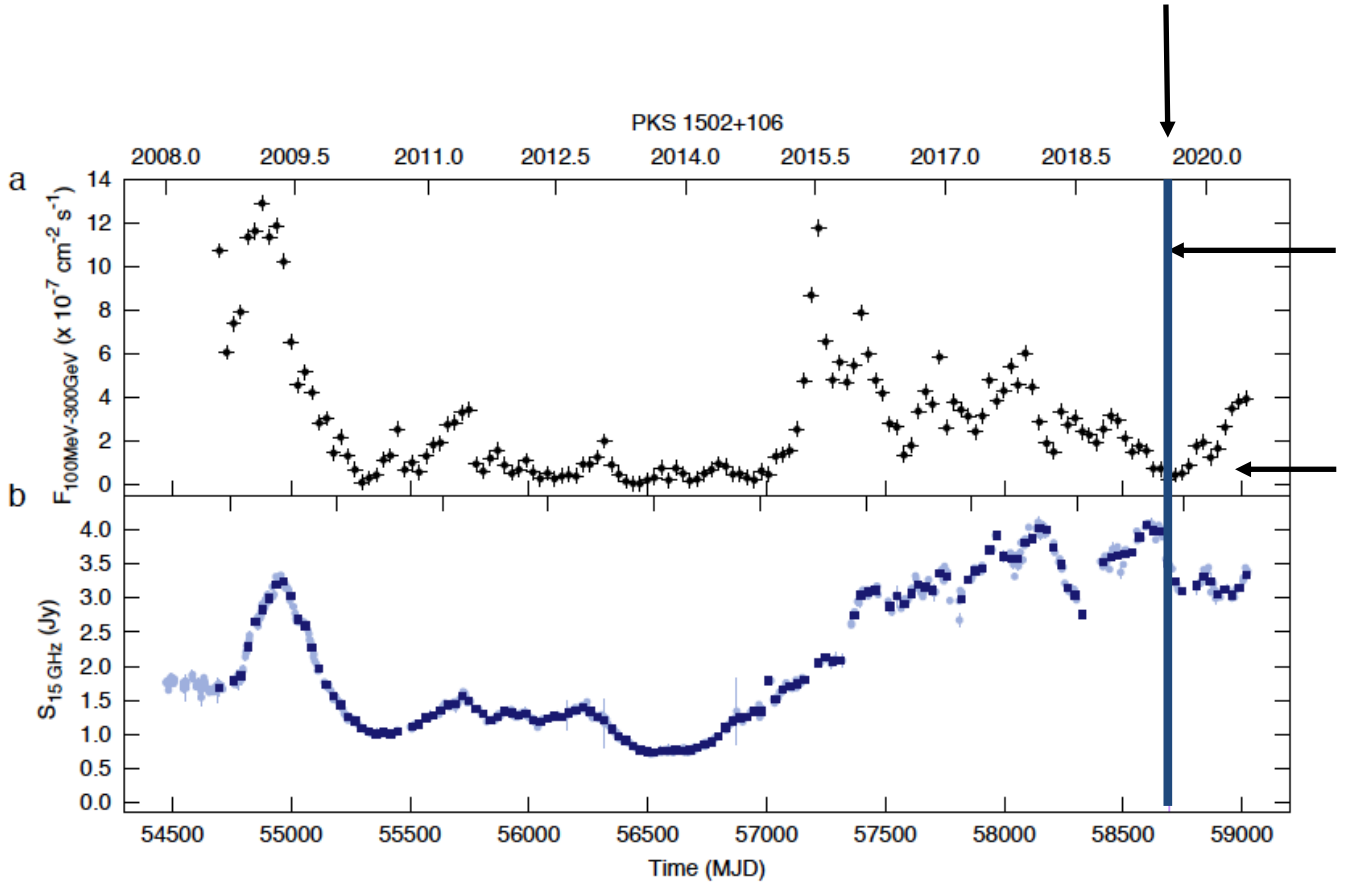
PKS 1502+106

neutrino



γ-ray

radio



300 TeV neutrino produced

thick target formed and neutrinos produced: photons absorbed

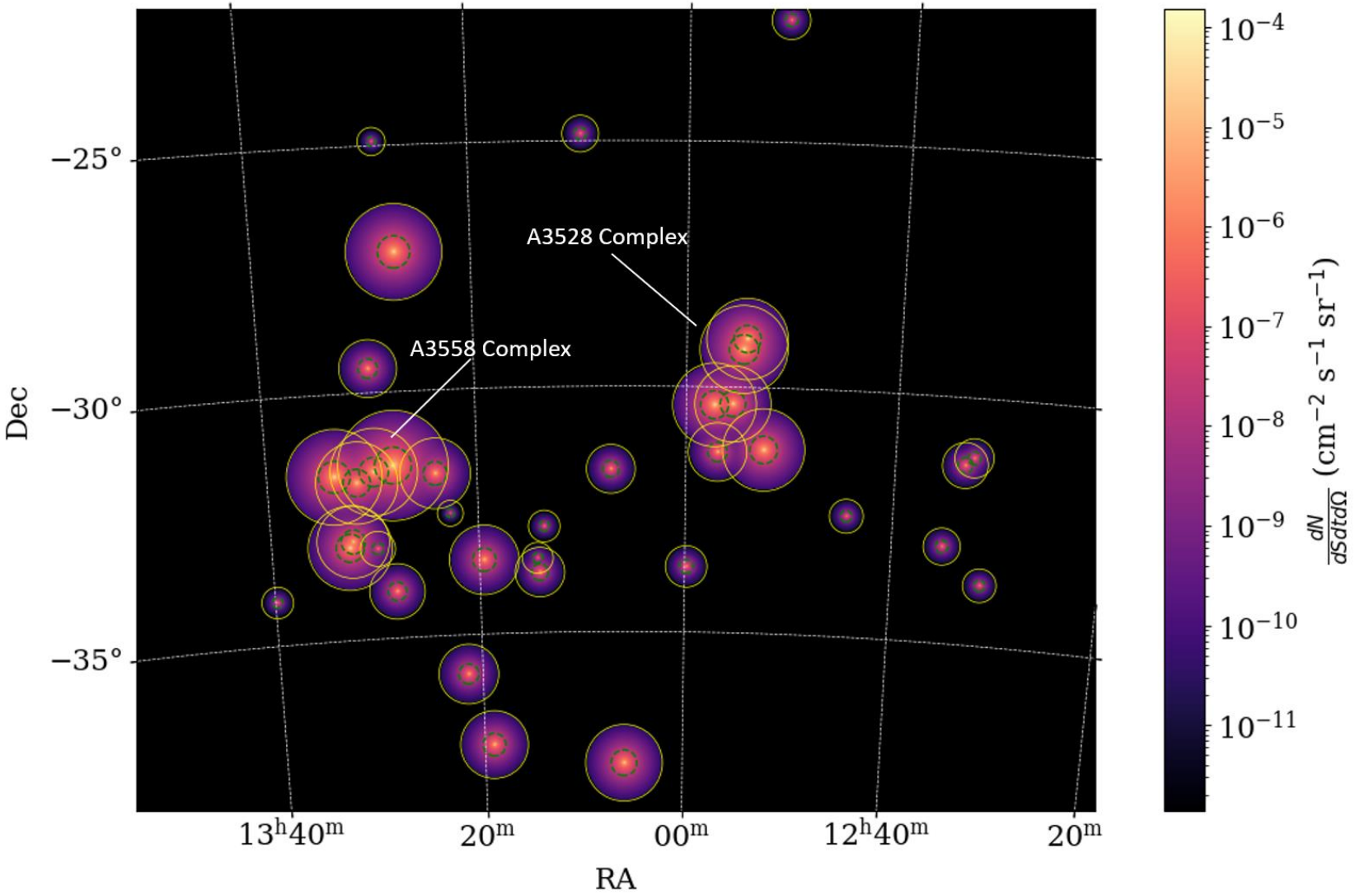
2009.09792 [astro-ph.HE]

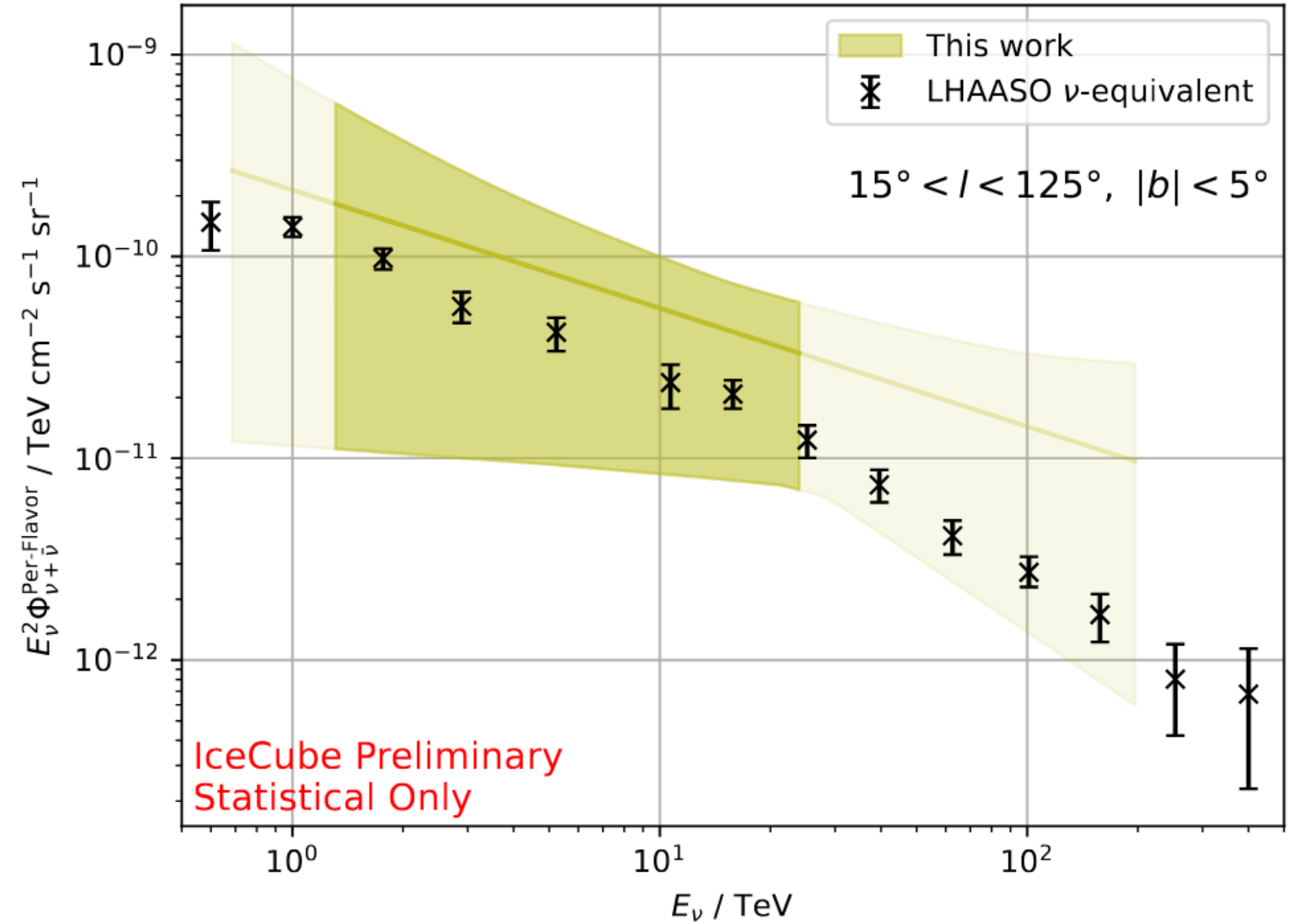
solution to the extragalactic cosmic ray problem?

clusters of superclusters

- neutrinos produced in individual active galaxies
- higher energy cosmic rays produced over the larger complex à la Hillas
- neutrino fluxes lower because of low target densities of larger scales
- need the next-generation detectors

the Shapley cluster of superclusters




This work
LHAASO ν-equivalent
15° < l < 125°, |b| < 5°
IceCube Preliminary
Statistical Only
$E_\nu^2 \Phi_{\nu+\bar{\nu}}^{\text{Per-Flavor}}$ / TeV cm^{-2} s^{-1} sr^{-1}
E_ν / TeV
10^{-9}
10^{-10}
10^{-11}
10^{-12}
10^{0}
10^{1}
10^{2}